

SEMESTER 3

MECHATRONICS ENGINEERING

SEMESTER S3

MATHEMATICS FOR ELECTRICAL SCIENCE AND PHYSICAL SCIENCE – 3

(Common to B & C Groups)

Course Code	GYMAT301	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Basic knowledge in complex numbers.	Course Type	Theory

Course Objectives:

1. To introduce the concept and applications of Fourier transforms in various engineering fields.
2. To introduce the basic theory of functions of a complex variable, including residue integration and conformal transformation, and their applications

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fourier Integral, From Fourier series to Fourier Integral, Fourier Cosine and Sine integrals, Fourier Cosine and Sine Transform, Linearity, Transforms of Derivatives, Fourier Transform and its inverse, Linearity, Transforms of Derivative. (Text 1: Relevant topics from sections 11.7, 11.8, 11.9)	9
2	Complex Function, Limit, Continuity, Derivative, Analytic functions, Cauchy-Riemann Equations (without proof), Laplace's Equations, Harmonic functions, Finding harmonic conjugate, Conformal mapping, Mappings of $w = z^2$, $w = e^z$, $w = \frac{1}{z}$, $w = \sin z$. (Text 1: Relevant topics from sections 13.3, 13.4, 17.1, 17.2, 17.4)	9
3	Complex Integration: Line integrals in the complex plane (Definition & Basic properties), First evaluation method, Second evaluation method, Cauchy's integral theorem (without proof) on simply connected domain, Independence of path, Cauchy integral theorem on multiply connected domain (without proof), Cauchy Integral formula (without proof). (Text 1: Relevant topics from sections 14.1, 14.2, 14.3)	9

4	Taylor series and Maclaurin series, Laurent series (without proof), Singularities and Zeros – Isolated Singularity, Poles, Essential Singularities, Removable singularities, Zeros of Analytic functions – Poles and Zeros, Formulas for Residues, Residue theorem (without proof), Residue Integration- Integral of Rational Functions of $\cos\theta$ and $\sin\theta$. (Text 1: Relevant topics from sections 15.4, 16.1, 16.2, 16.3, 16.4)	9
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Determine the Fourier transforms of functions and apply them to solve problems arising in engineering.	K3
CO2	Understand the analyticity of complex functions and apply it in conformal mapping.	K3
CO3	Compute complex integrals using Cauchy's integral theorem and Cauchy's integral formula.	K3
CO4	Understand the series expansion of complex function about a singularity and apply residue theorem to compute real integrals.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	-	2	-	-	-	-	-	-	-	2
CO2	3	3	-	2	-	-	-	-	-	-	-	2
CO3	3	3	-	2	-	-	-	-	-	-	-	2
CO4	3	3	-	2	-	-	-	-	-	-	-	2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Advanced Engineering Mathematics	Erwin Kreyszig	John Wiley & Sons	10 th edition, 2016

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Complex Analysis	Dennis G. Zill, Patrick D. Shanahan	Jones & Bartlett	3 rd edition, 2015
2	Higher Engineering Mathematics	B. V. Ramana	McGraw-Hill Education	39 th edition, 2023
3	Higher Engineering Mathematics	B.S. Grewal	Khanna Publishers	44 th edition, 2018
4	Fast Fourier Transform - Algorithms and Applications	K.R. Rao, Do Nyeon Kim, Jae Jeong Hwang	Springer	1 st edition, 2011

SEMESTER S3

ELECTRICAL MACHINES & DRIVES

Course Code	PCMRT302	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. This course aims the students to learn about the introduction of basics of electrical machines and basic concept of power electronics and machine drives.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	DC MACHINES & TRANSFORMERS DC MACHINES Essential parts of a DC machine – EMF equation – DC generator – working principle – types of DC generators – no load and load characteristics of DC generators – DC motor – working principle – back emf – types of DC motors – equations for torque and power (simple numerical problems) TRANSFORMERS Working principle - Construction - core and shell type - emf equation – voltage transformation ratio (simple numerical problems) – concept of ideal transformer- transformer on no-load, short circuit and open circuit test on transformer (basic concept only) – losses in transformer	11
2	THREE PHASE & SINGLE PHASE INDUCTION MOTORS THREE PHASE INDUCTION MOTOR Constructional details – operation – concept of rotating magnetic field – slip – torque equation - (simple numerical problems) - torque-slip characteristics – starting methods of 3- pahse induction motors SINGLE PHASE INDUCTION MOTOR Working principle – double revolving field theory – different types – split phase – capacitor start – capacitor start- run	11

3	SPECIAL ELECTRICAL MACHINES & INTRODUCTION POWER ELECTRONICS SPECIAL ELECTRICAL MACHINES Universal motor – stepper motor -different types servomotor (mechanism only) - Synchronous motor INTRODUCTION POWER ELECTRONICS Introduction – SCR -symbol, construction and modes of operation – V-I characteristics- Basic concepts of Rectifier – single phase half-wave controlled rectifier with R load – fully controlled bridge rectifier with R load – basic concept of inverter	11
4	ELECTRICAL DRIVES Electrical Drives – Advantages - Parts of electrical drives - Choice of electric drives - Status of DC and AC drives - Dynamics of Electric drives - Fundamental torque equations – Speed torque conventions and multi-quadrant operation - Components of load torque - Nature and classification of load torque – Steady state stability – load equalization	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the basic concept of dc generator, motor and transformer	K2
CO2	The basic concepts of 3 phase and 1 phase induction motors.	K2
CO3	The basic concept of special electrical machines & introduction to power electronics	K2
CO4	The basic concept of machine drives	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	-	-	2	2	1	-	-	-	3
CO2	3	3	2	1	-	2	2	1	-	-	-	3
CO3	3	3	2	1	-	2	2	1	-	-	-	3
CO4	3	1	1	1	-	2	2	1	-	-	-	3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Theory and Performance of Electrical machines	J B Gupta	S K Kataria & sons	First Edition 1967; Fifteenth Edition 2015
2	Power Electronics	P.S. Bimbhra	Khanna Publishers, New Delhi	Fifth Edition 1990
3	Fundamentals of Electrical Drives	Gopal K. Dubey	Narosa Publishing House	Second Edition 2002

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Principles Of Electrical Machines	V.K. Mehta, Rohit Mehta	S Chand Publication	Reprint Edition 2006
2	A Text Book of Electrical Technology, Vol.II “AC & DC Machines”	Theraja B.L., Theraja A.K	publication division of Nirja construction & development (p) Ltd., New Delhi	First Edition 2005
3	Power Electronics Circuits, Devices and Applications	M. H. Rashid	Pearson Education	Fourth Edition 2017
4	Electric Drives, Concepts & Applications	VedamSubrahmanyam,	Tata McGraw Hill Education Pvt. Ltd, New Delhi	Second Edition 2017

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/108105017
2	https://onlinecourses.nptel.ac.in/noc22_ee06/preview
3	https://nptel.ac.in/courses/108101038
4	https://archive.nptel.ac.in/courses/108/104/108104140/

SEMESTER S3

MECHANICS OF SOLIDS

Course Code	PCMRT303	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Engineering Mechanics/ (GCEST103)	Course Type	Theory

Course Objectives:

1. This course helps the students to understand the concept of stress and strain in different types of structure/machine under various loading conditions
2. The course also covers simple and compound stresses due to forces, stresses and deflection in beams due to bending, torsion in circular section, strain energy, different theories of failure, stress in thin cylinder and spheres

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Mechanical Properties of materials, stress, strain, Stress–Strain curves of Ductile and Brittle Materials, Poisson’s ratio. Constitutive equations- Hooke’s law, equations for linear elastic isotropic solids in terms of Young’s Modulus and Poisson’s ratio, Hooke’s law for Plane stress and plane strain conditions. Relations between elastic constants E, G, ν and K. Calculation of stress, strain and change in length in axially loaded members with single and composite materials. Effects of thermal loading – thermal stress and thermal strain. Thermal stress on a prismatic bar held between fixed supports.	11
2	Principal planes and principal stress, Mohr’s circle for 2D case. Circumferential and Longitudinal stress in a thin cylindrical vessel, stresses in a thin spherical vessel. Elastic strain energy for axial loading, transverse shear, bending and torsional loads. Expressions for strain energy in terms of load, geometry and material properties of the body for axial, shearing, bending and torsional loads.	11

3	Torsional deformation of circular shafts, assumptions for shafts subjected to torsion within elastic deformation range, derivation of torsion formula Torsional rigidity, Polar moment of inertia, basic design of transmission shafts. Simple problems to estimate the stress in solid and hollow shafts. Shear force and bending moment diagrams for cantilever and simply supported beams. Differential equations between load, shear force and bending moment. Normal and shear stress in beams.	11
4	Derivation of flexural formula, section modulus, flexural rigidity, numerical problems to evaluate bending stress, economic sections. Shear stress formula for beams: Derivation, shear stress distribution for a rectangular section. Deflection of beams, Simple problems to find the deflections. (using double integration method) Introduction to Theories of Failure, Rankine's theory for maximum normal stress, Guest's theory for maximum shear stress, Saint-Venant's theory for maximum normal strain, Hencky-von Mises theory for maximum distortion energy, Haigh's theory for maximum strain energy	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	3. Each question carries 9 marks. 4. Two questions will be given from each module, out of which 1 question should be answered. 5. Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (Cos)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Analyse the strength of materials using stress-strain relationships for structural and thermal loading	K3
CO2	Determine the stresses, strains and displacements of structures using Mohr's circle	K3
CO3	Perform basic design of shafts subjected to torsional loading and analyse beams subjected to bending moments	K3
CO4	Determine the deformation of structures subjected to various loading conditions using strain energy methods	K3
CO5	Estimate the strength of thin cylinders and spherical vessel, and appreciate the theories of failures and its relevance in mechanical design	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2									3
CO2	3	3	2									2
CO3	3	3	1									2
CO4	3	3	1									2
CO5	3	3	1									2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Mechanics of materials in S.I.units	R .C. Hibbeler	Pearson Higher Education	9e, 2018
2	Advanced Mechanics of Solids	L. S. Srinath	McGraw Hill	3e, 2017
3	Design of Machine Elements,	V. B Bhandari	McGraw Hill	4e, 2017

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Strength of Materials	Surendra Singh	S.K. Kataria & Sons	2011
2	Engineering Mechanics of Solids	Popov E	Pearson	2002
3	Mechanics of Materials S. I. units	Beer, Johnston, Dewolf	McGraw Hills	2017
4	Strength of Materials	Rattan	McGraw Hills	2011

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc23_me140
2	https://onlinecourses.nptel.ac.in/noc23_me140
3	https://onlinecourses.nptel.ac.in/noc23_me140
4	https://onlinecourses.nptel.ac.in/noc23_me140

SEMESTER S3

ANALOG ELECTRONICS

Course Code	PBMRT304	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:1	ESE Marks	60
Credits	4	Exam Hours	2 Hrs 30 Mins
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. To enable students to analyse, design and implement different analog circuits for various applications and functionalities

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Analog Electronics; BJT: Operating point of BJT- BJT as amplifier (CE configuration)-input chara-output chara-current amplification factor of BJT; Power amplifier (using Transistor): Class A- Class B- Class AB- Class C- (working, chara, power and efficiency calculation for Class A & Class B, comparison of power amplifiers) FET: Construction and characteristics of JFET & MOSFET –Enhancement & Depletion type-Comparison of BJT & FET UJT: Construction-working and characteristics of UJT- Overview of online & offline UPS- SMPS operation	13
2	Feedback- concept, positive and negative feedback Oscillator: Barkhausen criteria-difference between LC & RC oscillator-classification-working of Hartley and Colpitts Oscillator(using BJT)-analysis of Rc phase shift oscillator (Using BJT) 555 Timer IC:	10

	Functional diagram -Astable and Monostable Multivibrator using 555(working & derivation of time period) PLL: Block diagram- PLL IC 565(Pin diagram)-VCO(Analysis)-Capture range, lock in range -Applications of PLL	
3	OP-Amp: Block diagram of op-amp- Ideal & Practical OP- Amp parameters-frequency response- voltage transfer curve-inverting op-amp-non inverting op-amp-voltage follower-Integrator- Differentiator- comparator-Schmitt trigger-S/H- V to I & I to V converter, Instrumentation Amplifier Active Filters using op-Amp: Difference between active & passive Filters- Active filters: Analysis of First order HPF & LPF-working of Band pass and Band Stop Filters	12
4	Op- Amp Oscillators and Multivibrators: Phase shift and wein bridge oscillator-Triangular and sawtooth waveform generators- Astable and Monostable multivibrators Data convertors: DAC: Specifications-Weighted Resistor type, R-2R ladder type ADC: Specification-Flash type & Successive approximation type	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 2 marks (8x2 =16marks)	- Each question carries 6 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 2 sub divisions. (4x6 = 24 marks)	40

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	To understand the functionality and characteristics of different types of active electronics components	K2
CO2	To familiarise with the concept of feedback and to study about different types of oscillators, timers, multivibrators and PLL	K2
CO3	Familiarisation of op-Amp and its applications	K3
CO4	To understand about various types of code converters and to analyse their performance speed.	K2
CO5	Apply the concepts learnt in the course to develop a project and prepare a brief report.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	-	2	-	-	-	-	-	-	-	-	1
CO2	3	-	1	2	-	-	-	-	-	-	-	2
CO3	3	-	1	-	-	-	-	-	-	-	-	2
CO4	3	1	3	2	1	-	-	-	-	-	-	2
CO5	3	3	3		3				3	3	2	

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Micro Electronic circuits	Sedra A S& K C Smith	Oxford University Press	
2	Linear Integrated circuits	D Roy Choudhury, Shail B Jain	New Age International	
3	Electronic devices and circuit theory	Robert L Boylestad & Louis Nashelsky	Prentice Hall	

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Op-amps and Lnear Integrated Circuits	Ramakant A Gayakward	Pearson Education	4 th Edition,2012
2	Electroics devices and Circuits:An Introduction	Allen Mottershead	Prentice Hall of India	2013
3	Basic Electronics and linear circuits	NN Bhargava, DC Kulshreshtha, SC Gupta	McGraw Hill Education	2 nd Edition 2017
4	Millman's Electronic devices and circuits	Jacob Millman, Christos C Halkias, Satyabrata Jit	McGraw Hill Education	4 th Edition,2015

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/lpXNCwsnxjM?si=5uFIpXuzt4nbzGL , https://youtu.be/SpvmeG1hs7k?si=teZ0oZ70Q2LB8hQ9
2	https://youtu.be/1Uaz9QuHEVQ?si=-wGb46ZtmuulbsGw , https://youtu.be/WFsPI8_ZKbc?si=breDKci5X8Bk99Qs

PBL Course Elements

L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	Practical	Presentation
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation / Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		30

Project Assessment and Evaluation criteria (30 Marks)

1. Project Planning and Proposal (5 Marks)

- Clarity and feasibility of the project plan
- Research and background understanding
- Defined objectives and methodology

2. Contribution in Progress Presentation and Question Answer Sessions (4 Marks)

- Individual contribution to the presentation
- Effectiveness in answering questions and handling feedback

3. Involvement in the Project Work and Team Work (3 Marks)

- Active participation and individual contribution
- Teamwork and collaboration

4. Execution and Implementation (10 Marks)

- Adherence to the project timeline and milestones
- Application of theoretical knowledge and problem-solving
- Final Result

5. Final Presentation (5 Marks)

- Quality and clarity of the overall presentation
- Individual contribution to the presentation
- Effectiveness in answering questions

6. Project Quality, Innovation, and Creativity (3 Marks)

- Overall quality and technical excellence of the project
- Innovation and originality in the project
- Creativity in solutions and approaches

SEMESTER S3

INTRODUCTION TO ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

Course Code	GNEST305	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Demonstrate a solid understanding of advanced linear algebra concepts, machine learning algorithms and statistical analysis techniques relevant to engineering applications, principles and algorithms.
2. Apply theoretical concepts to solve practical engineering problems, analyze data to extract meaningful insights, and implement appropriate mathematical and computational techniques for AI and data science applications.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to AI and Machine Learning: Basics of Machine Learning - types of Machine Learning systems-challenges in ML- Supervised learning model example- regression models- Classification model example- Logistic regression-unsupervised model example- K-means clustering. Artificial Neural Network- Perceptron- Universal Approximation Theorem (statement only)- Multi-Layer Perceptron- Deep Neural Network- demonstration of regression and classification problems using MLP.(Text-2)	11
2	Mathematical Foundations of AI and Data science: Role of linear algebra in Data representation and analysis – Matrix decomposition- Singular Value Decomposition (SVD)- Spectral decomposition- Dimensionality reduction technique-Principal Component Analysis (PCA). (Text-1)	11
3	Applied Probability and Statistics for AI and Data Science: Basics of probability-random variables and statistical measures - rules in probability- Bayes theorem and its applications- statistical estimation-Maximum	11

	Likelihood Estimator (MLE) - statistical summaries- Correlation analysis- linear correlation (direct problems only)- regression analysis- linear regression (using least square method) (Text book 4)	
4	Basics of Data Science: Benefits of data science-use of statistics and Machine Learning in Data Science- data science process - applications of Machine Learning in Data Science- modelling process- demonstration of ML applications in data science- Big Data and Data Science. (For visualization the software tools like Tableau, PowerBI, R or Python can be used. For Machine Learning implementation, Python, MATLAB or R can be used.)(Text book-5)	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	6. Each question carries 9 marks. 7. Two questions will be given from each module, out of which 1 question should be answered. 8. Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply the concept of machine learning algorithms including neural networks and supervised/unsupervised learning techniques for engineering applications.	K3
CO2	Apply advanced mathematical concepts such as matrix operations, singular values, and principal component analysis to analyze and solve engineering problems.	K3
CO3	Analyze and interpret data using statistical methods including descriptive statistics, correlation, and regression analysis to derive meaningful insights and make informed decisions.	K3
CO4	Integrate statistical approaches and machine learning techniques to ensure practically feasible solutions in engineering contexts.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	3								
CO2	3	3	3	3								
CO3	3	3	3	3								
CO4	3	3	3	3								

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Linear Algebra	Gilbert Strang	Wellesley-Cambridge Press	6 th edition, 2023
2	Hands-on machine learning with Scikit-Learn, Keras, and TensorFlow	Aurélien Géron	O'Reilly Media, Inc.	2 nd edition, 2022
3	Mathematics for machine learning	Deisenroth, Marc Peter, A. Aldo Faisal, and Cheng Soon Ong	Cambridge University Press	1 st edition. 2020
4	Fundamentals of mathematical statistics	Gupta, S. C., and V. K. Kapoor	Sultan Chand & Sons	9 th edition, 2020
5	Introducing data science: big data, machine learning, and more, using Python tools	Cielen, Davy, and Arno Meysman	Simon and Schuster	1 st edition, 2016

Reference Books				
1	Data science: concepts and practice	Kotu, Vijay, and Bala Deshpande	Morgan Kaufmann	2 nd edition, 2018
2	Probability and Statistics for Data Science	Carlos Fernandez-Granda	Center for Data Science in NYU	1 st edition, 2017
3	Foundations of Data Science	Avrim Blum, John Hopcroft, and Ravi Kannan	Cambridge University Press	1 st edition, 2020
4	Statistics For Data Science	James D. Miller	Packt Publishing	1 st edition, 2019
5	Probability and Statistics - The Science of Uncertainty	Michael J. Evans and Jeffrey S. Rosenthal	University of Toronto	1 st edition, 2009
6	An Introduction to the Science of Statistics: From Theory to Implementation	Joseph C. Watkins	chrome-extension://efaidnbmnnnibpcajpcgglefndmkaj/ https://www.math.arizo	Preliminary Edition.

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/106/106/106106198/
2	https://archive.nptel.ac.in/courses/106/106/106106198/ https://ocw.mit.edu/courses/18-06-linear-algebra-spring-2010/resources/lecture-29-singular-value-decomposition/
3	https://ocw.mit.edu/courses/18-650-statistics-for-applications-fall-2016/resources/lecture-19-video/
4	https://archive.nptel.ac.in/courses/106/106/106106198/

SEMESTER S3/S4

ECONOMICS FOR ENGINEERS

Course Code	UCHUT346	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To provide students with an understanding of fundamental economic principles essential for effective decision-making in engineering contexts.
2. To enable students to apply economic analysis to production decisions, cost management, and market strategies in engineering practice.
3. To equip students with the ability to evaluate macroeconomic scenarios, financial methods, and investment decisions relevant to engineering projects.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basic economic problems – Production Possibility Curve – Utility – Law of diminishing marginal utility –Demand: Factors determining demand – Law of Demand – Demand curve- Price elasticity of demand- measurement of price elasticity and its applications – Supply: factors determining supply - Law of supply – Supply curve- Equilibrium price determination- Changes in demand and supply and its effects on equilibrium price and quantity Production: Production function - Law of variable proportion –Returns to scale- Cobb-Douglas Production Function	6
2	Cost: Cost concepts – Private cost and social cost – Sunk cost – Opportunity cost -Explicit and implicit cost –Short run cost curves –Long run average cost curve -Revenue concepts – Break-even point Market: Perfect Competition – Monopoly - Monopolistic Competition (features and equilibrium of a firm) - Oligopoly – Features – Kinked demand model	6

3	National income: Concepts (GDP, GNP and NNP)– Final goods and Intermediate goods - Methods of Estimation –output method – expenditure method-- Difficulties in the measurement of national income. Inflation: Causes and Effects – Measures to Control Inflation - Monetary and Fiscal policies – Repo and reverse repo rate	6
4	Value Analysis and value Engineering: Cost Value, Exchange Value, Use Value, Esteem Value - Aims, Advantages and Application areas of Value Engineering - Value Engineering Procedure Capital Budgeting: Time value of money - Net Present Value Method - Benefit Cost Ratio – Internal Rate of Return – Payback – Accounting Rate of Return.	6

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/Case Study/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
10	15	12.5	12.5	50

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• Minimum 1 and Maximum 2 Questions from each module. • Total of 6 Questions, each carrying 3 marks (6x3 =18 marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 2 sub divisions. Each question carries 8 marks. (4x8 = 32 marks)	50

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the fundamentals of various economic issues using laws and learn the concepts of demand, supply, elasticity and production function.	K2
CO2	Develop decision making capability by applying concepts relating to costs and revenue, and acquire knowledge regarding the functioning of firms in different market situations.	K3
CO3	Outline the macroeconomic principles of monetary and fiscal systems and national income.	K2
CO4	Make use of the possibilities of value analysis and engineering, and take investment decisions through capital budgeting techniques.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	-	-	-	-	-	1	-	-	-	-	1	-
CO2	-	-	-	-	-	1	1	-	-	-	1	-
CO3	-	-	-	-	1	-	-	-	-	-	2	-
CO4	-	-	-	-	1	1	-	-	-	-	2	-

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Managerial Economics	Geetika, Piyali Ghosh and Chodhury	Tata McGraw Hill,	2015
2	Engineering Economy	H. G. Thuesen, W. J. Fabrycky	PHI	1966
3	Engineering Economics	R. Paneerselvam	PHI	2012
4	Financial Management	I M Pandey	Vikas Publishing House	2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Engineering Economy	Leland Blank P.E, Anthony Tarquin P. E.	Mc Graw Hill	7 TH Edition
2	Indian Financial System	Khan M. Y.	Tata McGraw Hill	2011
3	Engineering Economics and analysis	Donald G. Newman, Jerome P. Lavelle	Engg. Press, Texas	2002
4	Contemporary Engineering Economics	Chan S. Park	Prentice Hall of India Ltd	2001
5	Financial Management: Theory and Practice	Prasanna Chandra	Mc Graw Hill	2007

MODEL QUESTION PAPER				
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY				
THIRD SEMESTER B. TECH DEGREE EXAMINATION, MONTH AND YEAR				
Course Code: UCHUT346				
Course Name: Economics for Engineers				
Max. Marks: 50			Duration: 2 hours 30 minutes	
	PART A			
		Answer all questions. Each question carries 3 marks	CO	Marks
1		What are the central problems of an economy?	CO1	(3)
2		Point out any three applications of price elasticity of demand.	CO1	(3)
3		What is the social cost of production?	CO2	(3)
4		What is repo rate?	CO3	(3)
5		What is esteem value?	CO4	(3)
6		Write a short note on time value of money.	CO4	(3)
PART B				
Answer any one full question from each module. Each question carries 8 marks				
Module 1				
9	a)	Suppose a country is producing at a point inside the production possibility curve. Draw a PPC and examine this situation.	CO1	(5)
	b)	State the law of demand. Point out any two exceptions of this law.	CO1	(3)
10	a)	A consumer purchases 10 units of a commodity when its price is Rs.100. Later when its price falls to Rs.90, he purchases 8 units only. Estimate price elasticity. What type of a commodity is this?	CO1	(5)
	b)	State the law of variable proportion.	CO1	(3)

Module 2				
11	a)	What is oligopoly? Why price is rigid under oligopoly?	CO2	(5)
	b)	The cost function of a firm is given as $TC=1000+10Q-6Q^2+Q^3$. Calculate fixed cost, variable cost and marginal cost when output is 10 units.	CO2	(3)
12	a)	Suppose a firm is earning super normal profit under monopolistic market condition. Explain this situation by drawing a diagram.	CO2	(5)
	b)	Suppose a firm sells its product at a price of Rs.10 per unit and its average variable cost is Rs.6. If the firm spend Ra.10000 as rent and pay Rs. 6000 as interest every month, estimate its break-even level of output.	CO2	(3)
Module 3				
13	a)	What is inflation? How does inflation affect investment and production.	CO3	(5)
	b)	How will you obtain NNP _{fc} from GDP _{mp} .	CO3	(3)
14	a)	From the data given below (In Rs. Crores) estimate GDP _{mp} and national income. Private final consumption expenditure = 1000, Government expenditure = 500, Invest expenditure = 700, Net exports = 300, Depreciation = 200, NFIA=(-200) and Net indirect tax = 100	CO3	(5)
	b)	What is bank rate? Examine the bank rate policy of the government during inflation.	CO3	(3)
Module 4				
15	a)	Examine the procedures of value engineering.	CO4	(5)
	b)	Examine the application areas of value engineering	CO4	(3)

16	a)	1. Suppose the initial investment of a project is Rs. 3000 (Crores) and the cost of capital or the opportunity cost of capital is 10 percent. Calculate NPV of the project based on the cash flows given below. <table><tr><td>Year</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>Cash flow</td><td>1000</td><td>900</td><td>800</td><td>700</td><td>600</td></tr></table> (In Crores)	Year	1	2	3	4	5	Cash flow	1000	900	800	700	600	CO4	(5)
	Year	1	2	3	4	5										
Cash flow	1000	900	800	700	600											
b)	Point out any three merits of NPV method.	CO4	(3)													

SEMESTER S3/S4

ENGINEERING ETHICS AND SUSTAINABLE DEVELOPMENT

Course Code	UCHUT347	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Equip with the knowledge and skills to make ethical decisions and implement gender-sensitive practices in their professional lives.
2. Develop a holistic and comprehensive interdisciplinary approach to understanding engineering ethics principles from a perspective of environment protection and sustainable development.
3. Develop the ability to find strategies for implementing sustainable engineering solutions.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamentals of ethics - Personal vs. professional ethics, Civic Virtue, Respect for others, Profession and Professionalism, Ingenuity, diligence and responsibility, Integrity in design, development, and research domains, Plagiarism, a balanced outlook on law - challenges - case studies, Technology and digital revolution-Data, information, and knowledge, Cybertrust and cybersecurity, Data collection & management, High technologies: connecting people and places-accessibility and social impacts, Managing conflict, Collective bargaining, Confidentiality, Role of confidentiality in moral integrity, Codes of Ethics. Basic concepts in Gender Studies - sex, gender, sexuality, gender spectrum: beyond the binary, gender identity, gender expression, gender stereotypes, Gender disparity and discrimination in education, employment and	6

	everyday life, History of women in Science & Technology, Gendered technologies & innovations, Ethical values and practices in connection with gender - equity, diversity & gender justice, Gender policy and women/transgender empowerment initiatives.	
2	Introduction to Environmental Ethics: Definition, importance and historical development of environmental ethics, key philosophical theories (anthropocentrism, biocentrism, ecocentrism). Sustainable Engineering Principles: Definition and scope, triple bottom line (economic, social and environmental sustainability), life cycle analysis and sustainability metrics. Ecosystems and Biodiversity: Basics of ecosystems and their functions, Importance of biodiversity and its conservation, Human impact on ecosystems and biodiversity loss, An overview of various ecosystems in Kerala/India, and its significance. Landscape and Urban Ecology: Principles of landscape ecology, Urbanization and its environmental impact, Sustainable urban planning and green infrastructure.	6
3	Hydrology and Water Management: Basics of hydrology and water cycle, Water scarcity and pollution issues, Sustainable water management practices, Environmental flow, disruptions and disasters. Zero Waste Concepts and Practices: Definition of zero waste and its principles, Strategies for waste reduction, reuse, reduce and recycling, Case studies of successful zero waste initiatives. Circular Economy and Degrowth: Introduction to the circular economy model, Differences between linear and circular economies, degrowth principles, Strategies for implementing circular economy practices and degrowth principles in engineering. Mobility and Sustainable Transportation: Impacts of transportation on the environment and climate, Basic tenets of a Sustainable Transportation design, Sustainable urban mobility solutions, Integrated mobility systems, E-Mobility, Existing and upcoming models of sustainable mobility solutions.	6
4	Renewable Energy and Sustainable Technologies: Overview of renewable energy sources (solar, wind, hydro, biomass), Sustainable technologies in energy production and consumption, Challenges and opportunities in renewable energy adoption. Climate Change and Engineering Solutions: Basics of climate change science, Impact of climate change on natural and human systems, Kerala/India and the Climate crisis, Engineering solutions to mitigate, adapt and build resilience to climate change. Environmental	6

	Policies and Regulations: Overview of key environmental policies and regulations (national and international), Role of engineers in policy implementation and compliance, Ethical considerations in environmental policy-making. Case Studies and Future Directions: Analysis of real-world case studies, Emerging trends and future directions in environmental ethics and sustainability, Discussion on the role of engineers in promoting a sustainable future.	
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Course Assessment Method

(CIE: 50 marks , ESE: 50)

Continuous Internal Evaluation Marks (CIE):

Continuous internal evaluation will be based on individual and group activities undertaken throughout the course and the portfolio created documenting their work and learning. The portfolio will include reflections, project reports, case studies, and all other relevant materials.

- The students should be grouped into groups of size 4 to 6 at the beginning of the semester. These groups can be the same ones they have formed in the previous semester.
- Activities are to be distributed between 2 class hours and 3 Self-study hours.
- The portfolio and reflective journal should be carried forward and displayed during the 7th Semester Seminar course as a part of the experience sharing regarding the skills developed through various courses.

Sl. No.	Item	Particulars	Group/ Individual (G/I)	Marks
1	Reflective Journal	Weekly entries reflecting on what was learned, personal insights, and how it can be applied to local contexts.	I	5
2	Micro project	1 a) Perform an Engineering Ethics Case Study analysis and prepare a report	G	8
	(Detailed documentation of the project, including methodologies, findings, and	1 b) Conduct a literature survey on 'Code of Ethics for Engineers' and prepare a sample code of ethics		
		2. Listen to a TED talk on a Gender-related topic, do a literature survey on that topic and make a report citing the relevant papers with a specific analysis of the Kerala context	G	5

	reflections)	3. Undertake a project study based on the concepts of sustainable development* - Module II, Module III & Module IV	G	12
3	Activities	2. One activity* each from Module II, Module III & Module IV	G	15
4	Final Presentation	A comprehensive presentation summarising the key takeaways from the course, personal reflections, and proposed future actions based on the learnings.	G	5
Total Marks			50	

*Can be taken from the given sample activities/projects

Evaluation Criteria:

- **Depth of Analysis:** Quality and depth of reflections and analysis in project reports and case studies.
- **Application of Concepts:** Ability to apply course concepts to real-world problems and local contexts.
- **Creativity:** Innovative approaches and creative solutions proposed in projects and reflections.
- **Presentation Skills:** Clarity, coherence, and professionalism in the final presentation.

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop the ability to apply the principles of engineering ethics in their professional life.	K3
CO2	Develop the ability to exercise gender-sensitive practices in their professional lives	K4
CO3	Develop the ability to explore contemporary environmental issues and sustainable practices.	K5
CO4	Develop the ability to analyse the role of engineers in promoting sustainability and climate resilience.	K4
CO5	Develop interest and skills in addressing pertinent environmental and climate-related challenges through a sustainable engineering approach.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1						3	2	3	3	2		2
CO2		1				3	2	3	3	2		2
CO3						3	3	2	3	2		2
CO4		1				3	3	2	3	2		2
CO5						3	3	2	3	2		2

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Ethics in Engineering Practice and Research	Caroline Whitbeck	Cambridge University Press & Assessment	2nd edition & August 2011
2	Virtue Ethics and Professional Roles	Justin Oakley	Cambridge University Press & Assessment	November 2006
3	Sustainability Science	Bert J. M. de Vries	Cambridge University Press & Assessment	2nd edition & December 2023
4	Sustainable Engineering Principles and Practice	Bhavik R. Bakshi,	Cambridge University Press & Assessment	2019
5	Engineering Ethics	M Govindarajan, S Natarajan and V S Senthil Kumar	PHI Learning Private Ltd, New Delhi	2012
6	Professional ethics and human values	RS Naagarazan	New age international (P) limited New Delhi	2006.
7	Ethics in Engineering	Mike W Martin and Roland Schinzinger,	Tata McGraw Hill Publishing Company Pvt Ltd, New Delhi	4" edition, 2014

Suggested Activities/Projects:**Module-II**

- Write a reflection on a local environmental issue (e.g., plastic waste in Kerala backwaters or oceans) from different ethical perspectives (anthropocentric, biocentric, ecocentric).
- Write a life cycle analysis report of a common product used in Kerala (e.g., a coconut, bamboo or rubber-based product) and present findings on its sustainability.

- Create a sustainability report for a local business, assessing its environmental, social, and economic impacts
- Presentation on biodiversity in a nearby area (e.g., a local park, a wetland, mangroves, college campus etc) and propose conservation strategies to protect it.
- Develop a conservation plan for an endangered species found in Kerala.
- Analyze the green spaces in a local urban area and propose a plan to enhance urban ecology using native plants and sustainable design.
- Create a model of a sustainable urban landscape for a chosen locality in Kerala.

Module-III

- Study a local water body (e.g., a river or lake) for signs of pollution or natural flow disruption and suggest sustainable management and restoration practices.
- Analyse the effectiveness of water management in the college campus and propose improvements - calculate the water footprint, how to reduce the footprint, how to increase supply through rainwater harvesting, and how to decrease the supply-demand ratio
- Implement a zero waste initiative on the college campus for one week and document the challenges and outcomes.
- Develop a waste audit report for the campus. Suggest a plan for a zero-waste approach.
- Create a circular economy model for a common product used in Kerala (e.g., coconut oil, cloth etc).
- Design a product or service based on circular economy and degrowth principles and present a business plan.
- Develop a plan to improve pedestrian and cycling infrastructure in a chosen locality in Kerala

Module-IV

- Evaluate the potential for installing solar panels on the college campus including cost-benefit analysis and feasibility study.
- Analyse the energy consumption patterns of the college campus and propose sustainable alternatives to reduce consumption - What gadgets are being used? How can we reduce demand using energy-saving gadgets?
- Analyse a local infrastructure project for its climate resilience and suggest improvements.
- Analyse a specific environmental regulation in India (e.g., Coastal Regulation Zone) and its impact on local communities and ecosystems.
- Research and present a case study of a successful sustainable engineering project in Kerala/India (e.g.,

sustainable building design, water management project, infrastructure project).

- Research and present a case study of an unsustainable engineering project in Kerala/India highlighting design and implementation faults and possible corrections/alternatives (e.g., a housing complex with water logging, a water management project causing frequent floods, infrastructure project that affects surrounding landscapes or ecosystems).

SEMESTER S3

ELECTRICAL TECHNOLOGY LAB

Course Code	PCMRL307	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:2:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Lab

Course Objectives:

1. This course aims the students to learn about the introduction, working and testing methods of main basics machines of electrical
2. To learn about the introduction of Power Electronics.

Expt. No.	Experiments
1	Brake test on DC series motor
2	Brake test on DC shunt motor.
3	Open circuit characteristics of dc shunt generator
4	Load test on dc shunt generator
5	Load test on single phase transformer
6	Three phase power measurement using two wattmeter method
7	Load test on three phase squirrel cage induction motor
8	OC and SC test on single phase transformer
9	Load test on three phase slip ring induction motor
10	Static characteristics of SCR.
11	Static characteristics of MOSFET.
12	Obtain output waveform of single phase bridge rectifier using SCR.

Minimum of 10 experiments from the above list are to be done.

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- 1. Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- 2. Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Test and validate different types of dc motors.	K3
CO2	Test and validate different types of dc generators.	K3
CO3	Validate and to do different test in transformer.	K3
CO4	To measure power in a three phase system	K3
CO5	Test and validate different types of three phase induction motors	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1	-	-	1	2	2	3	2	1	3
CO2	3	2	1	-	-	1	2	2	3	2	1	3
CO3	3	2	1	-	-	1	2	2	3	2	1	3
CO4	3	2	1	-	-	1	2	2	3	2	1	3
CO5	3	2	1	-	-	1	2	2	3	2	1	3

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Theory & performance of Electrical Machines	J B Gupta	S K Kataria & Sons	January 2013
2	Power Electronics	P S Bimbhra	Khanna	August 2018

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	V K Mehta & Rohit Mehta	Principles of Electrical Machines	S Chand	2006
2	M H Rashid	Power Electronics Circuits	Khanna	2018

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/108105017
2	https://archive.nptel.ac.in/courses/108/102/108102146/
3	https://www.youtube.com/watch?v=JPhCxbxPHbo
4	https://www.youtube.com/watch?v=4JsR4xfIPa4

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.

- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER S3

INSTRUMENTATION LAB

Course Code	PCMRL308	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:2:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Lab

Course Objectives:

1. To make students familiar with the techniques for measuring process parameters and techniques in metrology and evaluate the errors/uncertainties involved in the real time measurements as in the industry

Expt. No.	Experiments
1	Study and calibration of LVDT transducer for displacement measurement.
2	Calibration of pressure gauge using dead weight pressure gauge tester.
3	Calibration of thermocouple/ RTD for temperature measurement.
4	Calibration of strain gauge pressure cell
5	Measurement of vibration using Piezoelectric Accelerometers
6	Measurement of rotation speed using tachometer, tacho generator / stroboscopic tachometer
7	Acoustic measurement using Sound level meter
8	Study and use of linear measuring devices-vernier caliper, outside and inside micrometer, vernier depth gauge, vernier height gauge
9	Study and use of angular measuring devices, sine bar, clinometer angle gauge, bevel protractor- profile projector
10	Measurement of temperature using Radiation pyrometer and infrared pyrometer
11	Measurement of torque and force Measurement of cutting force during turning, drilling and milling using tool force dynamometer
12	Calibration of Load cell
13	Measurement of Air Humidity using sling psychrometer
14	Measurement of air velocity using anemometer
15	Measurements of parameters of gears and screw threads using profile projector
16	Measurement of surface finish using stylus type surface roughness measuring device

Course Assessment Method (CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- 1. Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- 2. Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop knowledge on the fundamental concepts and principles of metrology	K2
CO2	Understand the calibration and use of different measuring instruments & Evaluate the errors/uncertainties involved in any measurement	K5
CO3	Understand the operating principle and analyse the output characteristics of different electronics instruments	K2
CO4	Understand and analyse construction and operational aspects of different electromechanical measuring instruments along with their application domains	K2
CO5	Explain the need of various modern measuring instruments and precision measurement techniques	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2							2			
CO2	3	2							2			2
CO3	3	2	2						2			2
CO4	3	2							2			
CO5	3	2	2									2

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Engineering Metrology & Measurements	N.V. Raghavendra	Oxford University Press	1 st , 2013
2	Electrical Measurements in the Laboratory Practice	Bartirromo, Rosario, De Vincenzi, Mario	Springer International Publishing AG	1st ed. 2016
3	Basic Theory and Laboratory Experiments in Measurement and Instrumentation:by	Cataldo, A., Giaquinto, N., De Benedetto, E., Masciullo, A., Cannazza, G., Lorenzo, I., Nicolazzo, J., Meo, M.T., Monte, A.D., Parisi, G., Gaetani, F.	Springer	1st ed. 2020

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.

- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 4

MECHATRONICS ENGINEERING

SEMESTER S4
MATHEMATICS FOR PHYSICAL SCIENCE – 4
(C GROUP)

Course Code	GCMAT401	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Basic calculus.	Course Type	Theory

Course Objectives:

1. To familiarize students with the foundations of probabilistic and statistical analysis mostly used in varied applications in engineering and science.
2. To provide the students with the basics of various numerical methods to develop problem solving skills used in various engineering disciplines.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Random variables, Discrete random variables and their probability distributions, Cumulative distribution function, Expectation, Mean and variance, Binomial distribution, Poisson distribution, Poisson distribution as a limit of the binomial distribution, Joint pmf of two discrete random variables, Marginal pmf, Independent random variables, Expected value of a function of two discrete variables. [Text 1: Relevant topics from sections 3.1 to 3.4, 3.6, 5.1, 5.2]	9
2	Continuous random variables and their probability distributions, Cumulative distribution function, Expectation, Mean and variance, Uniform, Normal and Exponential distributions, Joint pdf of two Continuous random variables, Marginal pdf, Independent random variables, Expectation value of a function of two continuous variables. [Text 1: Relevant topics from sections 3.1, 4.1, 4.2, 4.3, 4.4, 5.1, 5.2]	9

3	Confidence Intervals, Confidence Level, Confidence Intervals and One-side confidence intervals for a Population Mean for large and small samples (normal distribution and <i>t</i> -distribution), Hypotheses and Test Procedures, Type I and Type II error, <i>z</i> Tests for Hypotheses about a Population Mean (for large sample), <i>t</i> Test for Hypotheses about a Population Mean (for small sample), Tests concerning a population proportion for large and small samples. [Text 1: Relevant topics from 7.1, 7.2, 7.3, 8.1, 8.2, 8.3, 8.4]	9
4	Newton-Raphson Method, Gauss Elimination Method, Gauss - Jordan Method, Numerical solution of ordinary differential equations-Euler's method, Modified Euler's method, Runge - Kutta method of 2 nd Order, Numerical solution of Laplace equation –Jacobi's Method, Curve Fitting by Method of Least Squares - Straight lines, Parabola. (Text 2: Relevant topics from sections 2.5, 4.2, 7.5, 8.4, 8.5, 9.4)	9

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the concept, properties and important models of discrete random variables and to apply in suitable random phenomena.	K3
CO2	Understand the concept, properties and important models of continuous random variables and to apply in suitable random phenomena.	K3
CO3	Estimate population parameters, assess their certainty with confidence intervals, and test hypotheses about population means and proportions using z -tests and the one-sample t -test.	K3
CO4	Apply numerical methods to find solutions of linear system of equations, ordinary differential equations and Laplace equations.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	-	-	-	-	-	-	-	2
CO2	3	3	2	2	-	-	-	-	-	-	-	2
CO3	3	3	2	2	-	-	-	-	-	-	-	2
CO4	3	3	2	2	-	-	-	-	-	-	-	2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Probability and Statistics for Engineering and the Sciences	Devore J. L	Cengage Learning	9 th edition, 2016
2	Introductory Methods of Numerical Analysis	S S Sastry	PHI Learning Pvt Limited	5 th edition, 2012

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Probability, Random Variables and Stochastic Processes,	Papoulis, A. & Pillai, S.U.,	McGraw Hill.	4 th edition, 2002
2	Introduction to Probability and Statistics for Engineers and Scientists	Ross, S. M.	Academic Press	6 th edition, 2020
3	Numerical methods for Engineers	Steven C. Chapra, Raymond P. Canale	McGraw Hill Education	8 th edition, 2021

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/117/105/117105085/
2	https://archive.nptel.ac.in/courses/117/105/117105085/
3	https://archive.nptel.ac.in/courses/117/105/117105085/
4	https://archive.nptel.ac.in/courses/111/107/111107105/

SEMESTER S4
DIGITAL ELECTRONICS

Course Code	PCMRT402	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To acquire basic and application knowledge of digital electronics circuits
2. To prepare students to perform analysis and design of digital circuits

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Number system and codes: Basics of Digital Electronics; Binary, octal, hexadecimal number system - Representations, base conversions and arithmetic; Signed representation of binary numbers-fixed and floating point ; 1's and 2's complement of binary numbers and their arithmetic. BCD number system- conversion of BCD to binary and Binary to BCD; Gray codes; Excess 3 codes; Basics of Verilog-basic language elements: identifiers, data objects, scalar data types, operators	9
2	Boolean Algebra and Logic gates Laws of Boolean algebra-postulates-Simplification of Boolean expressions; De-Morgan's law- minimization of Boolean expressions; Sum of product (SOP); Product of Sum (POS), Canonical form; Karnaugh Map Simplification (upto 4 variables); Quine MucCluskey method Basic and universal gates- Universality of NAND and NOR, ; design of simple logic circuits using logic gates TTL and CMOS logic families- comparison, Internal Diagram of TTL NAND gate and CMOS NOR gate	10

3	Combinational circuits: Difference between combinational and sequential circuits; Half adder/subtractor, Full adder/subtractor; Multiplexer (using logic gates)- 4:1,8:1,16:1; De-Multiplexer (using logic gates)-- 1:4,1:8,1:16; implementation of functions using Multiplexer IC74151; Encoder (4:2,8:3); Decoder (2:4,3:8), BCD Adder; Modelling and simulation of combinational circuits with Verilog codes at gate level (Adder, Subtractor, MUX,De-MUX)	12
4	Sequential circuits designs: Introduction to latch, Flip Flops, FSM; concept of Edge triggering and level triggering in FF; SR FF, JK FF, MS-JK FF, D FF, T FF-Truth table, Excitation table and characteristics equation; Counter designs-Asynchronous counter; Synchronous Counter; MOD counters ; Ring Counter , Johnson Counter, Shift registers using FFs- SISO,SIPO,PIPO,PISO; Programmable Logic Devices- PAL, PLA,FPGA (Basic concepts only) Modelling and simulation of JK and D flip flops in Verilog	13

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	To understand the basic concepts and representations of various number systems and design of codes	K2
CO2	To develop the ability to design and analyse digital logic circuits using Boolean expressions	K3
CO3	To create efficient combinational logic designs to meet specific functional requirements thereby developing problem solving skills among students	K3
CO4	To enhance problem solving skills for tackling complex design of sequential circuits	K3
CO5	To study the fundamentals of Verilog and design of various digital circuits	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1	1	2	1	-	-	-	-	-	-	3
CO2	3	2	2	2	-	-	-	-	-	-	-	3
CO3	3	3	3	2	2	-	-	-	-	-	-	3
CO4	3	3	3	2	2	-	-	-	-	-	-	3
CO5	3	1	3	2	3	-	-	-	-	-	-	3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Digital Fundamentals	Thomas L Floyd	Pearson Education	11 th Edition, 2011
2	Verilog HDL: A guide to Digital Design and Synthesis	Samir Palnikar	Pearson India	2 nd Edition, 2003
3	Digital Circuits and systems	D. V Hall	Tata McGraw Hill	1989
4	Fundamentals of Logic Design	C H Roth and L LKimney	Cengage Learning	7 th Edition, 2013

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fundamentals of Digital Electronics	A Anand Kumar	Maakzoo	4 th Edition, 2023
2	Digital Integrated Electronics	Taub& Schilling	McGraw Hill	1997
3	Digital Logic and Computer Design	M Morris Mano	Pearson Education	2002

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/108/105/108105132/
2	https://archive.nptel.ac.in/courses/108/105/108105132/
3	https://archive.nptel.ac.in/courses/108/105/108105132/
4	https://archive.nptel.ac.in/courses/108/105/108105132/

SEMESTER S4

COMPUTER ARCHITECTURE AND MICROCONTROLLER

Course Code	PCMRT403	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. To acquire basic and application knowledge of basic computer architecture
2. Understand the basic concepts and architectures of microcontrollers

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Algorithms for binary multiplication and division. Fixed and floating-point number representation. Functional units of a computer, Von Neumann and Harvard computer architectures, CISC and RISC architectures. Processor Architecture – General internal architecture, Address bus, Data bus, control bus. Register set – status register, accumulator, program counter, stack pointer, general purpose registers. Processor operation – instruction cycle, instruction fetch, instruction decode, instruction execute, timing response, instruction sequencing and execution (basic concepts, data path.	9
2	Microcontrollers and Embedded Processors. Architecture – Block diagram of 8051, Pin configuration, Registers, Internal Memory, Timers, Port Structures, Interrupts. Assembly Language Programming - Addressing Modes, Instruction set (Detailed study of 8051 instruction set is required). Simple programming examples in assembly language. Interfacing with 8051 using Assembly language programming: LED, Seven segment LED display.	11

3	Programming in C - Declaring variables, Simple examples – delay generation, port programming, code conversion. Interfacing of – Stepper Motor, DAC and ADC -- with 8051 and its programming. 8051 Timers/Counters - Modes and Applications. Serial Data Transfer – SFRs of serial port, working, Programming the 8051 to transfer data serially.	11
4	Introduction to ARM - ARM family, ARM 7 register architecture. ARM programmer's model. System software - Assembler, Interpreter, Compiler, Linker, Loader, Debugger. Types of memory - RAM, ROM. Memory Characteristics and Hierarchy. Cache memory – The basics of Caches, Mapping techniques, Improving Cache performance. Virtual memory – Overlay, Memory management, Address translation. Input/Output Organization – Introduction, Synchronous vs. asynchronous I/O, Programmed I/O, Interrupt driven I/O, Direct Memory Access.	13

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the functional units, I/O and memory management w.r.t a typical computer architecture.	K2
CO2	Distinguish between microprocessor and microcontroller.	K2
CO3	Develop simple programs using assembly language programming.	K3
CO4	Interface 8051 microcontroller with peripheral devices using ALP/Embedded C	K3
CO5	Familiarize system software and Advanced RISC Machine Architecture.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	1	2	1	1	1	1	2	1	2
CO2	3	3	1	1	2	1	1	1	1	1	1	1
CO3	3	2	2	2	3	1	1	1	1	2	1	2
CO4	3	3	3	3	3	1	1	1	2	2	2	2
CO5	3	2	2	2	3	1	1	1	1	2	1	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	The 8051 microcontroller and Embedded System	Muhammed Ali Mazidi & Janice Gilli Mazidi, R.D. Kinley	Pearson Education	2nd edition, 2007
2	Computer Architecture and Organization: From 8085 to Core2Duo and beyond	Subrata Ghoshal	Pearson	1 st edition, 2011
3	ARM System - on-chip Architecture	Steve Furber	Pearson Education	2014

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Computer System Architecture	Mano M M	Prentice Hall of India	3rd Edition, 2017
2	Computer organization and design: The Hardware/Software interface	David A. Patterson, John L. Hennessy	Elsevier	6th edition. 2021
3	Computer Organisation	V. Carl Hamacher, Zvonko G. Vranesic, Safwat G.Zaky	McGraw-Hill Education	5 th edition, 2017
4	8085 Microprocessor Architecture, Applications and Programming	Ramesh S Goankar	Penram International	5/e

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/106/105/106105163/
2	https://archive.nptel.ac.in/courses/106/105/106105163/
3	https://archive.nptel.ac.in/courses/106/105/106105163/
4	https://archive.nptel.ac.in/courses/106/105/106105163/

SEMESTER S4

MANUFACTURING PROCESS

Course Code	PBMRT404	CIE Marks	60
Teaching Hours/Week (L: T:P: R)	3:0:0:1	ESE Marks	40
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. To gain theoretical and practical knowledge in primary manufacturing processes
2. Provide a detailed discussion on the welding process. Introduce different welding processes and advanced processes to be able to appreciate the practical applications of welding.
3. To understand conventional machining and nonconventional manufacturing techniques

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Primary Processes: Casting- Characteristics of sand, patterns, cores, chaplets, riser, chills, Defects in castings. Forming- Forging - Extrusion- wire drawing process – Rolling. Selection of primary forming process. Defects in forging and rolling.	9
2	Joining processes –Welding: Classification, Gas welding: Oxyacetylene welding- types of flame.– Arc Welding: shielded metal arc welding, TIG, MIG, submerged arc welding, ultrasonic welding, thermit welding, friction welding, resistance spot welding, resistance seam welding. Brazing, Soldering, types of solders and fluxes- Adhesive bonding. Selection of joining process for various applications.	9
3	Machining – orthogonal and oblique cutting, rake and clearance angle, Types of chip, Tool life – Taylor’s tool life equation, Machinability. Lathe – types of operations, tool holders, inserts, work holding devices. Milling – types of operations, types of milling machines, milling cutters, simple indexing. Drilling. Introduction to Shaping, Planing and Slotting machines. Grinding – types of operations. NC and CNC machines, micromachining.	9

4	Non-conventional processes – EDM, ECM, USM, EBM, LBM, WJM, AWJM, PCM, process capabilities, applications. Additive manufacturing, Rapid prototyping, stereo-lithography, LIGA process.	9
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Course Assessment Method
(CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 2 marks (8x2 =16marks)	- Each question carries 6 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 2 sub divisions. (4x6 = 24 marks)	40

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the basics of the primary manufacturing processes, various joining processes and choose the appropriate mechanical and adhesive joining process for the parts.	K2
CO2	Understand the conventional machining operations	K2
CO3	Understand the various nonconventional machining techniques	K2
CO4	Understand Additive manufacturing and LIGA process	K2
CO5	Apply the concepts learnt in the course to develop a project and prepare a brief report.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2										3
CO2	3	2										3
CO3	3	3										3
CO4	3	2										3
CO5	3	3	3		3				3	3	2	3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Manufacturing Engineering and Technology	Serope Kalpakjian, Steven R. Schmid	Pearson	2013
2	Fundamentals of Modern Manufacturing	Mikell P. Groover,	Wiley	2010.
3	Manufacturing Processes for Engineering Materials	Kalpakjian and Schmid	Pearson	2018
4	Manufacturing Engineering	D.K. Singh	Ane Books India,	2008

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Manufacturing Technology: Foundry, Farming and Welding - Volume1	P. N. Rao	McGraw Hill	5e, 2018
2	Materials and Processes in Manufacturing	Paul Degarma E and Ronald A. Kosher	Prentice Hall College	8e, 1997

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc21_me81
2	https://archive.nptel.ac.in/courses/112/103/112103263
3	https://onlinecourses.nptel.ac.in/noc21_me29
4	https://onlinecourses.nptel.ac.in/noc24_me72

PBL Course Elements

L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	Practical	Presentation
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation / Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		30

Project Assessment and Evaluation criteria (30 Marks)

1. Project Planning and Proposal (5 Marks)

- Clarity and feasibility of the project plan
- Research and background understanding
- Defined objectives and methodology

2. Contribution in Progress Presentation and Question Answer Sessions (4 Marks)

- Individual contribution to the presentation
- Effectiveness in answering questions and handling feedback

3. Involvement in the Project Work and Team Work (3 Marks)

- Active participation and individual contribution
- Teamwork and collaboration

4. Execution and Implementation (10 Marks)

- Adherence to the project timeline and milestones
- Application of theoretical knowledge and problem-solving
- Final Result

5. Final Presentation (5 Marks)

- Quality and clarity of the overall presentation
- Individual contribution to the presentation
- Effectiveness in answering questions

6. Project Quality, Innovation, and Creativity (3 Marks)

- Overall quality and technical excellence of the project
- Innovation and originality in the project
- Creativity in solutions and approaches

SEMESTER S4

INDUSTRIAL ENGINEERING

Course Code	PEMRT 411	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. To enable students to understand, design, analyse about processes in various industries
2. To develop decision making, problem solving, project managing skill among students

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to industrial engineering- concepts and functions of industrial engineering- Manufacturing Vs purchase-problems-economic aspects- C-V-P analysis- simple problems Ergonomics man machine system- Anthropometry – value engineering	7
2	Work place design and ergonomics Work study: procedure-concept of work content- techniques to reduce work content Study method: steps-recording techniques-operation process chart- flow process chart- two hand process chart-multiple activity chart Diagrams: flow diagram- string diagram Principle of motion economy- SIMO chart Work measurement: techniques-time study- steps in time study- calculation of standard time (problems)	10
3	Plant: location, layout, material handling concepts Plant layout: types- tools and techniques-travel chart- REL chart-computer algorithms for layout design: CRAFT-ALDEP (methods only) system layout planning-line balancing- RPW (Problems) Principles of material handling: unit load concept- automated material	9

	handling system-AGVs	
4	Production cost concepts- production planning & control-types of production systems Demand forecasting:forecasting methods- Gantt's chart Need for new product-product life cycle Quality control:statistical quality control-causes and variations in quality-control charts for X and R. Reliability: causes of failure- bath tub curve-system reliability Project management:critical path method- PERT-determination of economic life- replacement policy- methods of replacement analysis (Simple problems)	10

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	To understand the basic concepts of industrial engineering and various process involved in it	K2
CO2	To study about the feasibility of good working environment, designing of work place, working measurements etc	K2
CO3	To enable students to get a deep knowledge about manufacturing plants, their designs and various algorithms used for preparing design layouts	K2
CO4	To provide knowledge about how production, quality control and project management is done	K2
CO5	To understand the basic concepts of industrial engineering and various process involved in it	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1	-	-	-	2	-	1	-	-	-	2
CO2	3	3	3	2	-	2	1	1	1	-	1	2
CO3	3	3	3	2	2	1	-	-	1	-	2	2
CO4	3	3	2	2	-	3	2	3	2	-	3	3
CO5	3	1	-	-	-	2	-	1	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Industrial Engineering & product management	Martand Telsang	S chand	Third revised edition,2018
2	Industrial Engineering	B Kumar	Khanna publisher	10 th Edition, 2015
3	Industrial engineering & production management	M Mahajan	Dhanpat Rai	2015
4	Manufacturing planning & control for supply chain management	Thomas E Vollmann, William L Berry, D Clay Whybark, F Robert Jacobs	McGraw Hill Education	5 th Edition, 2017

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Industrial Engineering and management	O P Khanna	Dhanpat Rai	2015
2	Statistical Quality control	Grant and Ieven worth	McGraw Hill	2000
3	Modern production management	E S Buffa	John Wiley	1983
4	Facility layout and location: An Analytical approach	Richard L Francis, F McGinnis Jr, John A White		2 nd Edition , 1991

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/yhywrCChJBQ?si=ZQVmu80Tv2002lGn
2	https://youtu.be/yhywrCChJBQ?si=ZQVmu80Tv2002lGn , https://youtu.be/zWQovrjB7Uc?si=dqf7k4WOyUvCJpEQ
3	https://youtu.be/yhywrCChJBQ?si=ZQVmu80Tv2002lGn , https://youtu.be/jdFrBtHeJbs?si=xYkca1jh6Ur6syr7
4	https://youtu.be/yhywrCChJBQ?si=ZQVmu80Tv2002lGn , https://youtu.be/jdFrBtHeJbs?si=xYkca1jh6Ur6syr7

SEMESTER S4
BIOMEDICAL INSTRUMENTATION

Course Code	PEMRT412	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. The course aims to give a brief introduction to human physiology.
2. The course presents various instrumentation systems used for measurement and analysis of physiological parameters.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Biomedical Instrumentation Overview of biomedical instrumentation – Basic principles and concepts –Source of bioelectric potentials - Resting and action potential – Propagation of action potentials - Bioelectric potentials example (ECG, EEG, EMG) - Electrode skin interface - Types of electrodes - Limb electrodes – Floating electrodes - Pre-gelled disposable electrodes - Micro, needle and surface electrodes.	9
2	Physiological measurements Heart and cardio vascular system - Electro conduction system of the heart-Electrocardiograph(ECG) – Electrodes and leads – Einthoven triangle - ECG read out devices - ECG machine – Block diagram - Measurement of blood flow and cardiac output - Measurement of heart sounds – Phonocardiography - Electroencephalogram (EEG) measurement - Electromyogram (EMG).	9
3	Biomedical Imaging Techniques X-rays - Principles of generation - Uses of X-rays - Diagnostic still picture – Fluoroscopy – Angiography – Endoscopy - Diathermy - Basic principle of computed tomography	

	(CT) - Magnetic resonance imaging (MRI) system and nuclear medicine imaging system (PET, SPECT) – Radiation therapy - Ultrasound imaging system - Introduction and basic principle	
4	Electrical safety and Therapeutic Devices Physiological effects of electric current – Shock hazards from electrical equipment, Leakage current - Electrical safety analyser – Testing of biomedical equipment – Method of accident prevention. Cardiac pacemakers – Internal and external pacemakers – Defibrillators –Artificial heart valves - Heart lung machine - Biotelemetry - Introduction to telemedicine - Cyber medicine.	9

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the fundamental principles and concepts of biomedical instrumentation, biopotential signals produced by the human body and electrodes used for biopotential signal acquisition.	K2
CO2	Understand the various physiological measurement techniques for cardiovascular, nervous and muscle systems.	K2
CO3	Acquire knowledge of the fundamental principles underlying medical imaging techniques	K2
CO4	Summarize the importance of electrical safety within medical environment	K2
CO5	Discuss the application of electronics in therapeutic applications and the field of biotelemetry	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2									
CO2	3	2	2									
CO3	3	2	2									
CO4	3	2	2			2						
CO5	3	2	2			2						

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Medical Instrumentation: Application and Design	John G. Webster	John Wiley and Sons	4th ed., 2010
2	Handbook of Biomedical Instrumentation	R. S. Khandpur	McGraw Hill Education (India) Private Limited	3rd ed., 2014

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Principles of Biomedical Instrumentation	Andrew G. Webb	Cambridge University Press	1 st ed., 2018
2	Biomedical Instrumentation and Measurements	L. Cromwell, F. J. Weibell and E. A. Pfeiffer	Prentice Hall	2 nd ed., 1980
3	Fundamentals of Biomedical Engineering	G.S. Sawhney	New Age International Publishers	1 st ed., 2007
4	Medical Imaging Signals and Systems	J. L. Prince and J. M. Links	Pearson Prentice Hall,	1 st ed., 2006

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.swayam2.ac.in/nou24_bt07/preview
2	https://onlinecourses.swayam2.ac.in/nou24_bt07/preview
3	https://archive.nptel.ac.in/courses/102/105/102105090/ https://onlinecourses.swayam2.ac.in/nou24_bt07/preview
4	https://onlinecourses.swayam2.ac.in/nou24_bt07/preview

SEMESTER S4

AUTOMOBILE ENGINEERING (Mechatronics Branch)

Course Code	PEMRT413	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	2:1:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Basics of Mechanical Engineering	Course Type	Theory

Course Objectives:

1. To know the anatomy of automobile in general
2. To understand the working of different automotive systems and subsystems
3. To update the latest developments in automobiles

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Components of an automobile. General classification. Conventional Chassis construction- Types of frames- Frameless constructions. Friction clutch: Principle, dry friction clutches- Pull type diaphragm clutch, multiple diaphragm clutch, multi-plate hydraulically operated automatic transmission clutch, semi centrifugal clutch, fully automatic centrifugal clutch, and integral single plate diaphragm clutch. Electromagnetic clutch operation. wet clutch	8
2	Manual transmission- Need of gear box, power to weight ratio, speed operating range-, constant mesh, and synchromesh gear boxes. Automatic transmission- Epicyclic gear box - torque convertor – Over drives. Automated manual transmission. Suspension: - suspension geometry, terminology- Macpherson strut friction and spring offset - suspension roll centers: -roll centers, roll axis, roll centre height, short swing arm suspension, transverse double wishbone, Macpherson strut suspension, body roll stability analysis: -	9

	body roll couple, body roll stiffness, body over turning couple.	
3	Brakes: mechanical and hydraulic brakes (review only) – Anti-Lock Braking system (ABS):- hydro-mechanical ABS - hydro-electric ABS - air-electric ABS. Pneumatic operated disc brakes – electronic-pneumatic brakes. Regenerative braking system. Steering: -basic principle of a steering system– Ackermann –over steer and under steer – slip angle, camber, king pin inclination, caster, toe-in and toe- out . Steering gear box:-worm and roller type steering gear box – need of power assisted steering.	9
4	Fuel injection systems: multiport fuel injection (MPFI) and common rail direct injection (CRDI) systems. Electric Vehicle Technology (EVT): EV Architecture, types of batteries, battery parameters, super capacitors. Fuel cells and its efficiency. EV Chassis – requirements, suspension for EVs. Recent Electric vehicles- Electric mobility aids. Future of electric vehicles –Tesla S, Maglev trains, Electric rail road systems.	9

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain different automotive systems and subsystems.	K2
CO2	Illustrate the principles of transmission, suspension systems of an automobile	K2
CO3	Illustrate the steering and braking systems of an automobile.	K2
CO4	Build a basic knowledge about the technology in electric vehicles.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											3
CO2	3											3
CO3	3											3
CO4	3											3
CO5	3											3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Vehicle and engine technology	Heinz Heisler	Butterworth-Heinemann	2nd edition, 1998
2	Auto design	R.B. Gupta.	Satya Prakashan Publishers	2016.
3	Electric vehicle technology explained	James Larminie and John Lowry	Wiley pubpublications	2nd edition, 2015.
4	Automobile Engineering Vol.1 & Vol.2	Kirpal Singh	Standard Publishers	13th edition, 2020

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fundamentals of modern vehicle technology	V.A.W. Hillier	Butterworth-Heinemann	2nd edition, 1998.
2	Electric and Hybrid Vehicles	Tom Denton.	Routledge Publishers	2nd edition, 2020.

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/hs7bABMtOMI
2	https://youtu.be/hs7bABMtOMI
3	https://youtu.be/hs7bABMtOMI
4	https://youtu.be/9JFSJmD3m1E

SEMESTER S4

NETWORK THEORY

Course Code	PEMRT414	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	2:1:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	GCEST204	Course Type	Theory

Course Objectives:

1. Gain a solid foundation in the principles of electrical networks, including the analysis and design of electrical circuits
2. Develop the ability to analyse complex AC and DC circuits

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Mesh and node analysis of network containing independent and dependent sources. Supermesh and Supernode analysis. Steady-state AC analysis using Mesh and Node analysis.	9
2	Thevenin's theorem, Norton's theorem, Superposition theorem, Reciprocity theorem, Maximum power transfer theorem. (applied to both dc and ac circuits having dependent source). Review of Laplace Transforms and Inverse Laplace Transforms, Initial value theorem & Final value theorem, Transformation of basic signals and circuits into s-domain.	9
3	Transient analysis of RL, RC, and RLC networks with impulse, step and sinusoidal inputs (with and without initial conditions). Network functions for the single port and two port network. Properties of driving point and transfer functions. Significance of Poles and Zeros of network functions, Time domain response from pole zero plot. Impulse Function & Response. Network functions in the sinusoidal steady state, Magnitude and Phase response.	9

4	Impedance, Admittance, Transmission and Hybrid parameters of two port network. Interrelationship among parameter sets. Series and parallel connections of two port networks. Reciprocal and Symmetrical two port network. Characteristic impedance, Image impedance and propagation constant (derivation not required).	9
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply Mesh / Node analysis or Network Theorems to obtain steady state response of the linear time invariant networks	K3
CO2	Apply Laplace Transforms to determine the transient behaviour of RLC networks.	K3
CO3	Apply Network functions and Network Parameters to analyse the single port and two port networks	K3
CO4	Conduct steady-state AC analysis using mesh and node analysis methods to determine voltages, currents, and power in AC circuits	K3
CO5	Understand and analyze network functions for single port and two port networks	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	2	-	-	-	1	2	-	2
CO2	3	3	2	2	2	-	-	-	1	1	-	2
CO3	3	3	2	2	2	-	-	-	1	2	-	2
CO4	3	3	2	2	2	-	-	-	1	2	-	2
CO5	3	3	2	2	2	-	-	-	1	2	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Network Analysis	Valkenburg V	Pearson	3/e, 2019
2	Circuits and Networks- Analysis and Synthesis	Sudhakar A, Shyammohan S. P	McGraw Hill	5/e, 2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Electric Circuits – Schaum's Outline Series	Edminister	McGraw-Hill	2009
2	Engineering Circuit Analysis	W. Hayt, J. Kemmerly, J. Phillips, S. Durbin	McGraw Hill	
3	Electric Circuits and Networks	K. S. Suresh Kumar	Pearson	2008
4	Network Analysis with Applications	William D. Stanley	Pearson	4/e 2006

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/108/105/108105159/
2	https://archive.nptel.ac.in/courses/108/105/108105159/
3	https://archive.nptel.ac.in/courses/108/105/108105159/
4	https://archive.nptel.ac.in/courses/108/105/108105159/

SEMESTER S4

OBJECT ORIENTED PROGRAMMING

Course Code	PEMRT415	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	UCEST105 Algorithmic Thinking with Python	Course Type	Theory

Course Objectives:

1. The purpose of this course is to enable learners to solve problems by breaking it down to object level while designing software and to implement it using Java.
2. This course covers the fundamental principles of object-oriented programming in Java, including classes, objects, inheritance, polymorphism, encapsulation and abstraction, and effectively applies them.
3. This course helps the learners to develop Java applications that utilize object-oriented programming principles to achieve modularity, reusability, and scalability in software design and implementation

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basic Object-Oriented concepts, Introduction to Java - Java programming and Runtime Environment, Development Platforms- Java Virtual Machine (JVM), Java compiler, Bytecode, Java Buzzwords, Java program structure, Comments. Primitive Data types - Integers, Floating Point Types, Characters, Boolean. Literals, Operators - Arithmetic Operators, Bitwise Operators, Relational Operators, Boolean Logical Operators, Assignment Operator, Conditional (Ternary) Operator. Control Statements - Selection Statements, Iteration Statements and Jump Statements.	11

2	Object Oriented Programming in Java - Class Fundamentals, Declaring Objects, Introduction to Methods, Constructors, this Keyword, Method Overloading, Using Objects as Parameters, Returning Objects. Static Members, Final Variables, Inner Classes. Inheritance - Super Class, Sub Class, The Keyword super, protected Members, Calling Order of Constructors, Method Overriding, the Object class, Abstract Classes and Methods, using final with Inheritance. Packages and Interfaces	11
3	Exception Handling - Checked Exceptions, Unchecked Exceptions, try Block and catch Clause, Multiple catch Clauses, Nested try Statements, throw. Java Library - String Handling – String Constructors, String Length, Special String Operations. Multithreaded Programming - The Java Thread Model, The Main Thread, Creating Thread, Creating Multiple Threads, Synchronization, Suspending, Resuming and Stopping Threads.	11
4	Event handling - Event Handling Mechanisms, Delegation Event Model, Event Classes, Sources of Events, Event Listener Interfaces. Java Database Connectivity (JDBC) - JDBC overview, Creating and Executing Queries – create table, delete, insert, select.	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand and apply fundamental Java programming concepts, including the runtime environment, primitive data types, operators, and control statements, to develop efficient and well-structured Java applications.	K3
CO2	Apply key object-oriented programming principles in Java, leveraging packages and interfaces effectively to design and implement Java applications.	K3
CO3	Understand the handling of Java exceptions, effectively manipulate strings, and implement multithreaded programming techniques.	K3
CO4	Understand the different JDBC queries to handle a data base.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3									2
CO2	3	3	3									2
CO3	3	3	3		2							2
CO4	3	3	3		2							2
CO5	3	3	3		2							2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Java: The Complete reference	Herbert Schildt	Tata McGraw Hill	8/e, 2011.
2	Programming JAVA a Primer	Balagurusamy E	Tata McGraw Hill	5/e, 2014

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Java Programming	Y. Daniel Liang	Pearson Education	7/e, 2013
2	Java How to Program, Early Objects	Paul Deitel, Harvey Deitel	Pearson Education	11/e, 2018
3	Core Java: An Integrated Approach	Nageswararao R	Dreamtech Press	2008
4	Java in A Nutshell	Flanagan D	O'Reilly	5/e, 2005
5	Core Java: Volume I & II– Fundamentals	Cay S. Horstmann and Gary Cornell	Pearson Education	2008

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/106/105/106105191
2	https://archive.nptel.ac.in/courses/106/105/106105191
3	https://archive.nptel.ac.in/courses/106/105/106105191
4	https://archive.nptel.ac.in/courses/106/105/106105191

SEMESTER S3/S4

ECONOMICS FOR ENGINEERS

Course Code	UCHUT346	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To provide students with an understanding of fundamental economic principles essential for effective decision-making in engineering contexts.
2. To enable students to apply economic analysis to production decisions, cost management, and market strategies in engineering practice.
3. To equip students with the ability to evaluate macroeconomic scenarios, financial methods, and investment decisions relevant to engineering projects.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basic economic problems – Production Possibility Curve – Utility – Law of diminishing marginal utility –Demand: Factors determining demand – Law of Demand – Demand curve- Price elasticity of demand- measurement of price elasticity and its applications – Supply: factors determining supply - Law of supply – Supply curve- Equilibrium price determination- Changes in demand and supply and its effects on equilibrium price and quantity Production: Production function - Law of variable proportion –Returns to scale- Cobb-Douglas Production Function	6
2	Cost: Cost concepts – Private cost and social cost – Sunk cost – Opportunity cost -Explicit and implicit cost –Short run cost curves –Long run average cost curve -Revenue concepts – Break-even point Market: Perfect Competition – Monopoly - Monopolistic Competition (features and equilibrium of a firm) - Oligopoly – Features – Kinked demand model	6

3	National income: Concepts (GDP, GNP and NNP)– Final goods and Intermediate goods - Methods of Estimation –output method – expenditure method-- Difficulties in the measurement of national income. Inflation: Causes and Effects – Measures to Control Inflation - Monetary and Fiscal policies – Repo and reverse repo rate	6
4	Value Analysis and value Engineering: Cost Value, Exchange Value, Use Value, Esteem Value - Aims, Advantages and Application areas of Value Engineering - Value Engineering Procedure Capital Budgeting: Time value of money - Net Present Value Method - Benefit Cost Ratio – Internal Rate of Return – Payback – Accounting Rate of Return.	6

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/Case Study/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
10	15	12.5	12.5	50

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• Minimum 1 and Maximum 2 Questions from each module. • Total of 6 Questions, each carrying 3 marks (6x3 =18 marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 2 sub divisions. Each question carries 8 marks. (4x8 = 32 marks)	50

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the fundamentals of various economic issues using laws and learn the concepts of demand, supply, elasticity and production function.	K2
CO2	Develop decision making capability by applying concepts relating to costs and revenue, and acquire knowledge regarding the functioning of firms in different market situations.	K3
CO3	Outline the macroeconomic principles of monetary and fiscal systems and national income.	K2
CO4	Make use of the possibilities of value analysis and engineering, and take investment decisions through capital budgeting techniques.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	-	-	-	-	-	1	-	-	-	-	1	-
CO2	-	-	-	-	-	1	1	-	-	-	1	-
CO3	-	-	-	-	1	-	-	-	-	-	2	-
CO4	-	-	-	-	1	1	-	-	-	-	2	-

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Managerial Economics	Geetika, Piyali Ghosh and Chodhury	Tata McGraw Hill,	2015
2	Engineering Economy	H. G. Thuesen, W. J. Fabrycky	PHI	1966
3	Engineering Economics	R. Paneerselvam	PHI	2012
4	Financial Management	I M Pandey	Vikas Publishing House	2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Engineering Economy	Leland Blank P.E, Anthony Tarquin P. E.	Mc Graw Hill	7 TH Edition
2	Indian Financial System	Khan M. Y.	Tata McGraw Hill	2011
3	Engineering Economics and analysis	Donald G. Newman, Jerome P. Lavelle	Engg. Press, Texas	2002
4	Contemporary Engineering Economics	Chan S. Park	Prentice Hall of India Ltd	2001
5	Financial Management: Theory and Practice	Prasanna Chandra	Mc Graw Hill	2007

MODEL QUESTION PAPER				
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY				
THIRD SEMESTER B. TECH DEGREE EXAMINATION, MONTH AND YEAR				
Course Code: UCHUT346				
Course Name: Economics for Engineers				
Max. Marks: 50			Duration: 2 hours 30 minutes	
	PART A			
		<i>Answer all questions. Each question carries 3 marks</i>	CO	Marks
1		What are the central problems of an economy?	CO1	(3)
2		Point out any three applications of price elasticity of demand.	CO1	(3)
3		What is the social cost of production?	CO2	(3)
4		What is repo rate?	CO3	(3)
5		What is esteem value?	CO4	(3)
6		Write a short note on time value of money.	CO4	(3)
PART B				
<i>Answer any one full question from each module. Each question carries 8 marks</i>				
Module 1				
9	a)	Suppose a country is producing at a point inside the production possibility curve. Draw a PPC and examine this situation.	CO1	(5)
	b)	State the law of demand. Point out any two exceptions of this law.	CO1	(3)
10	a)	A consumer purchases 10 units of a commodity when its price is Rs.100. Later when its price falls to Rs.90, he purchases 8 units only. Estimate price elasticity. What type of a commodity is this?	CO1	(5)
	b)	State the law of variable proportion.	CO1	(3)

Module 2				
11	a)	What is oligopoly? Why price is rigid under oligopoly?	CO2	(5)
	b)	The cost function of a firm is given as $TC=1000+10Q-6Q^2+Q^3$. Calculate fixed cost, variable cost and marginal cost when output is 10 units.	CO2	(3)
12	a)	Suppose a firm is earning super normal profit under monopolistic market condition. Explain this situation by drawing a diagram.	CO2	(5)
	b)	Suppose a firm sells its product at a price of Rs.10 per unit and its average variable cost is Rs.6. If the firm spend Ra.10000 as rent and pay Rs. 6000 as interest every month, estimate its break-even level of output.	CO2	(3)
Module 3				
13	a)	What is inflation? How does inflation affect investment and production.	CO3	(5)
	b)	How will you obtain NNP_{fc} from GDP_{mp} .	CO3	(3)
14	a)	From the data given below (In Rs. Crores) estimate GDP_{mp} and national income. Private final consumption expenditure = 1000, Government expenditure = 500, Invest expenditure = 700, Net exports = 300, Depreciation = 200, $NFIA=(-200)$ and Net indirect tax = 100	CO3	(5)
	b)	What is bank rate? Examine the bank rate policy of the government during inflation.	CO3	(3)
Module 4				
15	a)	Examine the procedures of value engineering.	CO4	(5)
	b)	Examine the application areas of value engineering	CO4	(3)

16	a)	1. Suppose the initial investment of a project is Rs. 3000 (Crores) and the cost of capital or the opportunity cost of capital is 10 percent. Calculate NPV of the project based on the cash flows given below. <table><tr><td>Year</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>Cash flow</td><td>1000</td><td>900</td><td>800</td><td>700</td><td>600</td></tr></table> (In Crores)	Year	1	2	3	4	5	Cash flow	1000	900	800	700	600	CO4	(5)
	Year	1	2	3	4	5										
Cash flow	1000	900	800	700	600											
b)	Point out any three merits of NPV method.	CO4	(3)													

SEMESTER S3/S4

ENGINEERING ETHICS AND SUSTAINABLE DEVELOPMENT

Course Code	UCHUT347	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Equip with the knowledge and skills to make ethical decisions and implement gender-sensitive practices in their professional lives.
2. Develop a holistic and comprehensive interdisciplinary approach to understanding engineering ethics principles from a perspective of environment protection and sustainable development.
3. Develop the ability to find strategies for implementing sustainable engineering solutions.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamentals of ethics - Personal vs. professional ethics, Civic Virtue, Respect for others, Profession and Professionalism, Ingenuity, diligence and responsibility, Integrity in design, development, and research domains, Plagiarism, a balanced outlook on law - challenges - case studies, Technology and digital revolution-Data, information, and knowledge, Cybertrust and cybersecurity, Data collection & management, High technologies: connecting people and places-accessibility and social impacts, Managing conflict, Collective bargaining, Confidentiality, Role of confidentiality in moral integrity, Codes of Ethics. Basic concepts in Gender Studies - sex, gender, sexuality, gender spectrum: beyond the binary, gender identity, gender expression, gender stereotypes, Gender disparity and discrimination in education, employment and everyday life, History of women in Science & Technology,	6

	Gendered technologies & innovations, Ethical values and practices in connection with gender - equity, diversity & gender justice, Gender policy and women/transgender empowerment initiatives.	
2	Introduction to Environmental Ethics: Definition, importance and historical development of environmental ethics, key philosophical theories (anthropocentrism, biocentrism, ecocentrism). Sustainable Engineering Principles: Definition and scope, triple bottom line (economic, social and environmental sustainability), life cycle analysis and sustainability metrics. Ecosystems and Biodiversity: Basics of ecosystems and their functions, Importance of biodiversity and its conservation, Human impact on ecosystems and biodiversity loss, An overview of various ecosystems in Kerala/India, and its significance. Landscape and Urban Ecology: Principles of landscape ecology, Urbanization and its environmental impact, Sustainable urban planning and green infrastructure.	6
3	Hydrology and Water Management: Basics of hydrology and water cycle, Water scarcity and pollution issues, Sustainable water management practices, Environmental flow, disruptions and disasters. Zero Waste Concepts and Practices: Definition of zero waste and its principles, Strategies for waste reduction, reuse, reduce and recycling, Case studies of successful zero waste initiatives. Circular Economy and Degrowth: Introduction to the circular economy model, Differences between linear and circular economies, degrowth principles, Strategies for implementing circular economy practices and degrowth principles in engineering. Mobility and Sustainable Transportation: Impacts of transportation on the environment and climate, Basic tenets of a Sustainable Transportation design, Sustainable urban mobility solutions, Integrated mobility systems, E-Mobility, Existing and upcoming models of sustainable mobility solutions.	6
4	Renewable Energy and Sustainable Technologies: Overview of renewable energy sources (solar, wind, hydro, biomass), Sustainable technologies in energy production and consumption, Challenges and opportunities in renewable energy adoption. Climate Change and Engineering Solutions: Basics of climate change science, Impact of climate change on natural and human systems, Kerala/India and the Climate crisis, Engineering solutions to mitigate, adapt and build resilience to climate change. Environmental	6

	Policies and Regulations: Overview of key environmental policies and regulations (national and international), Role of engineers in policy implementation and compliance, Ethical considerations in environmental policy-making. Case Studies and Future Directions: Analysis of real-world case studies, Emerging trends and future directions in environmental ethics and sustainability, Discussion on the role of engineers in promoting a sustainable future.	
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**Course Assessment Method
(CIE: 50 marks , ESE: 50)**

Continuous Internal Evaluation Marks (CIE):

Continuous internal evaluation will be based on individual and group activities undertaken throughout the course and the portfolio created documenting their work and learning. The portfolio will include reflections, project reports, case studies, and all other relevant materials.

- The students should be grouped into groups of size 4 to 6 at the beginning of the semester. These groups can be the same ones they have formed in the previous semester.
- Activities are to be distributed between 2 class hours and 3 Self-study hours.
- The portfolio and reflective journal should be carried forward and displayed during the 7th Semester Seminar course as a part of the experience sharing regarding the skills developed through various courses.

Sl. No.	Item	Particulars	Group/ Individual (G/I)	Marks
1	Reflective Journal	Weekly entries reflecting on what was learned, personal insights, and how it can be applied to local contexts.	I	5
2	Micro project (Detailed documentation of the project, including methodologies, findings, and reflections)	1 a) Perform an Engineering Ethics Case Study analysis and prepare a report 1 b) Conduct a literature survey on 'Code of Ethics for Engineers' and prepare a sample code of ethics	G	8
		2. Listen to a TED talk on a Gender-related topic, do a literature survey on that topic and make a report citing the relevant papers with a specific analysis of the Kerala context	G	5
		3. Undertake a project study based on the concepts of sustainable development* - Module II, Module III &	G	12

		Module IV		
3	Activities	2. One activity* each from Module II, Module III & Module IV	G	15
4	Final Presentation	A comprehensive presentation summarising the key takeaways from the course, personal reflections, and proposed future actions based on the learnings.	G	5
Total Marks			50	

*Can be taken from the given sample activities/projects

Evaluation Criteria:

- **Depth of Analysis:** Quality and depth of reflections and analysis in project reports and case studies.
- **Application of Concepts:** Ability to apply course concepts to real-world problems and local contexts.
- **Creativity:** Innovative approaches and creative solutions proposed in projects and reflections.
- **Presentation Skills:** Clarity, coherence, and professionalism in the final presentation.

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop the ability to apply the principles of engineering ethics in their professional life.	K3
CO2	Develop the ability to exercise gender-sensitive practices in their professional lives	K4
CO3	Develop the ability to explore contemporary environmental issues and sustainable practices.	K5
CO4	Develop the ability to analyse the role of engineers in promoting sustainability and climate resilience.	K4
CO5	Develop interest and skills in addressing pertinent environmental and climate-related challenges through a sustainable engineering approach.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1						3	2	3	3	2		2
CO2		1				3	2	3	3	2		2
CO3						3	3	2	3	2		2
CO4		1				3	3	2	3	2		2
CO5						3	3	2	3	2		2

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Ethics in Engineering Practice and Research	Caroline Whitbeck	Cambridge University Press & Assessment	2nd edition & August 2011
2	Virtue Ethics and Professional Roles	Justin Oakley	Cambridge University Press & Assessment	November 2006
3	Sustainability Science	Bert J. M. de Vries	Cambridge University Press & Assessment	2nd edition & December 2023
4	Sustainable Engineering Principles and Practice	Bhavik R. Bakshi,	Cambridge University Press & Assessment	2019
5	Engineering Ethics	M Govindarajan, S Natarajan and V S Senthil Kumar	PHI Learning Private Ltd, New Delhi	2012
6	Professional ethics and human values	RS Naagarazan	New age international (P) limited New Delhi	2006.
7	Ethics in Engineering	Mike W Martin and Roland Schinzinger,	Tata McGraw Hill Publishing Company Pvt Ltd, New Delhi	4" edition, 2014

Suggested Activities/Projects:**Module-II**

- Write a reflection on a local environmental issue (e.g., plastic waste in Kerala backwaters or oceans) from different ethical perspectives (anthropocentric, biocentric, ecocentric).
- Write a life cycle analysis report of a common product used in Kerala (e.g., a coconut, bamboo or rubber-

based product) and present findings on its sustainability.

- Create a sustainability report for a local business, assessing its environmental, social, and economic impacts
- Presentation on biodiversity in a nearby area (e.g., a local park, a wetland, mangroves, college campus etc) and propose conservation strategies to protect it.
- Develop a conservation plan for an endangered species found in Kerala.
- Analyze the green spaces in a local urban area and propose a plan to enhance urban ecology using native plants and sustainable design.
- Create a model of a sustainable urban landscape for a chosen locality in Kerala.

Module-III

- Study a local water body (e.g., a river or lake) for signs of pollution or natural flow disruption and suggest sustainable management and restoration practices.
- Analyse the effectiveness of water management in the college campus and propose improvements - calculate the water footprint, how to reduce the footprint, how to increase supply through rainwater harvesting, and how to decrease the supply-demand ratio
- Implement a zero waste initiative on the college campus for one week and document the challenges and outcomes.
- Develop a waste audit report for the campus. Suggest a plan for a zero-waste approach.
- Create a circular economy model for a common product used in Kerala (e.g., coconut oil, cloth etc).
- Design a product or service based on circular economy and degrowth principles and present a business plan.
- Develop a plan to improve pedestrian and cycling infrastructure in a chosen locality in Kerala

Module-IV

- Evaluate the potential for installing solar panels on the college campus including cost-benefit analysis and feasibility study.
- Analyse the energy consumption patterns of the college campus and propose sustainable alternatives to reduce consumption - What gadgets are being used? How can we reduce demand using energy-saving gadgets?
- Analyse a local infrastructure project for its climate resilience and suggest improvements.
- Analyse a specific environmental regulation in India (e.g., Coastal Regulation Zone) and its impact on local communities and ecosystems.

- Research and present a case study of a successful sustainable engineering project in Kerala/India (e.g., sustainable building design, water management project, infrastructure project).
- Research and present a case study of an unsustainable engineering project in Kerala/India highlighting design and implementation faults and possible corrections/alternatives (e.g., a housing complex with water logging, a water management project causing frequent floods, infrastructure project that affects surrounding landscapes or ecosystems).

SEMESTER S4

ANALOG AND DIGITAL ELECTRONICS LAB

Course Code	PCMRL407	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	GCESL208, PBMRT304	Course Type	Lab

Course Objectives:

1. To impart working knowledge, design and practical skills among students.

Expt. No.	Experiments
1	Study and verify the truth table of basic gates, half and full adder & subtractor
2	Design and implementation of binary to gray , gray to binary code converter, BCD to excess 3 code using logic gates
3	Design and implementation of MUX and De-Mux using a) logic gates (4x1 MUX, 1X 4 De-Mux) IC 74151 – Implementation of basic SOP functions (16x1 , 8x1)
4	Study and verify the truth table of flip flops (Implement using logic gates)
5	Design and Implementation of 3 bit Synchronous and Asynchronous counter using JK Flip Flop
6	Design and Implementation of a) 3 bit Synchronous and Asynchronous up counter using JK Flip Flop b) MOD counters (eg: MOD 10, MOD 12 Ripple counter)
7	Implementation of basic programs using verilog-Adder, Subtractor, Mux, De-MUX
8	CE characteristics of BJT (Input and output Chara)
9	Characteristics of JFET(Input and output Chara)
10	Design and implement Inverting op-amp, non inverting op-amp and voltage follower using op amp. Plot the frequency response curve.
11	Design and implement Integrator, Differentiator, Schmitt trigger (UTP=+3v & LTP= -3v) using IC 741. Verify the output waveform on CRO.

12	Design and implement RC phase shift oscillator/wein bridge oscillator, Triangular/ Saw tooth wave generator, Astable/ Monostable multivibrator using IC 741
13	Astable/ Monostable multivibrator using IC 555 timer
14	Design of PLL using IC 565.

Any 3 experiments (8 to 14) can be simulated using LT SPICE/ PROTEUS

Course Assessment Method **(CIE: 50 marks, ESE: 50 marks)**

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- ***Submission of Record:** Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- ***Endorsement by External Examiner:** The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	To acquire skills in designing, analysing, testing and implementing various digital and analog circuits	K3, K4, K5, K6
CO2	Analyse and interpret circuits using software and simulation tools	K3, K4, K5, K6
CO3	To enhance the ability to function effectively as an individual and in a team to accomplish the given task	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	3	2	1	1	1	3	2	2	2
CO2	2	2	2	3	3	1	1	1	3	2	2	2
CO3	1	2	2	2	2	1	1	3	3	2	2	2

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fundamentals of Logic Design	Charles H Roth Jr	CENGAGE Learning Custom Publishing	7 th /2013
2	Fundamentals of Digital Logic With Verilog Design	Stephen Brown, Zvonko Vranesic	McGraw Hill	3 rd /2013
3	Digital Fundamentals	Floyd and Jain	Pearson Education	11 th /2017
4	Laboratory Manual for Operational Amplifiers and Linear ICs	David A. Bell	OUP Canada	2 nd edition 2006

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Electronics Lab Manual- volume 1, 2	K A Navas	PHI Learning Private Ltd	2015

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER S4

MECHANICAL ENGINEERING LAB

Course Code	PCMRL408	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Lab

Expt. No.	Experiments
1, 2,3, 4	Exercises on Centre Lathe: - Facing, plain turning, step turning, taper turning and form turning, groove cutting, knurling and chamfering, multi-start thread, square thread and internal thread, Inspection of the work using measuring instruments
5, 6	Exercises on Milling Machine: - Face milling, end milling – spur and helical gear cutting, milling of key ways, T slot cutting, Inspection of the work using measuring instruments
7	Exercises on drilling machine: - drilling, boring, reaming, tapping and counter sinking etc.
8	Exercises on shaping machine / slotting machine - flat surfaces, grooves and key ways.
9	Exercises on Grinding machine: - Exercise on surface grinding, cylindrical grinding and tool grinding, Inspection of the work using measuring instruments
10	Exercises on Welding: - Exercises on arc and gas welding - butt welding and lap welding of M.S. sheets.
11	Preparation of program and Exercise on CNC lathe including simulation: -turning, step turning, taper turning, thread cutting, etc.
12	Preparation of program and Exercise on CNC milling machine including simulation: - surface milling, pocket milling, etc
13	Use of CAM software for generation of CNC code from CAD data.
14	Exercises on Rapid Prototyping: - 3D printing - simple box, helix shape, U bracket

A minimum of 10 experiments are to be carried out

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Get hands on various manual production machines and processes	K3
CO2	Identify the various operations and the required machines and attachments for various applications	K3
CO3	Preparing the CNC machines and programming.	K3
CO4	Use CAD/CAM for generating CNC code for production machines to realise parts	K3
CO5	Understand the properties of 3D printed parts and use the rapid prototyping effectively to make prototypes.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											3
CO2	3											3
CO3	3				2							3
CO4	3	2	2		3							3
CO5	3	2	2		3							3

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Production Technology,	HMT	Tata McGraw Hill.	1e, 1986
2	Machine Tool Design, Vol. I, II, III and IV,	Acherkan N. S.	MIR Publications.	1969

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Workshop Technology Part I	W. A. J. Chapman,	ELBS & Edward Arnold Publishers.	1e, 1972

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc21_me04

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 5

MECHATRONICS ENGINEERING

SEMESTER S5
LINEAR CONTROL SYSTEMS

Course Code	PCMRT501	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	GCMAT101, GCMAT201, GCMAT301	Course Type	Theory

Course Objectives:

1. To introduce the elements of control system and its modelling
2. To introduce methods for analyzing the time response, the frequency response and the stability of systems.
3. This course enables students to systematically study the principles of system modelling

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Control systems: Basic components of a Control System, Open-Loop Control Systems and Closed-Loop Control Systems, Examples of control system, Transfer functions, Block diagram reduction, Signal flow graph and Mason's gain formula. Differential equations for mass, spring and dashpot elements, Mathematical modelling of control systems - Electrical Systems, Translational and Rotational Mechanical systems, Analogous systems-force voltage & force-current analogy - torque-voltage & torque-current analogy.	11
2	Time domain analysis: Continuous systems -standard test signals - step, ramp, parabolic, impulse - transient and steady state response –first order systems – second order systems, step responses of first order systems and second order systems - under damped and over damped systems - time domain specifications - steady state error – static position, velocity &	11

	acceleration error constants.	
3	Stability of control systems and Frequency Domain Analysis: Concept of stability - stability & location of the poles in S-plane - Routh-Hurwitz stability criterion-Root Locus Method, Construction of root locus-Effect of poles and zeros and their location on the root locus. Frequency Response representation- Polar Plot- Relative stability: gain margin and phase margin, Stability analysis with Bode plot.	11
4	Compensation techniques and Case studies in Mechatronics: Need for Cascade compensation, Cascade Compensation- PI, PD and PID controllers – tuning of PID Controller- Lead, Lag and Lead- Lag compensation- Role of control system in mechatronics-case studies Automatic temperature control, automatic traffic light control-Automatic street light control	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	To introduce the elements of control system and its modelling	K2
CO2	To introduce methods for analyzing the time response.	K3
CO3	To introduce methods for analyzing the frequency response and the stability of systems.	K3
CO4	Apply the knowledge of control systems in real time control applications	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	2								2
CO2	3	3		2								2
CO3	3	2	2									2
CO4	3	2		2								2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Control Systems Engineering	I. J. Nagrath and M. Gopal	New Age International Pvt Ltd, 6/e	Seventh Edition 2021
2	Control Systems	A. Nagoorkani	RBA Publications	Fifth Edition 2020

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Modern Control Engineering	Katsuhiko Ogata	Pearson Education India, 5/e	Fifth Edition 2015
2	Automatic Control Systems	Kuo	Prentice Hall	First Edition 2012
3	Control Systems Engineering	Norman S. Nise	Wiley India Pvt. Ltd	First Edition 2018

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc20_ee90/preview
2	https://archive.nptel.ac.in/courses/107/106/107106081/
3	https://archive.nptel.ac.in/content/storage2/courses/112104040/lecture23/23_8.htm
4	https://archive.nptel.ac.in/courses/107/106/107106081/

SEMESTER S5
THERMODYNAMICS

Course Code	PCMRT502	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. To understand basic concepts and laws of thermodynamics
2. Determine change in properties of pure substances during phase change processes

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Role of Thermodynamics and its applications–Basic Concepts Macroscopic and Microscopic viewpoints, Concept of Continuum, Thermodynamic System and Control Volume, Surrounding, Boundaries, Types of Systems, Universe, Thermodynamic properties, Process, Cycle, Thermodynamic Equilibrium, Quasi – static Process State, Point and Path function. Zeroth Law of Thermodynamics, Measurement of Temperature, reference Points, Temperature Scales.	11
2	Energy - Work - Pdv work and other types of work transfer, free expansion work, heat and heat capacity. Joule's Experiment- First law of Thermodynamics - First law applied to Non flow Process- Enthalpy-specific heats- PMM1, First law applied to Flow Process, Mass and Energy balance in simple steady flow process, Limitations of First Law.	11
3	Second Law of Thermodynamics, Thermal Reservoir, Heat Engine, Heat pump – Kelvin- Planck and Clausius Statements, Equivalence of two statements, Reversibility, Irreversible Process, Causes of Irreversibility, PMM2, Carnot's theorem and its corollaries, Absolute Thermodynamic Temperature scale. Clausius Inequality, Entropy- Entropy changes in various thermodynamic processes, principle of increase of entropy, Entropy	11

	generation, Entropy and Disorder, isentropic process, Third law of thermodynamics.	
4	Pure Substances, Phase Transformations, Triple point, properties during change of phase, T-v, p-v and p-T diagram of pure substance, Saturation pressure and Temperature, T-s, h-s diagrams or Mollier Charts, Dryness Fraction, Property calculations using steam tables. The ideal Gas Equation, Characteristic an Universal Gas constants, Deviations from ideal Gas Model: Equation of state of real substances, Vander Waals Equation of State, Compressibility factor, General Thermodynamic Relations–Helmholtz and Gibb's functions - Maxwell's Relations, Tds equations, Throttling process, Joule Thomson Coefficient.	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand basic concepts and laws of thermodynamics	K2
CO2	Conduct first law analysis of open and closed systems	K3
CO3	Determine entropy and availability changes associated with different processes	K3
CO4	Understand the application and limitations of different equations of state	K2
CO5	Determine change in properties of pure substances during phase change processes	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2										2
CO2	3	3	2	2								2
CO3	3	3	2	2								2
CO4	2	2	2	2								2
CO5	3	3	2	2								2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation\

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Engineering Thermodynamics	P. K. Nag	McGraw Hill	2013
2	Fundamentals of Engineering Thermodynamics	E. Rathakrishnan	PHI	2005
3	Thermodynamics an Engineering Approach	Y. A. Cengel and M. A. Boles	McGraw Hill	2011

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fundamentals of Engineering Thermodynamics	Moran J., Shapiro N. M.	Wiley	2006
2	Fundamentals of Thermodynamics	R. E. Sonntag and C. Borgnakke	Wiley	2009
3	Thermodynamics	Holman J. P.	McGraw Hill	2004
4	Engineering Thermodynamics	M. Achuthan	PHI	2004

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/112/106/112106310/ https://www.youtube.com/watch?v=9GMBpZZtjXM&list=PLD8E646BAB3366BC8
2	https://archive.nptel.ac.in/courses/112/106/112106310/ https://www.youtube.com/watch?v=sUDfpFD0xX4&list=PLD8E646BAB3366BC8&index=4
3	https://archive.nptel.ac.in/courses/112/106/112106310/ https://www.youtube.com/watch?v=lvy8h-yWhRQ&list=PLD8E646BAB3366BC8&index=6
4	https://archive.nptel.ac.in/courses/112/106/112106310/ https://www.youtube.com/watch?v=QdnssaDSyZs&list=PLD8E646BAB3366BC8&index=14

SEMESTER S5

INDUSTRIAL HYDRAULICS & PNEUMATICS

Course Code	PCMRT503	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. To understand various components of pneumatic and fluid power systems and their advantages
2. Applying the fundamental concepts of fluid power and pneumatic to obtain solution to engineering problems.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basics of fluid power system, Components, Advantages and applications of Fluid power systems Fluid properties: specific weight, density and specific gravity, Pascal's Law and its application Continuity equation, Conservation Of Energy, Frictional losses in hydraulic pipelines - laminar and turbulent flow, Fluid power symbols.	11
2	Hydraulic pumps: Gear, Vane and Piston pumps, Pump performance, Characteristics and Selection - Sizing of pumps Hydraulic Motors- Limited Rotation Hydraulic Motors, Gear Motors, Vane Motors, Piston Motors, Hydraulic Motor Theoretical Torque, Power, And Flow Rate, Hydraulic Motor Performance	11
3	Hydraulic Valves- Direction control valves: Three way valve, Four way valves, Check valve and shuttle valve Actuation mechanisms in DCV Pressure control valves: Pressure relief, Pressure Reducing, Counter balance, Sequencing and Unloading Valves Flow control valves and its types Proportional Valves Servo valves: Mechanical type and	11

	Electrohydraulic servo valves.	
4	Pneumatics: Air compressors, air treatment: Air filters, Dryer, Lubricator, pressure regulation, Compressor types, Pneumatics and Hydraulics, Properties Of Air, The Perfect Gas Laws, Fluid Conditioners, Air Control Valves, Pneumatic Actuators, Fluid Logic Control Systems - Introduction To Boolean Algebra, Laws Of Boolean Algebra, Illustrative Examples Using Boolean Algebra	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand various components of fluid power systems, their advantages and select them for various applications.	K2
CO2	Design and simulate fluid power systems for automation of machine tools and other equipment's.	K3
CO3	Choose appropriate control valves for fluid power applications	K3
CO4	Select pneumatic components and fluid power actuators for low cost automation	K2
CO5	Apply the fundamental concepts of fluid power and to obtain solution to engineering problems.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

3	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1									3
CO2	3	2	2									3
CO3	3	2	2			2						3
CO4	3	2	2					1				3
CO5	3	2	2			2						3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Hydraulics and Pneumatics	A. Parr	Elsevier	1999
2	Fluid Power with Applications	Anthony Esposito	Pearson New International Edition-Pearson	2013
3	Pneumatic System-Principles and Maintenance	S.R.Majumdar	McGraw Hill	2017.
4	Hydraulic and Pneumatics control	R.Srinivasan	McGraw Hill	2008

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Hydraulic control systems	H. E. Merritt	John Wiley& Sons, NewYork	1967
2	Basic fluid power	D. A. Peace	Prentice Hall	1987
3	Power Hydraulic	M. J. Pinches	Longman	1988

SEMESTER S5

SIGNAL PROCESSING& COMMUNICATION

Course Code	PBMRT504	CIE Marks	60
Teaching Hours/Week (L: T:P: R)	4	ESE Marks	40
Credits	3:0:0:1	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	4	Course Type	Theory

Course Objectives:

1. This course aims to impart the basic concepts of continuous and discrete signals and systems.
2. This course aims to develop understanding about frequency domain approaches used for analysis of continuous and discrete time signals and systems.
3. This course introduces various schemes of analog and digital modulation used in communication systems.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Signals & Systems: Elementary Signals, Classification of signals, Basic operations on signals. Concept of System, Classification, Properties of Systems-Linearity, Time invariance, Causality, Stability. Linear Time Invariant (LTI) Systems – Convolution sum, Impulse response, Difference Equation representation of LTI Systems	9
2	Frequency Domain Approaches: Discrete Time Fourier Transform (DTFT), Discrete Fourier Transform (DFT)-Properties. Review of Laplace Transform, Region of Convergence (ROC), Properties of ROC, Transfer function, Stability and causality conditions. Analysis of LTI systems using Laplace transforms. Sampling of continuous time signals (definition and concept only), Sampling	9

	theorem for lowpass signals, aliasing. Basic Elements of a Digital Signal Processing (DSP) system, Typical DSP applications.	
3	Analog Communication: Block diagram of a communication system, Need for modulation. Principle of Amplitude Modulation (AM) – Frequency spectrum – power relations – generation of AM – DSB, DSB/SC, SSB. Angle Modulation - Frequency and Phase modulation - Theory and generation of Frequency Modulation (FM). Super heterodyne receivers (AM and FM).	9
4	Digital Communication: Sampling Process-Pulse modulation Techniques-PAM, PWM and PPM, Principles of digital communication & advantages – PCM encoder and decoder, Applications of data communication. Digital modulation schemes – ASK, FSK, PSK.	9

Suggestion on Project Topics

Course Assessment Method (CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Ex-1	Internal Ex-2	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 2 marks (8x2 =16 marks)	2 questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 2 sub divisions. - Each question carries 6 marks. (4x6 = 24 marks)	40

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply properties of signals and systems to classify them.	K3
CO2	Apply frequency domain approaches on continuous and discrete time signals and systems.	K3
CO3	Summarize analog modulation techniques and the need for modulation	K2
CO4	Explain analog and digital pulse modulation schemes.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

[illegible]

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Signals & Systems	Simon Haykin	Wiley	2/e, 2007
2	Signals and Systems	Alan V. Oppenheim and Alan Willsky	Pearson Education India	2/e, 2015
3	Electronic Communication Systems	Kennedy G.	McGraw-Hill Education	5/e, 2011
4	Communication Systems	Simon Haykin	Wiley	4/e, 2006

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Signals and Systems	Anand Kumar	Prentice Hall India	3/e, 2013
2	Digital Signal Processing	Proakis J. G. and Manolakis D. G	Pearson Education India	4/e, 2007
3	Electronic Communications	Roody and Coolen	Pearson Education India	4/e, 2008
4	Digital Communications	John G Proakis	McGraw-Hill Education	5/e, 2018

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
Module - I	https://nptel.ac.in/courses/108104100
Module - II	https://nptel.ac.in/courses/108104100
Module - III	https://nptel.ac.in/courses/117102059
Module - IV	https://nptel.ac.in/courses/117102059

PBL Course Elements

L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	Practical	Presentation
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation/ Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		30

1. Project Planning and Proposal (5 Marks)

- Clarity and feasibility of the project plan
- Research and background understanding
- Defined objectives and methodology

2. Contribution in Progress Presentation and Question Answer Sessions (4 Marks)

- Individual contribution to the presentation
- Effectiveness in answering questions and handling feedback

3. Involvement in the Project Work and Team Work (3 Marks)

- Active participation and individual contribution
- Teamwork and collaboration

4. Execution and Implementation (10 Marks)

- Adherence to the project timeline and milestones
- Application of theoretical knowledge and problem-solving
- Final Result

5. Final Presentation (5 Marks)

- Quality and clarity of the overall presentation
- Individual contribution to the presentation
- Effectiveness in answering questions

6. Project Quality, Innovation, and Creativity (3 Marks)

- Overall quality and technical excellence of the project
- Innovation and originality in the project
- Creativity in solutions and approaches

SEMESTER S5

POWER ELECTRONICS

Course Code	PE44T521	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PC44T302	Course Type	Theory

Course Objectives:

1. Identify and describe the characteristics and operations of various power semiconductor devices.
2. Comprehend the operation and design of basic power converter circuits.
3. Implement control strategies in power electronic converters.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Power semiconductor devices: Power diodes, power transistors, thyristor family, SCRs, Triac, GTOs, power MOSFETs, IGBTs, MCTs-static and dynamic characteristics, protection circuits, series and parallel connections, turn-on characteristics, turn off characteristics.	9
2	DC-DC Converters Step down and Step up choppers – Single-quadrant and Two-quadrant chopper – control schemes Switching regulators – Buck, Boost & Buck-boost –Operation with continuous conduction mode – Waveforms	9
3	DC-AC Converters (Inverters) Basic principles of inverters and types (square wave, sine wave, PWM inverters). Performance parameters and harmonic analysis. Applications in UPS, motor drives, and grid-tied renewable energy systems AC-DC Converters (Rectifiers) Operation of half-wave and full-wave rectifiers. Bridge rectifiers and performance parameters (ripple, efficiency).	9

4	Control Techniques in Power Electronics Pulse Width Modulation (PWM): Introduction to PWM techniques, Types of PWM (single-pulse, multiple-pulse, sinusoidal PWM). Applications in voltage regulation and motor control. Industrial Applications and Practical Considerations Power Electronics in Renewable Energy, Role of power electronics in solar PV and wind energy systems. Role of power electronics in motor control. Variable frequency drives (VFD).	9
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the fundamental concepts and significance of power electronics in modern technology.	K2
CO2	Design and analyse DC-DC converters.	K4
CO3	Design and analyse DC-AC and AC to DC converters.	K4
CO4	Apply pulse-width modulation and feedback control techniques in power electronic systems and understand the applications of power electronics in renewable energy systems and motor drives.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											3
CO2	3											3
CO3	3											3
CO4	3											3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Power Electronics: Converters, Applications, and Design	Ned Mohan, Tore M. Undeland, and William P. Robbins.	WILEY INDIA	3RD EDITION 2002
2	Power Electronics Handbook	Muhammad H. Rashid.	BH ELSEVIER	5TH EDITION 2023

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Power Electronics: Circuits, Devices and Applications	Muhammad H. Rashid	BH ELSEVIER	5TH EDITION 2023

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/108/102/108102145/
2	https://archive.nptel.ac.in/noc/courses/noc20/SEM2/noc20-ee97/
3	https://archive.nptel.ac.in/noc/courses/noc20/SEM2/noc20-ee97/
4	https://archive.nptel.ac.in/noc/courses/noc20/SEM2/noc20-ee97/

SEMESTER S5

DESIGN OF MANUFACTURE

Course Code	PEMRT522	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PBMRT404	Course Type	Theory

Course Objectives:

1. Equip students with a comprehensive understanding of engineering design principles, including material selection, manufacturability, process capability, and form design.
2. Enable students to apply sustainable and environmentally responsible design practices, focusing on recyclability, energy efficiency, and minimizing environmental impact.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Engineering design: Concept – Kinds of design – Design process steps – Factors influencing design – Concurrent Engineering – Material selection process – Evaluation methods for material selection. General design principles for manufacturability: strength and mechanical factors, mechanisms selection, evaluation method	9
2	Factors Influencing form Design: Working principle, Material, Manufacture, Design- Possible solutions, Materials choice, Influence of materials on form design. Process capability: Feature tolerances, Geometric tolerances, Assembly limits, Datum features, and Tolerance stacks. Process capability analysis – Centrality analysis – Compound assembly – Selective and Interchangeable assembly – Grouped Datum systems.	9
3	Design for machining: Design features to facilitate machining using drills, milling cutters, keyways, Doweling procedures, Reduction of machined	9

	area, simplification by separation, simplification by amalgamation, Design for machinability, Design for economy, Design for accessibility, Design for assembly, Design for disassembly.	
4	Design for the Environment: Introduction, Basic DFE methods, Design guide lines, Example application, Lifecycle assessment, Basic method, Environmentally responsible product assessment, Weighted sum assessment method, Lifecycle assessment method, Techniques to reduce environmental impact, Design to minimize material usage, Design for recyclability, Design for remanufacture, Design for energy efficiency.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply the knowledge of General design principles for manufacturability	K3
CO2	Identify Factors Influencing form Design, improve process capability and use Selective and Interchangeable assembly to simplify designs.	K2
CO3	Identify uneconomical design of casting and welding, Modify the design, and design better castings, welded joints and brazing and soldering joints.	K3
CO4	Simplify Design features to facilitate simplicity in different machining processes and apply design principles for machinability, assembly and disassembly.	K2
CO5	Apply the knowledge of Design for energy efficiency and less environmental impact	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

2	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3		3									2
CO2	3		3									2
CO3	2		2									2
CO4	2		3				2					2
CO5	3	2				2						2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Engineering Design: A Materials and processing Approach	Dieter, G.E.	McGraw Hill Co. Ltd	2000
2	Assembly, Automation and product design	Boothroyd, G.	CRC press	2005
3	Product Design	Kevien Otto and Kristin Wood	Pearson Publication	2004
4	Product design and development	Ulrich, K.T. and Eppinger, S.D.	Tata McGraw Hill	
5	Design for Manufacturability Handbook	James G. Bralla	McGraw-Hill companies, New York, USA	Second Edition, 1998
6	Design for the Environment	Fixel, J.	McGraw Hill	1996

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Product Design for Manufacture	Boothroyd, G, Hertz and Nike	Marcel Dekker	1994
2	Engineering Design and Design for Manufacture and Structural Approach	Dickson, John. R, and Corroda Poly	Field Stone Publisher, USA	1995
3	Design for the Environment	Graedel T. Allen By. B	Reason Pub.	1996
4	Engineering Design	Eggert, R.J.	Pearson Education, Inc. New Jersey	2005
5	Engineering Design for Manufacture	KalandarSaheb, S.D and Prabhakar, O.	ISPE	1999.

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/IYvyQ0sXrrU
2	https://youtu.be/IYvyQ0sXrrU
3	https://youtu.be/IYvyQ0sXrrU
4	https://youtu.be/IYvyQ0sXrrU

SEMESTER S5
OPERATIONAL RESEARCH

Course Code	PEMRT523	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. To provide students with a problematic understanding of linear programming and transportation.
2. To familiarize students with different algorithms in network theory and game theory.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Linear programming problems Convex functions - Linear programming problem- Graphical method -Basic feasible solutions- Optimal solutions-additional topics of LPP-The Simplex Method-The simplex method in tabular form- Tie breaking in simplex method.	9
2	Transportation and assignment problems Transportation problem - Formulation - Existence of solutions of Transportation problems - Solutions of Transportation problem - Finding an initial basic solution- Vogel's approximation method-assignment problem- A special algorithm for the assignment problem.	9
3	Network optimization models The terminology of networks-The shortest path problem- Dijkstras Algorithm-The minimum spanning tree problem- The maximum flow problem- Ford Fulkerson algorithm- The minimum cost flow problem – The network simplex method.	9
4	Theory of games Introduction to game theory-Matrix(rectangular) games- Problem of game	9

	theory-Minimax theorem- Saddle point- Strategies and pay off- Theorems of matrix games- Graphical solutions- Notion of dominance- Rectangular game as an Linear programming problem.	
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand different methods for solving a linear programming problem	K2
CO2	Illustrate the graphical methods for solving LPP	K3
CO3	Apply various methods in transportation problem	K3
CO4	Understand and utilize different algorithms for network theory	K2
CO5	Apply various techniques for solving rectangular games	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	2	2					2		3
CO2	3	3	3	2	2					2		3
CO3	2	3	3	2	2					2		3
CO4	2	3	3	2	2					2		3
CO5	2	3	3	2	2					2		3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to operations research concepts and cases	Frederick S. Hillier Gerald J. Lieberman	Mc Graw Hill	8 th and 2011
2	Optimization methods in operations research and systems analysis	K.V. Mital C. Mohan	New age international (P) Ltd	3 rd and 1996

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Operations Research (SIE)	Frederick S. Hillier Gerald J. Lieberman Bodhibroto Nag <u>Preetam Basu</u>	Mc Graw Hill	11th Edition and 2021
2	Operations Research	R Panneerselvam	PHI	5 th and 2004
3	Linear programming	G Hadley	Adison	1975

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc22_ma48/preview
2	https://onlinecourses.nptel.ac.in/noc22_ma48/preview
3	https://onlinecourses.nptel.ac.in/noc22_ma48/preview
4	https://onlinecourses.nptel.ac.in/noc22_ma48/preview

SEMESTER S5
NUMERICAL COMPUTATION USING PYTHON

Course Code	PEMRT524	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. To provide students with a solid foundation in numerical methods for solving engineering problems using Python.
2. To enable students to implement numerical techniques for nonlinear equations, interpolation, integration, differentiation, and ordinary differential equations.
3. To develop students' skills in applying numerical methods to analyse and interpret engineering data and solve real-world problems.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Root Finding Methods (6 Hours) (Derivation or proof not required) Bisection method; Newton-Raphson method; Secant method; False Position Method; Fixed-point iteration; Convergence and error analysis of root-finding methods. Solving nonlinear equations using the above methods in Python (2 Hours)	8
2	Interpolation (9 Hours) (Derivation or proof not required) Finite difference interpolation: Newton's finite and backward difference interpolation methods. Polynomial Interpolation: Lagrange interpolation and Newton's divided difference interpolation. Spline Interpolation: Cubic splines and B-splines. Error analysis in interpolation. Implementing interpolation methods in Python (2 Hours)	11

3	Numerical Integration and Differentiation (6 Hours) Numerical Differentiation: Finite difference methods; Error analysis in numerical differentiation. Numerical Integration: Trapezoidal rule, Simpson's rule, Gaussian quadrature, Adaptive quadrature methods. Writing Python programs to perform numerical differentiation and integration (2 Hours)	8
4	Solution of Ordinary Differential Equations (7 Hours) Initial Value Problems: Euler's method; Runge-Kutta methods; Multistep methods (Adams-Bashforth and Adams-Moulton methods). Boundary Value Problems: Shooting method; Finite difference method. Solving ODEs using Python (2 Hours)	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Implement root-finding algorithms in Python to solve nonlinear equations and analyse the convergence behaviour of these methods.	K3
CO2	Apply interpolation techniques in Python for data fitting and solving engineering problems and perform error analysis.	K3
CO3	Develop Python programs to implement numerical differentiation and integration techniques.	K3
CO4	Implement numerical methods in Python to solve ordinary differential equations.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3				3							3
CO2	3				3							3
CO3	3				3							3
CO4	3				3							3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Numerical Methods in Engineering with Python 3	Jaan Kiusalaas	Cambridge University Press	3 & 2013
2	Numerical Methods for Engineers	Steven C. Chapra and Raymond P. Canale	McGraw-Hill Education	7 & 2014

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Numerical Analysis	Richard L. Burden and J. Douglas Faires	Cengage Learning	10 & 2015
2	Applied Numerical Methods with MATLAB for Engineers and Scientists	Steven C. Chapra	McGraw-Hill Education	4 & 2017
3	Introductory Methods of Numerical Analysis	S.S.Sastry	Prentice Hall India Learning Private Limited	5 & 2012
4	Numerical Method and Programming	Bikas Chandra Bhui & Dipak Chatterjee	Vikas	2 & 2016
5	Numerical Methods Using Matlab	Mathews and Fink	Pearson Education India	4 & 2015

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	NPTEL - MATLAB Programming for Numerical Computation, IIT Madras NPTEL - Numerical methods, IIT Roorkee NPTEL - Applied Numerical Methods, IIT Kanpur
2	NPTEL - MATLAB Programming for Numerical Computation, IIT Madras NPTEL - Numerical methods, IIT Roorkee NPTEL - Applied Numerical Methods, IIT Kanpur
3	NPTEL - MATLAB Programming for Numerical Computation, IIT Madras NPTEL - Numerical methods, IIT Roorkee NPTEL - Applied Numerical Methods, IIT Kanpur
4	NPTEL - MATLAB Programming for Numerical Computation, IIT Madras NPTEL - Numerical methods, IIT Roorkee NPTEL - Applied Numerical Methods, IIT Kanpur

SEMESTER S5

METALLURGY AND MATERIAL ENGINEERING

Course Code	PEMRT526	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Engineering Physics and Engineering Chemistry	Course Type	Theory

Course Objectives:

1. Provide students with a deep understanding of crystal imperfections, diffusion, phase diagrams, and heat treatment processes, and their impact on material properties and mechanical behaviour.
2. Enable students to explore and apply advanced materials and strengthening mechanisms, including fatigue, creep, composites, superalloys, ceramics, and various alloy steels, to improve material performance in engineering applications.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Classification of crystal imperfections - forest of dislocation, role of surface defects on crack initiation- Burgers vector –Frank Read source - Correlation of dislocation density with strength and nano concept - high and low angle grain boundaries– driving force for grain growth and applications - Polishing and etching - X – ray diffraction, simple problems –SEM and TEM – Diffusion in solids, fick’s laws, mechanisms, applications of diffusion in mechanical engineering, simple problems.	9
2	Phase diagrams: - need of alloying - classification of alloys - Hume Rothery’s rule – equilibrium diagram of common types of binary systems: five types - Coring - lever rule and Gibb’s phase rule- Reactions- Detailed discussion on Iron-Carbon equilibrium diagram with microstructure and properties -Heat treatment: - TTT, CCT diagram, applications -	9

	Tempering- Hardenability, Jominy end quench test, applications- Surface hardening methods.	
3	Strengthening mechanisms - cold and hot working - alloy steels: how alloying elements affecting properties of steel - nickel steels - chromium steels - high speed steels -cast irons - principal non ferrous alloys.	9
4	Fatigue: - creep -DBTT - super plasticity - need, properties and applications of composites, super alloy, intermetallics, maraging steel, Titanium - Ceramics:- structures, applications.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the basic chemical bonds, crystal structures (BCC, FCC, and HCP), and their relationship with the properties	K2
CO2	Analyze the microstructure of metallic materials using phase diagrams and modify the microstructure and properties using different heat treatments.	K3
CO3	Quantify mechanical integrity and failure in materials.	K2
CO4	Apply the basic principles of ferrous and non-ferrous metallurgy for selecting materials for specific applications.	K3
CO5	Define and differentiate engineering materials on the basis of structure and properties for engineering applications.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											2
CO2	3	3										2
CO3	3			2								2
CO4	3				3							2
CO5	3											2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Material Science and Engineering	Callister William. D.	John Wiley	2014
2	Engineering Metallurgy part - I	Higgins R. A.	ELBS	1998

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Physical Metallurgy	E Avner H Sidney	Tata McGraw Hill	2009
2	Material Science for Engineers	Anderson J.C. et.al.	Chapman and Hall	1990
3	Physical metallurgy for Engineers	Clark and Varney	Van Nostrand	1964
4	Mechanical Metallurgy, Tata	Dieter George E	McGraw Hill	1976
5	Material Science and Engineering	Raghavan V	Prentice Hall	2004
6	Physical metallurgy principles	Reed Hill E. Robert	Cengage Learning	4th edition, 2009
7	Mechanical behavior of materials	Myers Marc and Krishna Kumar Chawla	Cambridge University press	2008
8	Elements of Material Science	Van Vlack	Addison Wesley	1989

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/113/106/11310603s
2	https://nptel.ac.in/courses/113/106/11310603s
3	https://nptel.ac.in/courses/113/106/11310603s
4	https://nptel.ac.in/courses/113/106/11310603s

SEMESTER S5

IOT AND APPLICATIONS

Course Code	PEMRT525	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs.30 Min.
Prerequisites (if any)	None	Course Type	

Course Objectives:

1. Equip students with a comprehensive understanding of IoT technology, including its architecture, enabling technologies, communication models, and protocols, to effectively design and deploy IoT systems.
2. Enable students to address IoT security, privacy, and vulnerability issues, and apply practical IoT solutions using platforms like Raspberry Pi for real-world applications such as smart cities.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to IoT technology: Definitions and Characteristics of IoT, IoT Architectural View, Physical Design of IOT, Logical Design of IoT- IoT Functional blocks, IoT communication models, IoT Enabling Technologies, IoT Levels & Deployment Templates.	9
2	IoT and M2M- M2M, Difference between IoT and M2M, SDN and NFV for IoT, Smart Objects: The “Things” in IoT: Sensors, Actuators, and Smart Objects, Sensor Networks- Wireless Sensor Networks (WSNs), Communication Protocols for Wireless Sensor Networks- Connecting Smart Objects- Communication Criteria.	9
3	Unified Data Standards –Protocols –IEEE 802.15.4 -The Physical Layer, The Media-Access	9

	Control Layer, Uses of 802.15.4, The Future of 802.15.4: 802.15.4e and 802.15.4g–Modbus– ZigBee-Zigbee Architecture- LoRaWAN -Standardization and Alliances, Physical Layer, MAC Layer, Topology, LTE-M, NB-IoT-Network layer –The next generation: IP-based protocols	
4	Raspberry-Pi-Raspberry-Pi Interfaces (serial, SPI, I2C). Raspberry Pi interfacing and programming examples using python (LED, switch, sensor, serial, SPI, I2C devices). IoT privacy, security and vulnerabilities solutions, vulnerabilities, security requirements, threat analysis, security tomography, layered attacker model, Identity management, access control, secure message communication. Smart City IoT Architecture-Street Layer, City Layer, Data Center Layer, Services Layer.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

<i>Attendance</i>	<i>Internal Ex</i>	<i>Evaluate</i>	<i>Analyse</i>	<i>Total</i>
5	15	10	10	40

Criteria for Evaluation(Evaluate and Analyse): 20 marks

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	2 questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. - Each question carries 9 marks. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the IoT fundamentals and architecture modelling	K2
CO2	Understand the smart things in IoT and functional blocks	K2
CO3	To understand the communication networks and protocols used in IoT.	K2
CO4	To understand the cloud resources, data analysis and applications.	K2
CO5	To apply the IoT processes in embedded applications.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2		1			2				2
CO2	3	3	3		3			2				2
CO3	3	3	3		3			2	3			2
CO4	3								3			2
CO5	3								3			2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Internet of Things (A Hands - on-Approach)	Vijay Madiseti and ArshdeepBahga	VPT	Ist Edition,2014
2	IoT Fundamentals: Networking Technologies, Protocols and Use Cases for Internet of Things	David Hanes, Gonzalo Salgueiro, Patrick Grossetete, Rob Barton and Jerome Henry,	Cisco Press	Ist Edition,2017
3	Internet of Things: Architecture and Design Principles	Rajkamal	McGraw Hill (India)	2nd edition, 2022
4	Raspberry Pi Cookbook, Software and Hardware Problems and solutions	Simon Monk, O'Reilly (SPD)	ISBN	2016

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	“The Internet of Things –Key applications and Protocols”,	Olivier Hersent, David Boswarthick, Omar Elloumi	Wiley,	2012
2	Internet of things: A survey on enabling technologies, protocols, and applications.	Al-Fuqaha <i>et.al</i>	IEEE Communications Surveys & Tutorials	(2015), pp. 2347-2376.
3	The Internet of Things: Converging Technologies for Smart Environments and Integrated Ecosystems	Ovidu Vermesan and Peter Friess	River Publishers	1st edition, 2013
4	Internet of Things - From Research and Innovation to Market Deployment-	Ovidu Vermesan	River Publishers	1st edition, 2014

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc22_cs53/preview
2	https://nptel.ac.in/courses/106105081
3	https://nptel.ac.in/courses/108108098
4	https://onlinecourses.nptel.ac.in/noc21_cs14/preview

SEMESTER S5

CAD LAB

Course Code	PCMRL507	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Lab

Course Objectives:

1. To introduce students to the basics and standard of engineering design related to machine components.
2. To make students familiarize with different solid modelling software's
3. To convey the principle and requirements of modelling of machine elements
4. To introduce the preparation of part modelling and assembly modelling of machineries
5. To introduce standard CAD package to perform animations of machine parts

Expt. No.	Experiments
1	Introduction to computer aided design
2	Creation of 2D drawings (At least 8)
3	Creation of 3D objects from 2D drawings (At least 4)
4	Creation of part drawings (At least 4)
5	Making of assembly drawings (at least 4) (Universal joints, Flanged coupling, Cotter joint, Old ham's joint, Plumber block, Screw jack, knuckle joint, Clutch etc.)
6	Creation of 3D animation models (at least 1 drawing) (Industrial robot, Robotic arm, Robotic gripper, Piston cylinder & Connecting rod, etc.

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Gain working knowledge in Computer Aided Design and modelling procedures	K2
CO2	Gain knowledge in creating solid machinery parts	K3
CO3	Gain knowledge in assembling machine elements	K3
CO4	Gain knowledge in creating solid machinery parts with animation	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	-	-	-	3	-	-	-	3	-	-	3
CO2	3	-	3	-	3	-	-	-	3	3	-	3
CO3	3	-	3	-	3	-	-	-	3	3	-	3
CO4	3	-	3	-	3	-	-	-	3	3	-	3

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	CAD/ CAM Theory and Practice	Ibrahim Zeid	McGraw Hill	2007
2	CAD/ CAM – Computer Aided Design and Manufacturing	Mikell P. Groover and Emory W. Zimmer	Pearson Education	1987

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Autodesk Fusion 360	Sandeep Dogra, Cadartifex	A power guide for beginners and intermediate users by John Willis	2018
2	Beginner's guide to solid works 2022 – Level I: Parts, Assemblies, Drawings, Photo view 360 and simulation Xpress		1st edition by Alejandro Reyes	2022
3	Machine Drawing	P. I. Varghese, K. C. John	VIP publications	2001
4	A Textbook of machine drawing	K. C. John	PHI publications	2009

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.

- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER S5
MICROCONTROLLER LAB

Course Code	PCMRL508	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCMRT403	Course Type	Lab

Course Objectives:

1. Familiarize the students with Assembly Language Programming of modern microcontrollers.
2. Impart the skills for interfacing the microcontroller with the help of Embedded C/Assembly Language Programming.

Expt. No.	Experiments
1	Data transfer/exchange between specified memory locations.
2	Largest/smallest from a series.
3	Sorting (Ascending/Descending) of data.
4	Addition / subtraction / multiplication / division of 8/16 bit data.
5	Sum of a series of 8 bit data.
6	Multiplication by shift and add method.
7	Code conversion – Hex to Decimal/ASCII to Decimal and vice versa.
8	Time delay generation and relay interface.
9	Display (LED/Seven segments/LCD) and keyboard interface.
10	DAC interface with wave form generation.
11	Stepper motor and DC motor interface.
12	Realization of Boolean expression through port.

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Demonstrate the ability to transfer, manipulate, and exchange data between specified memory locations, and perform arithmetic operations on 8-bit and 16-bit data using a microcontroller.	K3
CO2	Develop and implement algorithms for sorting data in ascending and descending order, finding the largest and smallest elements in a series, and calculating the sum of a series of 8-bit data efficiently on a microcontroller.	K6
CO3	Design and implement interfaces for various peripherals, such as LED, seven-segment displays, LCDs, keyboards, DACs for waveform generation, and motors (stepper and DC motors), and generate appropriate time delays and relay control signals.	K6
CO4	Execute code conversion between different number systems (Hex to Decimal, ASCII to Decimal, and vice versa) and understand the significance of various data representations in microcontroller programming.	KK3
CO5	Realize and implement Boolean expressions through microcontroller ports, demonstrating the ability to design and control logical operations in embedded systems.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2			2							
CO2	3	3	2		2							
CO3	3	2	3	2	3				2			
CO4	3				2							2
CO5	3	2	2		2				2			

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	The 8051 microcontroller and Embedded System	Muhammed Ali Mazidi & Janice Gilli Mazidi, R.D. Kinley	Pearson Education	2nd edition, 2007
2	Computer Architecture and Organization: From 8085 to Core2Duo and beyond	Subrata Ghoshal	Pearson	1st edition, 2011
3	ARM System - on-chip Architecture	Steve Furber	Pearson Education	2014

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Computer System Architecture	Mano M M	Prentice Hall of India	3rd Edition, 2017
2	Computer organization and design: The Hardware/Software interface	David A. Patterson, John L. Hennessy	Elsevier	6th edition. 2021
3	Computer Organisation	V. Carl Hamacher, Zvonko G. Vranesic, Safwat G.Zaky	McGraw-Hill Education	5th edition, 2017
4	8085 Microprocessor Architecture, Applications and Programming	Ramesh S Goankar	Penram International	5/e

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
Module - I	https://archive.nptel.ac.in/courses/106/108/106108100/
Module - II	https://archive.nptel.ac.in/courses/106/108/106108100/
Module - III	https://archive.nptel.ac.in/courses/106/108/106108100/
Module - IV	https://archive.nptel.ac.in/courses/106/108/106108100/

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.

- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 6

MECHATRONICS ENGINEERING

SEMESTER S6

ROBOTICS & MACHINE VISION

Course Code	PCMRT601	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. To impart knowledge about the robotics as well as the concept of machine vision.
2. This includes basics of robots, sensors, kinematics as well as details of machine vision, an advanced sensor for robotics and automation

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Robotics –Introduction –Basic Structure-Classification of Robot and Robotic System-Law of Robotics-Robot Motion-Wrist Configuration-Motion – Roll –Pitch-Yaw-Drives-Hydraulic Motors-DC Motor--Stepper Motor-Power Transmission Systems	11
2	Sensors in Robotics: Position Sensor-Potentiometer-Encoders-LVDT-Velocity Sensor-Acceleration Sensor-Force-Pressure and Torque Sensor-Touch and Tactile Sensor-Proximity –Range and Sniff Sensor-Robot End Effectors-Types of End Effectors- Mechanical Gripper –Types of Gripper Mechanism	11
3	Position Orientation-Frames-Mapping-Changing Description from Frames to Frames. Transformation arithmetic's -Translation-rotation-transformation- transforms equations- transformation of the vectors-	11

	Introduction to manipulations Forward Kinematics and inverse Kinematics- D-H representation-Method of Robotic Programming (Qualitative Treatment Only).	
4	Machine Vision: Machine Vision as a sensor - low and higher level vision- 1D and 3D vision-image acquisition and digitization- cameras- CCD- CID- CPD illumination and types- image processing and analysis- feature extraction- Applications of vision in industrial automation – Control systems using vision for decision making and feedback-Visual Servoing	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the basics of robotics & its various classifications and configurations	K2
CO2	Understand the robot actuators and controls	K2
CO3	Get knowledge on robot sensors & actuators for robotic application	K2
CO4	Understand the kinematics of robots and adaptive control	K3
CO5	Identify machine vision as an advanced sensor for automation.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1										
CO2	3	2		2								
CO3	3	2	2	2								2
CO4	3	3										
CO5	3	1		1								2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Industrial Robotics- Technology, Programming and Applications	M.P Groover	McGraw-Hill USA	1986
2	Introduction to Robotics	John Craig	Macmillan	1985
3	Machine Vision – Automated Visual Inspection and Robot Vision	Vernon D	Prentice Hall, International Ltd	1991
4	Machine Vision	Ramesh Jain, Rangachar Kasturi, Brain G. Schunk	McGraw Hill International	-

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Robotics Technology and Flexible Automation	Deb S. R, Deb S	Tata McGraw Hill Education Pvt. Ltd	2010

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc24_me117/preview
2	https://onlinecourses.nptel.ac.in/noc24_me117/preview
3	https://onlinecourses.nptel.ac.in/noc24_me88/preview
4	https://onlinecourses.nptel.ac.in/noc24_me88/preview

SEMESTER S6

AUTOTRONICS

Course Code	PCMRT602	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. To understand the fundamentals and control of automobiles using electronic equipment.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Automotive fundamentals: The engine, components, drive train, starting & charging systems operation, ignition system, suspension systems, brakes, steering system. Sensors and actuators for power train and chassis systems: Introduction, working principle, throttle position, Manifold absolute pressure (MAP), lambda sensor, detonation, emission.	11
2	SI Engine Management: Feedback carburettor system, throttle body injection and multi point fuel injection system, injection system controls, advantage of electronic ignition systems, three way catalytic converter, group and sequential injection techniques. Working of the fuel system components. Advantages of electronic ignition systems CI Engine Management: Parameters affecting combustion, noise and emissions in CI engines. Pilot, main, advanced, post injection and retarded post injection layout of the Common Rail Fuel Injection system, working of components like fuel injector, fuel pump, rail pressure limiter, flow limiter, EGR valve control in electronically controlled systems.	11
3	Safety and comfort : Antilock Braking System (ABS), Electronic Stability	11

	Control (ESC), Roll Stability Control (RSC), traction control system, adaptive cruise control, electric power steering, electronic clutch, electronic suspension system, airbags. Electric vehicles and hybrid vehicles: Introduction, electric vehicle development, system layout, basic system components, fuel cell, electric vehicle, hybrid vehicle: series hybrid vehicle, parallel hybrid vehicle, CNG-electric hybrid vehicle.	
4	Vehicle Intelligence: Introduction, basic structure, vision based autonomous road vehicles, architecture for dynamic vision system, features, applications, application of mobile robot vision to a vehicle information system, object detection, collision warning and avoidance system, low tire pressure warning system.	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Acquire knowledge on various electronic controls, sensors and actuators used in automobiles	K2
CO2	Obtain an overview of components, sub-systems, and basics of engine management employed in today's automotive industry.	K2
CO3	Apply fundamental knowledge for design of safety systems like airbags, ABS, etc. in automobiles.	K3
CO4	Comprehend the basic system components and their functions of electric and hybrid vehicles. Acquire a fundamental knowledge on intelligence vehicle systems.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2										
CO2	3	2										
CO3	3	2	2						1			
CO4	3	2	2						1			2
CO5												

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Automobile electrical and electronic systems	Tom Denton	BH Publications	3rd Ed., 2004
2	Automobile Electrical & Electronic Equipments	Young, Griffiths	Butterworths	
3	Automotive Electronics Design Fundamental	Najamuz Zaman	Springer,	1st Ed., 2015
4	Fundamentals of Motor Vehicle Technology on Chassis and Body Electronics	Hillier's	Nelson	5th Ed.

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Automotive Technology	Jack Erjavec, Robert Scharff	Delmar publications Inc	1992
2	Electric and Hybrid-electric vehicles	Ronald K. Jorgen	SAE	2002
3	Vision-based Vehicle Guidance	Ichiro Masaki	Springer Verlag, Newyork	1992
4	Class room manual for automotive service and system	Jay Webster	Delmer	1995

SEMESTER S6

NETWORK & DATA SECURITY

Course Code	PEMRT631	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. To understand the fundamental principles and architectures of computer networks, including physical structures, network models, and the interconnection of networks, along with an in-depth exploration of the OSI model and TCP/IP protocol suite.
2. To gain comprehensive knowledge of network security mechanisms, including symmetric and public-key cryptography, authentication protocols, and firewall technologies, while mastering advanced topics in reliable data transfer, congestion control, and routing algorithms in the Internet.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Networks: Network criteria, physical structures, network models, categories of networks, Interconnection of Networks: Internetwork. Network models: Layered tasks, OSI model, Layers in OSI model, TCP/IP protocol suite. Application Layer: Principles of Network Applications: Network Application Architectures, Processes Communicating, Transport Services Available to Applications, Transport Services Provided by the Internet, Application-Layer Protocols. Guided and unguided transmission media - Co-axial cable, UTP, STP, Fiber optic cable	9
2	Transport Layer : Introduction and Transport-Layer Services: Relationship Between Transport and Network Layers, Overview of the	

	Transport Layer in the Internet. Multiplexing and Demultiplexing: Connectionless Transport: UDP,UDP Segment Structure, UDP Checksum Principles of Reliable Data Transfer: Building a Reliable Data Transfer Protocol, Pipelined Reliable Data Transfer Protocols, Go-Back-N, Selective repeat, Connection-Oriented Transport TCP: The TCP Connection, TCP Segment Structure, Reliable Data Transfer, Flow Control, TCP Connection Management, Principles of Congestion Control: The Causes and the Costs of Congestion, Approaches to Congestion Control, Network-assisted congestion-control example, ATM ABR Congestion control, TCP Congestion Control: Fairness.	9
3	The Network layer: Router: Input Processing, Switching, Output Processing, Routing: Routing and Forwarding, Static routing and Dynamic routing, Routing control plane, IPv6, A Brief foray into IP Security, Routing Algorithms: The Link-State (LS) Routing Algorithm, The Distance-Vector (DV) Routing Algorithm, Hierarchical Routing, Routing in the Internet, Intra-AS Routing in the Internet: RIP, Intra-AS Routing in the Internet: OSPF, Inter/AS Routing: BGP, Broadcast Routing Algorithms and Multicast.	9
4	Network Security: Overview of Network Security: Elements of Network Security, Classification of Network Attacks, Security Methods, Symmetric-Key Cryptography: Data Encryption Standard (DES), Advanced Encryption Standard (AES), Public-Key Cryptography: RSA Algorithm, Diffie-Hellman Key-Exchange Protocol, Authentication: Hash Function, Secure Hash Algorithm (SHA), Digital Signatures, Firewalls and Packet Filtering, Proxy Server.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Demonstration of application layer protocols	K2
CO2	Discuss transport layer services and understand UDP and TCP protocols	K2
CO3	Explain routers, IP and Routing Algorithms in network layer	K2
CO4	Explain network security, and attacks and its security methods	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3									2
CO2	3	3	3									2
CO3	3	3	3									2
CO4	3	3	3									2
CO5												

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Computer Networking, A Top-Down Approach	James F Kurose and Keith W Ross	Pearson	Eighth Edition, 2021
2	Computer and Communication Networks	Nader F Mir	Pearson	2nd Edition, 2014
3	CRYPTOGRAPHY AND NETWORK SECURITY	Behrouz A Forouzan	McGraw Hill, Indian Edition	2007
4	Data Networks	D. Bertsekas, RG Gallager,	Prentice Hal	2nd edition, 1991

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Data and Communications and Networking	Behrouz A Forouzan	McGraw Hill, Indian Edition	Fifth Edition
2	Computer Networks	Larry L Peterson and Bruce S Davie	ELSEVIER 3	Fifth Edition
3	Computer Networks	Andrew S Tanenbaum	Pearson 4	Fifth Edition
4	Computer Networks	Mayank Dave	Cengage Learning	Second edition

SEMESTER S6

RENEWABLE ENERGY

Course Code	PEMRT632	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. Inculcate in students an awareness of new and renewable energy sources.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction, Classification of Energy Resources- Conventional Energy Resources - Availability and their limitations- Non-Conventional Energy Resources – Classification, Advantages, Limitations; Comparison. SOLAR THERMAL SYSTEMS- Principle of Conversion of Solar Radiation into Heat – Solar thermal collectors. – Flat plate collectors. SOLAR ELECTRIC SYSTEMS- Solar Thermal Electric Power Generation – Solar Photovoltaic – Solar Cell fundamentals - characteristics, classification, construction. Solar PV Systems – stand-alone and grid connected- Applications.	9
2	ENERGY FROM OCEAN- Ocean Thermal Energy Conversion (OTEC)- Principle of OTEC system- Open Cycle (Claude cycle), Closed Cycle TIDAL ENERGY – Principle of Tidal Power- Components of Tidal Power Plant (TPP)- Classification-single basin- double basin types –Limitations - Environmental impacts. WIND ENERGY- Introduction- Basic principles of Wind Energy Conversion Systems (WECS). Types of rotors. wind power equation -Betz limit. Advantages and Disadvantages of WECS -site selection criteria.	9

3	BIOMASS ENERGY- Introduction- Biomass Fuels-Biomass conversion technologies -Urban waste to Energy Conversion- Biomass Gasification- Biomass to Ethanol Production- Biogas production from waste biomass-factors affecting biogas generation-types of biogas plants – KVIC and Janata model-Biomass program in India.	9
4	SMALL HYDRO POWER- Classification as micro, mini and small hydro projects - Basic concepts and types of turbines- selection considerations. EMERGING TECHNOLOGIES: Fuel Cell-principle of operation – classification- conversion efficiency and losses - applications. Hydrogen energy -hydrogen production -electrolysis thermo chemical methods - hydrogen storage and utilization.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Choose the appropriate energy source depending on the available resources and the concepts of solar thermal and solar electric systems.	K2
CO2	Illustrate the operating principles of wind, and ocean energy conversion systems.	K2
CO3	Outline the features of biomass and small hydro energy resources	K2
CO4	Describe the concepts of fuel cell and hydrogen energy technologies	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3					1	2					3
CO2	3					1	1					3
CO3	3					1	1					3
CO4	3					1	1					3
CO5												

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Non-Conventional Energy Sources	G. D. Rai	Khanna Publishers	2010
2	Energy Technology	Rao S. and B. B. Parulekar	Khanna Publishers	1999
3	Non-Conventional Energy Resources	Sawhney G. S	PHI Learning	2012

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Renewable and Novel Energy Sources	Sab S. L	MI. Publications	1995
2	Renewable Energy Sources and Their Environmental Impact	Abbasi S. A. and N. Abbasi	Prentice Hall of India	2001
3	Renewable Energy - Power for Sustainable Future	Boyle G	Oxford University Press	1996

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
For all module	https://nptel.ac.in/courses/103103206

SEMESTER S6

ADVANCED AUTOMATION

Course Code	PEMRT633	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. Make students familiar with automation and control technologies in modern manufacturing.
2. Determine the modern application of automation systems in manufacturing industry.
3. Provide knowledge on the elements of modern manufacturing systems.
4. Examine the mechanisms of CMM and FMS.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Production system facilities-Manufacturing support systems - Automation in production systems-manual labor in production systems - automation principles and strategies USA principle-ten strategies of Automation and Production Systems - Automation Migration strategy-manufacturing industries and products-manufacturing operations-processing and assembly operations-product / production relationships. Classification of manufacturing systems	9
2	Elements of an automated system-advanced automation functions - safety monitoring-maintenance and repair diagnostics - capabilities of computer control and levels of industrial process control-computer process monitoring-direct digital control-numerical control and robotics-PLC. Components of a manufacturing system - production machines - material handling system.	9

3	Automation levels - part or product variety - Type I type II and type III manufacturing systems. Part families - parts classification and coding-features and examples of part classification and coding systems - production flow analysis-cellular manufacturing - application of group technology. Inspection metrology-coordinate measuring machines - CMM construction - CMM operation and planning - CMM applications and benefits.	9
4	Machine vision - image acquisition and digitizing - machine vision applications. Flexible manufacturing systems - types of FMS - FMS components - workstations - material handling and storage systems - computer control systems - human resources - FMS applications and benefits - FMS planning and implementation issues - FMS planning and design issues - FMS operational issues. Comparison of lean and agile manufacturing.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the principles of automation systems and to determine the relationship between product and production.	K2
CO2	Analyse the different elements of automation system and to find the importance of control systems in automation.	K3
CO3	Classify the manufacturing systems, what are the components of manufacturing systems and understand about CMM.	K2
CO4	Explain about Machine vision in manufacturing system and FMS.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1		1								
CO2	3	2	1	2								2
CO3	3	3										
CO4	3	2	3	2								2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Automation, Production Systems and Computer – Integrated Manufacturing	Mikell P Groover,	Pearson Education	2001

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	CAD/CAM and CIM	Radhakrishnan, P Subramanian	Wiley Eastern	2003
2	Advanced Automation Techniques in Adaptive material processing	XiaoQi Chen, Rajagopalan Devanathan, Aik Meng Fong	World Scientific, first edition	2002
3	Principles of automation and Advanced Manufacturing System	Dr. KC Jain, Sanjay Jain	Khanna Publishers	2004

SEMESTER S6

WIRELESS & SENSORS NETWORKS

Course Code	PEMRT634	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:3:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. To make students understand the basics of Wireless sensor Networks.
2. To familiarize with learning of the Architecture of WSN.
3. To understand the concepts of Networking and Networking in WSN.
4. To study the design consideration of topology control and solution to the various problems.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	OVERVIEW OF WIRELESS SENSOR NETWORKS Single-Node Architecture Hardware Components Network Characteristics unique constraints and challenges, Enabling Technologies for Wireless Sensor Networks Types of wireless sensor networks.	9
2	ARCHITECTURES Network Architecture Sensor Networks Scenarios Design Principle, Physical Layer and Transceiver Design Considerations, Optimization Goals and Figures of Merit, Gateway Concepts, Operating Systems and Execution Environments introduction to Tiny OS and nesC Internet to WSN Communication.	9
3	NETWORKING SENSORS MAC Protocols for Wireless Sensor Networks, Low Duty Cycle Protocols And Wakeup Concepts SMAC, BMAC Protocol, IEEE 802.15.4 standard and ZigBee, the Mediation Device Protocol, Wakeup Radio Concepts,	10

	Address and Name Management, Assignment of MAC Addresses, Routing Protocols Energy Efficient Routing, Geographic Routing.	
4	INFRASTRUCTURE ESTABLISHMENT Topology Control, Clustering, Time Synchronization, Localization and Positioning, Sensor Tasking and Control.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand challenges and technologies for wireless networks	K4
CO2	Understand architecture and sensors	K4
CO3	Describe the communication, energy efficiency, computing, storage and transmission	K6
CO4	Establishing infrastructure and simulations	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	2	1	2					3	3
CO2	3	3	3	2	1	2					3	3
CO3	3	3	3	2	1	2					3	3
CO4	3	3	3	2	1	2					3	3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Protocols And Architectures for Wireless Sensor Networks	Holger Karl & Andreas Willig	John Wiley	1 st and 2005
2	Wireless Sensor Networks An Information Processing Approach	Feng Zhao & Leonidas J.Guibas,	Elsevier	1 st and 2007
3	Fundamentals of Wireless Sensor Networks Theory and Practice	Waltenegus Dargie , Christian Poellabaue	John Wiley & Sons Publications	1 st and 2011

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Wireless Sensor Networks Technology, Protocols, and Applications	Kazem Sohraby, Daniel Minoli, & Taieb Znati	John Wiley	1 st and 2007
2	Wireless Sensor Network Designs	Anna Hac	John Wiley	1 st and 2003

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/GUSrkWJ_Z2g?feature=shared
2	https://nptel.ac.in/courses/106/105/106105160/
3	https://onlinecourses.swayam2.ac.in/arp19_ap52/preview
4	https://cse.iitkgp.ac.in/~smisra/course/wasn.html

SEMESTER S6

EMBEDDED SYSTEMS

Course Code	PE44T 636	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PC44T403	Course Type	Theory

Course Objectives:

1. The purpose of this course is to mould students into high calibre embedded real time application designers by enhancing their knowledge and skills in various system design aspects of embedded real time system.
2. This course helps to make students familiar with the architecture, hardware and software elements, programming models, tools for embedded system design and implementation of embedded system.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Embedded Systems: Application domain of embedded systems, features and characteristics, System model, Microprocessor Vs Microcontroller, current trends and challenges, hard and soft real time systems, Embedded product development, Life Cycle Management (water fall model), Tool Chain System, Assemblers, Compilers, linkers, Loaders, Debuggers Profilers & Test Coverage Tools-cross compilation.	9
2	Embedded System Design: Introduction to Embedded Systems and Computer Systems Terminology. Modular approach to Embedded System Design using Six-Box model: Input devices, output devices, embedded computer, communication block, host and storage elements and power supply. Microcontroller Based Embedded System Design. Salient Features of Modern Microcontrollers. Elements of Microcontroller Ecosystem and their	9

	significance.	
3	ATMEGA 8 Microcontroller: Architecture of ATMEGA 8, Pin Description, Memory organization: Program memory, Data Memory, I/O Ports, Timers, Counters, Analog comparator, Serial Peripheral Interface, USART, External Hardware Interrupts, ADC.	9
4	ATMEGA 8 Embedded C Programming: I/O ports, Register configuration, Programming – Timers: Modes – Programming – Counters – ADC: Configuration registers – Programming – External Hardware Interrupts: Types – Programming.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand Embedded Systems enabling them to analyze application domains, navigate the development lifecycle using the waterfall model, and proficiently utilize tool chain components for efficient embedded system development.	K2
CO2	Understand the basic concept of microcontroller based embedded system design	K2
CO3	Interpret architecture and interfacing concepts of ATMEGA 8 microcontroller.	K2
CO4	Develop embedded programming using ATMEGA 8 microcontrollers.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	2								3
CO2	3	2	2	2								3
CO3	3	2	2	2	2							2
CO4	3	3	2	3	3							2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Embedded System Design – A Unified Hardware/ Software Introduction	Frank Vahid And Tony Givargis	John Wiley & Sons	2002
2	Embedded Systems – An Integrated Approach	Lyla B Das	Pearson Publication	6 th edition, 2014
3	The Avr Microcontroller and Embedded Systems Using Assembly and C - using Arduino Uno and Atmel Studio	Sarmad Naimi, Muhammad Ali Mazidi, Sepehr Naimi	MicroDigitalEd	2017

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Designing Embedded Hardware.	John Catsoulis	Shroff Publishers and Distributors	2 nd edition
2	Embedded System- Architecture, programming, Design.	Rajkamal	Tata McGraw Hill	2011
3	Embedded Systems	Rao B. Kanta	PHI Learning Pvt. Ltd.	2011
4	An embedded software primer	David E.Simon	Pearson Education Asia	2001
5	Data sheet – ATMEGA 8		ATMEGA	

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/106/105/106105193
2	https://onlinecourses.nptel.ac.in/noc20_ee98/preview https://archive.nptel.ac.in/courses/108/102/108102169
3	https://nptel.ac.in/courses/108105102
4	https://nptel.ac.in/courses/108105102

SEMESTER S6

MECHANICS OF MACHINERY

Course Code	PEMRT 637	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	GCEST104	Course Type	Theory

Course Objectives:

1. Have fundamental knowledge of Kinematics and Dynamics of Machines.
2. To study basics of power transmission.
3. To Study the essentiality of balancing.
4. To acquaint with working principles of CAM Mechanism.
5. To gain knowledge on gyroscope and types of vibration on machines.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to kinematics and mechanisms - various mechanisms, kinematic diagrams, degree of freedom- Grashof's criterion, inversions, coupler curves mechanical advantage, transmission angle. Straight line mechanisms exact, approximate. Displacement, velocity analysis– relative motion - relative velocity. Instantaneous centre -Kennedy's theorem.	9
2	Belt Drivers: Flat Belt Drives, Ratio of Belt Tensions, Centrifugal Tension, power Transmitted. Gears and Gear Trains: Gear terminology, law of gearing, Path of contact Arc of contact, Contact ratio of spur gears. Simple gear trains, Compound gear trains for large speed. Reduction, Epicyclic gear trains. Related Problems. Study of various power transmission methods used in robotic and automation applications	9

3	Static and Dynamic Balancing -Balancing of single rotating mass by a single mass rotating in the same plane -Balancing of several masses rotating in the same plane-Problems on above (Analytical and Graphical methods). Cams: classification of cam and followers - displacement diagrams Cams: velocity and acceleration analysis of SHM, uniform velocity, uniform acceleration, cycloidal motion, Graphical cam profile synthesis, pressure angle.	9
4	Gyroscope: Vectorial Representation of Angular Motion, Gyroscopic Couple. Effect of Gyroscopic Couple on Ship, Plane Disc, Aircraft. Vibration: Terms used in Vibrations-Types of Vibrations-Free Vibrations- Forced Vibrations-Damped Vibrations-Types of Free Vibrations-Longitudinal, Transverse and Torsional- Critical or Whirling speed of a shaft.	9

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the fundamentals of kinematics, various planar mechanisms and interpret the basic principles of mechanisms and machines	K2
CO2	Select appropriate power transmission mechanisms	K2
CO3	Appreciate the essentiality of balancing in Rotating Parts	K2
CO4	To acquaint with working principles of CAM Mechanism.	K2
CO5	To study the different types of vibration and to understand gyroscopic effects.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2											2
CO2	3	3	3	2	2							2
CO3	3	3	2	2	2							2
CO4	3	2	1	1	1							2
CO5	3	2	2	1	2							2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Theory of Machines and Mechanisms	Ballaney P. L	Khanna Publishers	25 th Edition 2018
2	Theory of Machines	S. S. Rattan	Tata Mc Graw Hill	4 th edition 2017
3	Theory of machines	Khurmi R S, Gupta J.K	S.Chand and company	2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Kinematics and Dynamics of Machinery	.C E. Wilson, P. Sadler,	Pearson Education	2005
2	Machines and Mechanisms Applied Kinematic Analysis	D.H. Myskza	Pearson Education	2013
3	Mechanism Design: Analysis and synthesis Vol I & II	G. Erdman, G. N. Sandor	Prentice Hall of India,.	1984
4	Kinematics and Dynamics of Machinery,	Norton	Tata McGraw Hill	2009

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/112/105/112105268/
2	https://archive.nptel.ac.in/courses/112/105/112105268/
3	https://www.youtube.com/watch?v=oQrcPiQuCHI
4	https://www.youtube.com/watch?v=OyqHoJ2Wd8Q&list=PLAC668A0566953FB5&index=2

SEMESTER S4

SOFT COMPUTING

Course Code	PEMRT635	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To introduce students to the fundamental concepts and techniques of Soft Computing, including Neural Networks, Fuzzy Logic, and Genetic Algorithms.
2. To enable students to apply Soft Computing methods to real-world problems through project-based learning.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Overview of Soft Computing Definition and importance of Soft Computing - Comparison with Hard Computing - Applications in various domains. Introduction to Neural Networks Biological inspiration and McCulloch-Pitts model - Activation functions: Types and uses - Neural Network Architectures: Single layer feedforward networks - Multilayer feedforward networks - Recurrent networks. Learning Process in Neural Networks Supervised vs. Unsupervised learning - Learning rules: Perceptron, Widrow-Hoff, Delta rule, Hebbian and Competitive learning - Applications: Pattern association, recognition, and function approximation.	9
2	Single Layer Networks Perceptron and ADALINE - Adaptive filtering and optimisation techniques - Perceptron convergence theorem.	9

	Multilayer Networks Architecture and learning algorithms - Backpropagation: Detailed working, issues, and solutions - XOR problem and decision rules - Introduction to MADALINE. Radial Basis Neural Networks Concept and applications - Network training and regularisation theory - Comparison with Multilayer Perceptron.	
3	Basics of Fuzzy Logic Introduction to Fuzzy Sets - Properties, operations, and relations - Membership functions and fuzzy rules. Fuzzy Rule-Based Systems Canonical rule forms - Composite rules and their decomposition - Aggregation of fuzzy rules and decision-making. Introduction to Genetic Algorithms Evolutionary concepts and GA features - Coding, selection, crossover, and mutation - Basic GA structure and operators.	9
4	Hybrid Systems Concept of Neuro-Fuzzy Systems - Integration of neural networks and fuzzy logic - Introduction to Neuro-Genetic Systems. Advanced Topics and Cross-Validation Cross-validation methods and model selection - Early stopping methods and their importance - Introduction to Boltzmann Machines and Mean Field Theory (brief overview)	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the working principles and architectures of various neural networks and learning rules to train different neural network models.	K2
CO2	Apply and implement single-layer and multilayer perceptron networks to solve complex problems, and Radial Basis Function (RBF) networks for classification tasks.	K3
CO3	Design and implement fuzzy logic systems for real-world problems.	K3
CO4	Design and implement hybrid systems combining neural networks, fuzzy logic, and genetic algorithms, and apply advanced techniques for model validation and optimisation.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2											2
CO2	3	2										3
CO3	3	2										3
CO4	3	2										3
CO5	3	3	2	1	3				3	3	2	3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Neural Networks, Fuzzy Logic and Genetic Algorithm (Synthesis and Application)	S.Rajasekaran & G.A.Vijayalakshmi Pai	Prentice Hall of India	2 & 2017
2	Neural Nrtworks (A Comprehensive Foundation)	Simon Haykins	Pearson Education	1 & 1997
3	Fuzzy Logic With Engineering Applications	Ross.T.J	McGraw Hill	3 & 2011
4	Neural Networks for Pattern Recognition	Christopher M. Bishop	Oxford	1 & 1995

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Neural Networks and Fuzzy Systems	Bart Kosko	Prentice Hall of India	1 & 1994
2	Introduction to Artificial Neural Systems	Jacek M.Zurada	Jaico Publishing House	1 & 1994
3	An Introduction to Fuzzy Control.	Driankov.D, Flellendoom.H & Reinfrank.M	Springer-Verlag Berlin and Heidelberg GmbH & Co. K	2 & 1996
4	Genetic Algorithms	David E.Goldberg	Pearson Education	1 & 1989
5	Neural Networks. A comprehensive foundation	Simon Haykins	Macmillan College, Proc, Con, Inc, New York	1 & 1997

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	Swayam - Introduction To Soft Computing - Prof. Debasis Samanta, IIT Kharagpur

SEMESTER S6

PLC & DATA ACQUISITION SYSTEM

Course Code	PBMRT604	CIE Marks	60
Teaching Hours/Week (L: T:P: R)	3:0:0:1	ESE Marks	40
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)		Course Type	Theory

Course Objectives:

1. To study about PLC and the different programming languages
2. To understand the various PLC instructions
3. To understand the need of computer control in automation
4. To study the data acquisition systems.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	PLC PROGRAMMING METHODS Architecture, Study of different types of pneumatic and hydraulic valves, PLC Programming languages - Relay Logic-Ladder Logic-Programming of Gates, PLC Timers-ON delay, OFF delay, Retentive Timers, Programming Counters-up/down counter-Combining counter, PLC Arithmetic functions	11
2	PLC DATA HANDLING FUNCTIONS PLC Program control instructions: Master Control Reset - Skip -Jump & Move Instruction, Sequencer instructions, Types of PLC analog modules and systems, PLC analog signal processing - Case study of Tank level control system, bottle filling system and sequential switching of motors	11
3	COMPUTER CONTROL Need of computer in a control system-Functional block diagram of a computer control system-Data Loggers - Supervisory computer control-	11

	Direct Digital Control - SCADA -Programming of Robot (Pick and Place)	
4	DATA ACQUISITION SYSTEMS Sampling Theorem - Sampling and digitizing - Aliasing - Sample and Hold Circuit -signal conditioning of Strain gauge, LVDT, Thermocouple, Pressure Transducer. Interfacing ADC and DAC with microprocessor / Multiplexer- Microprocessor/PC based acquisition systems	11

Suggestion on Project Topics

Course Assessment Method

(CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Ex-1	Internal Ex-2	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 2 marks (8x2 =16 marks)	2 questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 2 sub divisions. - Each question carries 6 marks. (4x6 = 24 marks)	40

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Study about programming of PLC and Understand the various PLC instruction	K2
CO2	Design specification using PLC	K3
CO3	Understand the need of computer control in Automation	K2
CO4	Study data acquisition systems	K1
CO5	Apply the concepts learnt in the course to develop a project and prepare a brief report.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2			2				3		3
CO2	3	3	3	3	3	2			3	3		3
CO3	3	2	2		3	2			3	3		3
CO4	3	2	2			2				3		3
CO5	3	3	3		3				3	3	2	3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Programmable Logic Controllers	Petrezeulla	McGraw Hill	1989
2	Process Control Instrumentation Technology	Curtis D Johnson	Prentice Hall	8th Edition June 2005
3	Linear Integrated Circuits	D Roy Choudhury & Shail B Jain	New age International Pvt.Ltd	6th Edition, 2021

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Programmable Logic Controllers	Hughes.T	ISA Press	1989
2	Data Converters	G B Clayton	The Mac Millian Press Ltd.	1982
3	Programmable Logic Controllers- Principles and Applications	John W Webb & Ronald A Reis	PHI Learning Pvt.Ltd.,New Delhi	5 th edition 2010

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/MS3qJq2jvu0
2	https://youtu.be/UQ16Cous_tY
3	https://youtu.be/a0lCy5Q7n88
4	https://youtu.be/nX_Xp2hVc0s

PBL Course Elements

L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	Practical	Presentation
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation/ Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		30

1. Project Planning and Proposal (5 Marks)

- Clarity and feasibility of the project plan
- Research and background understanding
- Defined objectives and methodology

2. Contribution in Progress Presentation and Question Answer Sessions (4 Marks)

- Individual contribution to the presentation
- Effectiveness in answering questions and handling feedback

3. Involvement in the Project Work and Team Work (3 Marks)

- Active participation and individual contribution
- Teamwork and collaboration

4. Execution and Implementation (10 Marks)

- Adherence to the project timeline and milestones
- Application of theoretical knowledge and problem-solving
- Final Result

5. Final Presentation (5 Marks)

- Quality and clarity of the overall presentation
- Individual contribution to the presentation
- Effectiveness in answering questions

6. Project Quality, Innovation, and Creativity (3 Marks)

- Overall quality and technical excellence of the project
- Innovation and originality in the project
- Creativity in solutions and approaches

SEMESTER S6

DESIGN THINKING AND PRODUCT DEVELOPMENT

(Common to Group C & Group D)

Course Code	GZEST605	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	60
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To guide students through the iterative stages of design thinking, including empathizing with users, defining problems, ideating solutions and developing Proof of Concepts (PoC) and technical feasibility studies.
2. To promote the development of critical thinking skills by engaging students in integrative inquiry, where they ask meaningful questions that connect classroom knowledge with real-world applications.
3. To equip students with the ability to involve in product design considering the sustainability, inclusivity, diversity and equity aspects.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamentals of design thinking and product development: Overview of stages of product development lifecycle; Design thinking -Definition-Design thinking for product innovation; Bringing social impact in ideation-Identifying societal needs-understanding multi-faceted issues-community engagement and empathetic design-technological innovation meeting societal needs; Understanding and Bridging the divide using Human Centered Design (HCD); Designing for inclusivity in product development-embracing user diversity - Long term impact - sustainability encompassing environmental,economic and social dimensions; Technology Readiness Level in the Innovation Life-cycle; Performing a self-check on innovative ideas - Originality of idea- understanding innovation landscape - patentability - understanding the economic landscape - Unique Selling Proposition (USP) - Repeatability and Manufacturability - Sustainability - Leveraging business models	6

	for comprehensive analysis	
2	Empathize: Design thinking phases; Role of empathy in design thinking; Methods of empathize phase - Ask 5 Why/ 5 W+H questions; Empathy maps - Things to be done prior to empathy mapping - Activities during and after the session; Understanding empathy tools - Customer Journey Map - Personas. Define: Methods of Define Phase: Storytelling, Critical items diagrams, Define success.	6
3	Ideation : Stages of ideation; Techniques and tools - Divergent thinking tools - Convergent thinking tools - Idea capturing tools; Cross-industry inspiration; Role of research in ideation - Market research - consumer research - leveraging research for informed ideation; Technological trends - navigating the technological landscape - Integrating emerging technologies; Feasibility studies - technical, economic, market, operational, legal, and ethical feasibility; Ideation session- techniques and tips. Proof of Concept (PoC) : Setting objectives; Risk assessment; Technology scouting; Document and process management; Change management; Knowledge Capture; Validating PoC; Story telling in PoC presentation	6
4	Design: Navigating from PoC to detailed design; Developing Specification Requirement Document (SRD); Design for manufacturability; Design to cost; Pre-compliance; Design Failure Mode and Effects Analysis (DFMEA); Forecasting future design changes. Prototyping: Alpha prototypes; Beta prototypes; Transition from design to prototype; Goals and expectations for Alpha and Beta prototypes; Effective strategies for maintaining timeline in prototyping; Testing and refining Alpha prototypes; Transitioning to Beta prototypes. Pilot build: Definition and purpose of a pilot build; setting objectives; Identification and selection of manufacturing partner for pilot build; Testing procedures in pilot build; Scaling from pilot build to full-scale production / implementation.	6

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Empathize to capture the user needs and define the objectives with due consideration of various aspects including inclusivity, diversity and equity	K5
CO2	Ideate using divergent and convergent thinking to arrive at innovative ideas keeping in mind the sustainability, inclusivity, diversity and equity aspects.	K6
CO3	Engage in Human Centric Design of innovative products meeting the specifications	K5
CO4	Develop Proof of Concepts (PoC), prototypes & pilot build of products and test their performance with respect to the Specification Requirement Document.	K4
CO5	Reflect on professional and personal growth through the learnings in the course, identifying areas for further development	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2		2	3	3	3	2	2		3
CO2	3	2	3		2	3	3	3	2	2		3
CO3	3	2	3		2	3	3	2	2	2		3
CO4	3	2	2		3	3	3	2	2	2		3
CO5	3					3	3	2	2	2		3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Product Sense: Engineering your ideas into reality	Dr. K R Suresh Nair	NotionPress.com	2024
2	Change by Design: How Design Thinking Transforms Organizations and Inspires Innovation	Tim Brown	HarperCollins Publishers Ltd.	2009
3	Design Thinking for Strategic Innovation	Idris Mootee	John Wiley & Sons Inc.	2013

Sample Assignments:

1. Evaluate and prepare a report on how the aspects including inclusivity, diversity and equity are taken into consideration during the empathize and define phases of the Miniproject course.
2. Evaluate and prepare a report on how the aspects including sustainability, inclusivity, diversity and equity are taken into consideration during the ideate phase of the Miniproject course.
3. Evaluate and prepare a report on how User-Centric Design (UCD) is used in the design and development of PoC of the product being developed in the Miniproject course.
4. Prepare a plan for the prototype building of the product being developed in the Miniproject course.
5. Report on the activities during the empathize phase including the maps & other materials created during the sessions.
6. Report on the activities during the define phase including the maps & other materials created during the sessions.

SEMESTER S6

TRANSDUCERS & MEASUREMENTS

Course Code	OEMRT611	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. To provide students with a comprehensive understanding of various types of transducers used in engineering applications, including their principles of operation and characteristics
2. To familiarize students with different measurement techniques and instruments commonly used in engineering, enabling them to apply these techniques effectively for accurate measurement and analysis in practical scenarios

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to transducers and sensors Definition of sensors and transducers –Mechanical devices as Primary Detectors-Pressure Sensitive Primary Devices- Temperature Detectors- Hydro pneumatic Devices. Electric Transducers. Advantages of Electrical Transducers. Classification of Electrical Transducers- Primary and Secondary Transducers, Active and Passive Transducers, Analog and Digital Transducers. Electrical Phenomena Used in Transducers.	9
2	Resistive Transducers- Potentiometers, Strain gauges-gauge factor- Resistance Thermometers, Thermistors. Capacitance transducers -piezo electric transducers – Inductive transducers: LVDT characteristics- photoelectric sensors – Hall Effect transducers -measuring circuits- calibration- Optical transducers. Ionization transducers. Digital Transducer. Shaft Encoder.	9

3	Measurements and Measurement Systems Measurements- Significance of Measurements- Methods of Measurement: Direct & Indirect Methods-Instruments-Mechanical, Electrical and Electronic instruments- Classification of Instruments-Analog and Digital Modes of Operation-Functions of Instruments and Measurement Systems- Applications of Measurement Systems- Elements of a Generalized Measurement System- Characteristics of Instruments and Measurement Systems: Measurement system Performance- Static Calibration- Static Characteristics- Errors in Measurements-True Value-Static Error- Static Correction -Scale Range and Scale Span- Error Calibration Curve- Reproducibility and Drift- Accuracy and Precision, Linearity. Hysteresis-type of errors- classification of errors	9
4	DC bridges: introduction, sources and detectors-General equation-Types of bridges: Wheatstone, Kelvin bridge- AC bridges: introduction, sources and detectors- General equation. Maxwell's inductance and Maxwell's inductance -capacitance bridge. Cathode ray oscilloscopes: principles, construction and limitations- Digital storage oscilloscopes: principles- Measurements using CRO s and DSOs- Recording instruments: Strip chart recorder, X-Y Plotter, LCD displays.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Summarize the concepts of sensors, transducers and classify various trasducers.	K2
CO2	Apply the principles and functions of various types of Transducers in measuring systems.	K3
CO3	Illustrate the working principles of electronic measuring instruments and Identify various types of errors in measuring systems and choose methods for minimization of the errors.	K3
CO4	Explain the concepts of CRO, DSO, various recording devices.	K2
CO5	Understand and utilize various measurement systems, enhancing their capability to apply theoretical knowledge in practical scenarios	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3				1							3
CO2	3				1							3
CO3	3	2										3
CO4	2											3
CO5	3											3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	A Course in Electronic Measurements and Instrumentation	A. K. Sawhney	Dhanpat Rai & Co.	2015 Edition
2	Electronic Instrumentation and Measurements	David A Bell	Oxford University Press	3rd Edition 2013
3	Electronic Instrument Design	Kim R Fowler	Oxford reprint	2 nd Edition, 2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Sensors and Transducers,	D .Patranabis	PHI	2nd edition, 2003
2	Electronic Instrumentation and Measurements	Kalsi HS	Mc Graw hill	4th edition, 2019.
3	Electrical Measurements and Measuring systems	Golding E W and Widdis F C	Wheeler & co	1993

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://www.youtube.com/watch?v=1uPTyJxZzyo
2	https://onlinecourses.nptel.ac.in/noc23_ee105/preview
3	https://nptel.ac.in/courses/112107242
4	https://onlinecourses.nptel.ac.in/noc23_ee112/preview https://www.youtube.com/watch?v=0FVzYLEdSA8

SEMESTER S6

PLC & DATA ACQUISITION SYSTEM

Course Code	OEMRT612	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)		Course Type	Theory

Course Objectives:

1. To study about PLC and the different programming languages
2. To understand the various PLC instructions
3. To understand the need of computer control in automation
4. To study the data acquisition systems.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	PLC PROGRAMMING METHODS Architecture, Study of different types of pneumatic and hydraulic valves, PLC Programming languages - Relay Logic-Ladder Logic-Programming of Gates, PLC Timers-ON delay, OFF delay, Retentive Timers, Programming Counters-up/down counter-Combining counter, PLC Arithmetic functions	9
2	PLC DATA HANDLING FUNCTIONS PLC Program control instructions: Master Control Reset - Skip -Jump & Move Instruction, Sequencer instructions, Types of PLC analog modules and systems, PLC analog signal processing - Case study of Tank level control system, bottle filling system and sequential switching of motors	9
3	COMPUTER CONTROL Need of computer in a control system-Functional block diagram of a computer control system-Data Loggers - Supervisory computer control-	9

	Direct Digital Control - SCADA -Programming of Robot (Pick and Place)	
4	DATA ACQUISITION SYSTEMS Sampling Theorem - Sampling and digitizing - Aliasing - Sample and Hold Circuit -signal conditioning of Strain gauge, LVDT, Thermocouple, Pressure Transducer. Interfacing ADC and DAC with microprocessor / Multiplexer-Microprocessor/PC based acquisition systems	9

Suggestion on Project Topics

Course Assessment Method (CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Ex-1	Internal Ex-2	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 2 marks (8x2 =16 marks)	2 questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 2 sub divisions. - Each question carries 6 marks. (4x6 = 24 marks)	40

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Study about programming of PLC and Understand the various PLC instruction	K2
CO2	Design specification using PLC	K3
CO3	Understand the need of computer control in Automation	K2
CO4	Study data acquisition systems	K1
CO5	Apply the concepts learnt in the course to develop a project and prepare a brief report.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2			2				3		3
CO2	3	3	3	3	3	2			3	3		3
CO3	3	2	2		3	2			3	3		3
CO4	3	2	2			2				3		3
CO5	3	3	3		3				3	3	2	3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Programmable Logic Controllers	Petrezeulla	McGraw Hill	1989
2	Process Control Instrumentation Technology	Curtis D Johnson	Prentice Hall	8th Edition June 2005
3	Linear Integrated Circuits	D Roy Choudhury & Shail B Jain	New age International Pvt.Ltd	6th Edition, 2021

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Programmable Logic Controllers	Hughes.T	ISA Press	1989
2	Data Converters	G B Clayton	The Mac Millian Press Ltd.	1982
3	Programmable Logic Controllers- Principles and Applications	John W Webb & Ronald A Reis	PHI Learning Pvt.Ltd.,New Delhi	5 th edition 2010

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/MS3qJq2jvu0
2	https://youtu.be/UQ16Cous_tY
3	https://youtu.be/a0lCy5Q7n88
4	https://youtu.be/nX_Xp2hVc0s

PBL Course Elements

L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	Practical	Presentation
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation/ Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10

5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		30

1. Project Planning and Proposal (5 Marks)

- Clarity and feasibility of the project plan
- Research and background understanding
- Defined objectives and methodology

2. Contribution in Progress Presentation and Question Answer Sessions (4 Marks)

- Individual contribution to the presentation
- Effectiveness in answering questions and handling feedback

3. Involvement in the Project Work and Team Work (3 Marks)

- Active participation and individual contribution
- Teamwork and collaboration

4. Execution and Implementation (10 Marks)

- Adherence to the project timeline and milestones
- Application of theoretical knowledge and problem-solving
- Final Result

5. Final Presentation (5 Marks)

- Quality and clarity of the overall presentation
- Individual contribution to the presentation
- Effectiveness in answering questions

6. Project Quality, Innovation, and Creativity (3 Marks)

- Overall quality and technical excellence of the project
- Innovation and originality in the project
- Creativity in solutions and approaches

SEMESTER: S6

MECHATRONICS SYSTEMS & PLC LAB

Course Code	PCMRL607	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:2:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None / Course Code	Course Type	Lab

Course Objectives:

1. Giving experience to students in the practical aspects of hydraulic and pneumatic kits and operation of robots as well as sensors. System modelling using software like MATLAB and LABVIEW provides simulation studies.
2. PLC Programming gives introduction to automation.

Expt. No.	Experiments
1	Design and assembly of pneumatic and hydraulic valve
2	Study of different type of pneumatic and hydraulic valve
3	Study of reciprocating movement of double acting cylinder using pneumatic direction control valve
4	Programming robot (pick and place)
5	Sensors for automotive
6	Automatic door opening and closing
7	Data acquisition for virtual instrumentation
8	Image acquisition techniques
9	Control of stepper motor
10	Control of servo motor

11	A/D and D/A conversion
12	Image processing examples using python
13	Image processing examples using matlab
14	Implementation of logic gates and flip - flops using plc
15	Implementation of timers and counters using plc
16	Tank level control using plc
17	Simulation sequential switching of motors using plc simulation

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Acquire the basic knowledge in using electro – pneumatic kits and assemble them	K2
CO2	Operate a robot for applications like pick and place	K2
CO3	Experimentally apply the virtual instrumentation technique using LABVIEW	K3
CO4	Develop automation modules using PLC	K3
CO5	Use platforms like MATLAB or python for data processing and acquire basic knowledge in various sensors used in automotive	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	2					2			2
CO2	2					2						
CO3	3	2	2		3				2			3
CO4	3	2	2	2					2			2
CO5	3	2	2	2	2				2			2

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Mechatronics System Design	Devadas Shetty, Richard A Kolkm	PWS Publishing company	2009

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Engineering Metrology	R K Jain	Khanna Publishers	1994
2	Hydraulics and pneumatics	Andrew Parr	Butterworth-Heinemann	Third edition, 2011

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER S7

MECHATRONICS ENGINEERING

SEMESTER S7

FLUID MECHANICS & MACHINERY

Course Code	PEMRT741	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. Gain fundamentals knowledge of fluid physical properties, pressure and its measurements, fluid statics, kinematics, dynamics and turbo machines.
2. Understand the concepts of Fluid statics, Fluid dynamics, Fluid kinematics, Dimensional analysis, turbo machines, Hydraulic turbines.
3. Apply the principles of turbo machines, fluid mechanics and machines

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamental concepts: Properties of fluid - density, specific weight, viscosity, surface tension, capillarity, vapour pressure, bulk modulus, compressibility, velocity, rate of shear strain, Newton's law of viscosity, Newtonian and non-Newtonian fluids, real and ideal fluids, incompressible and compressible fluids. Fluid statics: Atmospheric pressure, gauge pressure and absolute pressure. Pascal's Law, measurement of pressure - piezo meter, manometers, pressure gauges, energies in flowing fluid, head - pressure, dynamic, static and total head, forces on planar and curved surfaces immersed in fluids, centre of pressure, buoyancy, equilibrium of floating bodies, metacentre and metacentric height.	9
2	Fluid Kinematics: Types of fluid flow, continuity equation in 2D and 3D (Cartesian Co-ordinates only), velocity and acceleration, velocity potential function and stream function, problems. Fluid Dynamics: Introduction, Euler's equation of motion, Bernoulli's equation from first principles and also from Euler's equation, limitations of	9

	Bernoulli's equation, problems.	
3	Equations of fluid dynamics: (Euler's equation), Navier-Stokes equations Bernoulli's equation and its applications: Venturimeters and Orifice meters, Hydraulic coefficients, Flow through pipes: Laminar and Turbulent flow, Reynold's experiment, loss of head due to friction, Darcy – Weishbach Equation, Chezy's formula, Minor losses in pipes. Turbomachines: Definition of a Turbomachine, parts of a Turbomachine Hydraulic Turbines: Classification; Constructional features.	9
4	Pumps: Classifications-types of casing and impellers, Velocity triangle for pumps, Head of pump, Losses and efficiency, Minimum starting speed, Specific speed, NPSH, Multistage pump, Pumps in parallel and series, selection of pumps. Positive displacement pumps – working principle, types of reciprocating pumps, Indicator diagram- work done, effect of acceleration and frictional resistance, slip and coefficient of discharge., separation in suction and delivery pipes, Air vessel, Gear pump, Lobe pump, Vane pump, Screw pump.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the concept of fluid statics	K2
CO2	Calculate pressure variations in accelerating fluids using Euler's and Bernoulli's equations and become conversant with the concepts of flow measurements and flow through pipes	K2
CO3	Understand applications of Fluid flow measurements, dimensional analysis, turbo machines, hydraulic turbines	K2
CO4	Discuss the characteristics of different type of pumps.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	1	-	-	1	-	-	-	-	-	2
CO2	2	-	-	-	-	1	-	-	-	-	-	2
CO3	2	1	-	-	-	1	-	-	-	-	-	2
CO4	3	2	2	-	-	1	-	-	-	-	-	2
CO5												

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Hydraulics & Fluid Mechanics	Modi P. N. and S. M. Seth	Standard Book House	22 nd Edition 2019
2	Fluid Mechanics and Fluid Power Engineering	Kumar D. S	S. K. Kataria & Sons, New Delhi	2013

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fluid Mechanics and Fluid Machines	Dr. Bansal, R.K.	Lakshmi Publications	2019
2	A Textbook of Turbomachines	M S Govinde Gowda	M M Publishers	2018
3	Fluid Mechanics (SI Units)	Yunus A. Cengel John M.Oimbala	Tata McGraw Hill	2 nd Ed. 2006
4	Fluid Mechanics	John F.Douglas, Janul and M.Gasiosek and John A.Swaffield	Pearson Education Asia	5th ed., 2006

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/112/105/112105171/
2	https://www.youtube.com/watch?v=5H0euuo1PGQ https://youtu.be/vXPtNNLEOUc
3	https://www.youtube.com/watch?v=5aDa30ym8RI
4	https://archive.nptel.ac.in/courses/112/103/112103249/

SEMESTER S7

MACHINE LEARNING

Course Code	PEMRT742	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. Equip students with the skills to understand, apply, and evaluate various machine learning algorithms and techniques, and to address real-world problems using these tools while considering ethical implications

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Overview of machine learning Machine learning paradigms-supervised, semi-supervised, unsupervised, reinforcement learning. Basics of parameter estimation - maximum likelihood estimation (MLE) and maximum a posteriori estimation(MAP). Introduction to Bayesian formulation.	9
2	Supervised Learning Regression - Linear regression with one variable, Linear regression with multiple variables, solution using gradient descent algorithm and matrix method, basic idea of overfitting in regression. Linear Methods for Classification- Logistic regression, Naive Bayes, Decision tree algorithm ID3.	9
3	Unsupervised Learning Clustering - Similarity measures, Hierarchical Agglomerative Clustering, K-means partitional clustering, Expectation maximization (EM) for soft clustering. Dimensionality reduction – Principal Component Analysis.	9
4	Neural Networks (NN) and Support Vector Machines (SVM) Perceptron, Neural Network - Multilayer feed forward network, Activation functions (Sigmoid, ReLU, Tanh), Backpropagation algorithm.	9

	SVM - Introduction, Maximum Margin Classification, Mathematics behind Maximum Margin Classification, Maximum Margin linear separators, soft margin SVM classifier, non-linear SVM, Kernels for learning non-linear functions, polynomial kernel, Radial Basis Function(RBF)	
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Illustrate Machine Learning concepts and basic parameter estimation methods.	K2
CO2	Demonstrate supervised learning concepts (regression, linear classification).	K2
CO3	Illustrate the concepts of Multilayer neural network and Support Vector Machine	K2
CO4	Describe unsupervised learning concepts and dimensionality reduction techniques.	K2
CO5	Solve real life problems using appropriate machine learning models and evaluate the performance measures	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	2	2								3
CO2	2	2	2	2	2							3
CO3	2	2	2	2	2							3
CO4	2	2	2	2	2							3
CO5	2	2	2	2	2							3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Machine Learning	Ethem Alpaydin	MIT Press	2nd Edition, 2010
2	Data Mining and Analysis: Fundamental Concepts and Algorithms,	Mohammed J. Zaki and Wagner Meira	Cambridge University Press	First South Asia edition, 2016
3	Python Data Science Handbook	Jake Vander Plas	O'Reilly Media	1st Edition, 2016
4	Machine Learning	Tom Mitchel	McGraw-Hill	1st Edition, 1997

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introducing Data Science: Big Data, Machine Learning, and More, Using Python Tools	Davy Cielen, Arno D. B. Meysman, and Mohamed Ali	Dreamtech Press	1st Edition, 2016
2	Machine Learning: A Probabilistic Perspective	Kevin P. Murphy	MIT Press	1st Edition, 2012
3	Introducing Data Science: Big Data, Machine Learning, and More, Using Python Tools	Davy Cielen, Arno D.B. Meysman, and Mohamed Ali	Dreamtech Press	2016
4	Building Machine Learning Systems with Python	Willi Richert and Luis Pedro Coelho	Packt Publishing	2nd Edition, 2015
5	Elements of Machine Learning	Pat Langley	Morgan Kaufmann	1995

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc23_cs18/preview
2	https://onlinecourses.nptel.ac.in/noc23_cs87/preview
3	https://onlinecourses.nptel.ac.in/noc19_ee56/preview
4	https://onlinecourses.nptel.ac.in/noc20_cs62/preview

SEMESTER S7
ENTREPRENEURSHIP

Course Code	PEMRT743	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To equip students with comprehensive knowledge of entrepreneurship, including different types of entrepreneurs, forms of business, and the role of entrepreneurship in economic development, while fostering an understanding of ethical and social responsibilities.
2. To develop practical skills in entrepreneurial ideation, business planning, financial management, and the operation of MSMEs, with a focus on leveraging government schemes and understanding contemporary challenges and opportunities in both Indian and global contexts.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Entrepreneurship-Types of entrepreneurs, Forms of business, Entrepreneurial Traits and competencies, Entrepreneur vs. Manager, Entrepreneur vs. Intrapreneur. Role of Entrepreneurship in Economic Development, Ethics and Social responsibility of Entrepreneurs. Opportunities for Entrepreneurs in India and abroad. Start-up India, Stand up India, National Skill Development Program, PMEGP, Mudra Yojana, and KVIC schemes.	9
2	The Entrepreneurial ideation and decision process. Incubation procedures and processes, Business plan preparation, DPR preparation. Managing Finance and Growth- Sources of capital, Venture capitalists, Angel Investors, Crowd Funding, Institutional financial assistance- Role of Financial Institutions and Commercial Banks for loans and financial controls. Features and evaluation of joint ventures, acquisitions, mergers, franchising, public issues, rights issues, bonus issues and stock splits..	9

3	Micro, Small and Medium Enterprises - Importance, Evolution, Organisational Structure, Decision Making, Starting an MSME unit-phases, training requirements Extension Training Institute, Legal frameworks, Registration procedures, concessions and reliefs by Government. Production in MSME, Optimum Size of plant, Factors affecting production process, Production planning.	9
4	Monitoring and evaluation of enterprise, Reasons for failures, Sickness in Small Scale Industries. Rehabilitation of sick units, Effective management of the enterprise. E-commerce and Entrepreneurship. Rural entrepreneurship, Social entrepreneurship. Challenges in entrepreneurship. Successful Entrepreneurs from the contemporary Indian business world- Success stories of great Indian Tycoons who have contributed to build the nation.	9

**Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)**

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Imbibe the spirit, roles, functions and fundamentals of entrepreneurship in a developing economy.	K2
CO2	Develop Proficiency in business plan preparation and detailed project report (DPR) preparation and ensure all round development of them	K3
CO3	Familiarize with the ground realities of starting MSME units and opportunities available in the country	K2
CO4	Analyze the operation and management of MSME units and develop motivation and entrepreneurial competency to start and run an enterprise successfully.	K3
CO5	Evaluate the scope of e-commerce and the challenges in entrepreneurship	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		2	1	1								2
CO2		3		1			1					2
CO3		3	3	1								2
CO4			2	1	2	2						2
CO5		2		2								1

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Entrepreneurial Development.	Khanka SS	S. Chand & Company Pvt.Ltd	Revised edition, 2012
2	Entrepreneurship	Robert D.Hisrich, Mathew J Manimala, Michael P. Peters and Dean A. Shepherd:	McGraw Hill Education India (P) Ltd, Chennai,	9e (2014)

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Competing through Innovation	Bellon Whittington	Prentice Hall	2006
2	The Origin and Evolution of New Business	Bhide, Amar V	Oxford University Press	2000
3	Entrepreneurship development small business enterprises	Charanthimath	Pearson Education	2008
4	Entrepreneurship strategies and resources	Dollinger M J	Pearson Education, New Delhi	2006

SEMESTER S7

MICRO ELECTRO MECHANICAL SYSTEMS

Course Code	PEMRT744	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. To develop knowledge on Micro electro mechanical systems and materials used in the manufacture
2. To understand various micro fabrication procedures, techniques and packaging considerations

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Micro Electronic Mechanical Systems: Materials for MEMS- silicon- structure and orientations, properties. Other materials- shape memory alloy (SMA), polymers, silicon compounds etc, material properties. Commercial MEMS structures and systems: Overview of commonly used mechanical structures in MEMS - Beams, Cantilevers, Plates, Diaphragms – Typical applications. Passive MEMS structures- fluid nozzles- ink jet nozzles. Sensors- piezoresistive pressure sensors- mass flow sensors- acceleration sensors, actuators- micro mechanical valves-micro pumps.	9
2	Process of micromachining: Process of micromachining- pattern transfer using additive technique. Silicon growth- oxidation of silicon-molecular beam epitaxy. Physical vapour deposition (PVD)- sputtering- laser deposition- ion plating- cluster beam technology- Chemical vapour deposition (CVD)- step coverage- plasma enhanced CVD, Electron cyclotron resonance CVD (ECRCVD)- sol-gel deposition technique.	9

3	Lithography Photolithography- Introduction- masks- spinning- resist and soft backing- exposure and post exposure treatment- development- descumming and post baking- resists- wafer priming, wafer cleaning and contamination- clean room- resist striping. Next generation lithography-extreme ultra violet lithography (EUVL)- x-ray lithography- LIGA- charged particle beam lithography- electron beam lithography- ion beam lithography. Etching: Introduction- dry etching- physical etching- ion etching or sputtering- ion beam etching, plasma etching- deep reactive ion etching (DRIE)- isotropic wet etching- anisotropic wet etching.	9
4	MEMS packaging: Key design and packaging consideration- wafer stack thickness- wafer dicing concerns, thermal management- stress isolation- protective coating and media isolation. Wafer level packaging and die level packaging. Hermetic packaging- calibration and compensation- die attach process- solder alloy- bonding process using solder alloy- organic adhesive wiring and inter connections- Wire bonding- flip chip bonding- micro fluidics interconnections. Packaging material properties- mechanical- moisture penetration- interfacial adhesion, thermal- chemical- electrical.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Define MEMS and elaborate on various materials used for MEMS fabrication and to describe different kinds of commercial applications like micropumps, valves and sensors.	K2
CO2	Describe the various techniques involved in micromachining and discuss techniques for deposition of materials to create structures.	K2
CO3	Discuss lithography and etching techniques.	K2
CO4	Develop a deep understanding of design and packaging considerations during manufacture of MEMS devices	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											3
CO2	3											3
CO3	3											3
CO4	3											3
CO5												

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	An Introduction to Micro electromechanical Systems Engineering	Nadim Maluf	Artech house Boston London	6/e, 2006
2	Fundamentals of Micro-fabrication	Mark Madou	CRC Press, New York	3/e, 2011
3	Foundations of MEMS	Chang Liu	Pearson	2/e, 2011
4	MEMS and Microsystems Design and Manufacture	Tai Ran Hsu	McGraw Hill	4/e, 2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Electromechanics and MEMS	Thomas B. Jones, Nenad G. Nenadic	Cambridge University Press	Illustrated edition, 2013
2	Microsystem design	Stephen D. Senturia,	Springer (India)	2nd edition, 2006
3	Smart Sensors and MEMS	Stoyan Nihtianov, Antonio Luque	Woodhead Publishing Series	2nd edition, 2006

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/117105082
2	https://nptel.ac.in/courses/117105082
3	https://nptel.ac.in/courses/117105082
4	https://nptel.ac.in/courses/117105082

SEMESTER S7

ENERGY MANAGEMENT & AUDITING

Course Code	PEMRT746	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. This course introduces basic knowledge about energy management and audit
2. Energy management opportunities in electrical and mechanical systems are discussed.
3. Economic analysis of different energy conservation measures is also described.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Energy Management - General Principles and Planning: General principles of energy management and energy management planning Energy Audit: Definition, need, types and methodologies. Instruments for energy audit, Energy audit report - Power quality audit. Energy conservation in buildings: ECBC code (basic aspects), Building Management System (BMS).	9
2	Energy management in Electricity Utilization: Energy management opportunities in Lighting and Motors, Electrolytic Process and Electric heating. Types of industrial loads. Peak demand controls and methodologies	9
3	Energy management in HVAC systems: HVAC system: Coefficient of performance, Capacity, Factors affecting Refrigeration and Air conditioning system performance and savings opportunities. Classification and Advantages of Waste Heat Recovery system, analysis of waste heat recovery for Energy saving opportunities Cogeneration-Types and Schemes, Optimal operation of cogeneration plants- Case study. Computer	9

	aided energy management	
4	Energy Economics: Economic analysis methods-cash flow model, time value of money, evaluation of proposals, payback method, average rate of return method, internal rate of return method, present value method, life cycle costing approach, Case studies.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the significance and procedure for energy management and audit.	K2
CO2	Discuss the energy efficiency and management of electrical loads.	K2
CO3	Explain the energy management opportunities in HVAC systems	K2
CO4	Compute the economic feasibility of the energy conservation measures.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2					1	1					2
CO2	2					1	1					2
CO3	2					1	1					2
CO4	2					1	1					2
CO5												

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Energy management principles	Craig B. Smith	Pergamon Press	2007
2	Optimizing energy efficiencies in industry	G.G. Rajan	Tata McGraw Hill	2001
3	Handbook of Energy Audits	Albert Thumann, William J. Younger,	CRC Press	2003
4	Intro to Energy Conservation and manag.	M Jayaraju and Premlet,	Phasor Books	2008

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Industrial energy conservation	Charles M. Gottschalk	John Wiley & Sons	1996
2	Energy management Hand Book	Wayne C. Turner	The Fairmount Press	1997

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://www.youtube.com/watch?v=WwBquDjDGOA&list=PLImNQubhYtnAmyPNwO-nPU-VQIIXH0xqM
2	https://www.youtube.com/watch?v=WwBquDjDGOA&list=PLImNQubhYtnAmyPNwO-nPU-VQIIXH0xqM
3	https://www.youtube.com/watch?v=WwBquDjDGOA&list=PLImNQubhYtnAmyPNwO-nPU-VQIIXH0xqM
4	https://www.youtube.com/watch?v=WwBquDjDGOA&list=PLImNQubhYtnAmyPNwO-nPU-VQIIXH0xqM

SEMESTER 7

SMART MANUFACTURING

Course Code	PEMRT745	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. This course introduces Smart Manufacturing, encompassing the integration of advanced manufacturing techniques with digital technologies to create intelligent, interconnected systems.
2. Students will learn about the principles, technologies, and applications of SM, preparing them to implement and innovate within the modern manufacturing landscape.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Smart Manufacturing: Definition and evolution of Smart Manufacturing, Key components and principles, Comparison with traditional manufacturing. Industry 4.0 and Digital Transformation: Overview of Industry 4.0, Digitalization in manufacturing, Cyber-Physical Systems (CPS), Internet of Things (IoT) in Manufacturing: IoT fundamentals and architecture, IoT devices and sensors.	9
2	Data-Driven Manufacturing: Internet of Things (IoT) in Manufacturing (contd.) Applications of IoT in manufacturing, Data Analytics and Big Data in Manufacturing: Introduction to data analytics, Big Data in manufacturing, Predictive analytics and maintenance, Artificial Intelligence and Machine Learning: AI and ML basics, Applications of AI in manufacturing, Case studies of AI-driven manufacturing.	9

3	Advanced Manufacturing Technologies: Robotics and Automation: Robotics in manufacturing, Autonomous systems and collaborative robots, Automation technologies, Additive Manufacturing and 3D Printing: Principles of additive manufacturing, 3D printing technologies, Applications in manufacturing.	9
4	Smart Manufacturing Systems and Integration: Integration of digital and physical systems, Smart factories and digital twins. Case studies of integrated systems, Cybersecurity in Smart Manufacturing: Cybersecurity challenges in manufacturing, Protection strategies and best practices, Case studies of cybersecurity breaches, Sustainability and Future Trends in Smart Manufacturing: Sustainable manufacturing practices, Future trends.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

<i>Attendance</i>	<i>Internal Ex</i>	<i>Evaluate</i>	<i>Analyse</i>	<i>Total</i>
5	15	10	10	40

Criteria for Evaluation (Evaluate and Analyse): 20 marks

A 'Micro-Project' or Detailed presentation of 'Case Study' with a maximum of 3 students related to the application of smart manufacturing in modern industry should be done.

By using following rubrics table, evaluators can systematically assess students' performance on the "Evaluate and Analyse" component based on their understanding, application, critical thinking, and quality of analysis in the context of Smart Manufacturing.

Rubrics Table: (Evaluate and Analyse: 20 Marks)

Criteria	Excellent (5 Marks)	Good (4 Marks)	Satisfactory (3 Marks)	Needs Improvement (2 Marks)	Unsatisfactory (1 Mark)	Marks
1.Understanding of Concepts	Demonstrates a thorough and comprehensive understanding of the concepts related to Smart Manufacturing	Shows a good understanding of the core concepts, with minor gaps in knowledge.	Has a basic understanding of the concepts, but with noticeable gaps.	Demonstrates limited understanding of the concepts.	Shows little to no understanding of the concepts.	

Criteria	Excellent (5 Marks)	Good (4 Marks)	Satisfactory (3 Marks)	Needs Improvement (2 Marks)	Unsatisfactory (1 Mark)	Marks
	g, including IoT, AI, data analytics, and Industry 4.0.					
2.Application of Knowledge	Applies theoretical knowledge effectively to real-world scenarios and case studies, providing insightful analysis and solutions.	Applies knowledge to real-world scenarios with some insightful analysis.	Applies knowledge to real-world scenarios, but the analysis is basic and lacks depth.	Struggles to apply knowledge effectively to real-world scenarios.	Fails to apply knowledge to real-world scenarios.	
3.Critical Thinking and Problem-Solving	Demonstrates strong critical thinking and problem-solving skills, identifying and addressing complex issues in Smart Manufacturing.	Shows good critical thinking and problem-solving skills, but with minor gaps in addressing complex issues.	Demonstrates basic critical thinking and problem-solving skills, but with noticeable gaps.	Shows limited critical thinking and problem-solving skills.	Demonstrates little to no critical thinking or problem-solving skills.	

Criteria	Excellent (5 Marks)	Good (4 Marks)	Satisfactory (3 Marks)	Needs Improvement (2 Marks)	Unsatisfactory (1 Mark)	Marks
4.Quality of Analysis	Provides a comprehensive and in-depth analysis, backed by relevant data and references, demonstrating a strong grasp of the subject matter.	Offers a solid analysis with good use of data and references, though with minor gaps in depth or relevance.	Delivers a basic analysis with some use of data and references, but lacking depth and comprehensiveness.	Provides limited analysis with minimal use of data and references.	Offers little to no analysis, with few or no relevant data or references.	

Total: 20 Marks

Score Range	Performance Level
18-20 Marks	Excellent
14-17 Marks	Good
10-13 Marks	Satisfactory
6-9 Marks	Needs Improvement
0-5 Marks	Unsatisfactory

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	2 questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. - Each question carries 9 marks. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the fundamentals of Smart Manufacturing and Industry 4.0.	K2
CO2	Explore the key technologies driving Smart Manufacturing, such as IoT, AI, and data analytics.	K2
CO3	Learn about the integration of digital and physical systems in manufacturing.	K2
CO4	Analyze the impact of Smart Manufacturing on productivity, quality, and sustainability.	K4
CO5	Develop skills to design, implement, and manage Smart Manufacturing systems.	K5

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1	1	-	-	-	-	-	-	-	-
CO2	3	2	-	-	2	-	-	-	-	-	-	3
CO3	3	-	2	-	2	-	-	-	-	1	-	-
CO4	3	-	-	-		-	3	2	-	-	-	3
CO5	-	-	2	3	2	-	-	-	-	-	1	-

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Smart Manufacturing: Concepts and Methods	Shahrukh A. Irani	CRC Press	1st Edition, 2020
2	Industry 4.0: The Industrial Internet of Things	Alasdair Gilchrist	Apress	1st Edition, 2016
3	Big Data in Practice	Bernard Marr	Wiley	1st Edition, 2016
4	Artificial Intelligence: A Guide for Thinking Humans	Melanie Mitchell	Farrar, Straus and Giroux	1st Edition, 2019
5	Robotics and Automation Handbook	Thomas R. Kurfess	CRC Press	1st Edition, 2004

Additional References:

1. Research papers and articles from journals such as "Journal of Manufacturing Systems" and "International Journal of Advanced Manufacturing Technology."
2. Industry reports and whitepapers from organizations like McKinsey & Company, Deloitte, and PwC.

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Smart Manufacturing: An Overview	Rajat Kumar Roy	Tata McGraw-Hill Education	1st Edition, 2020
2	Industry 4.0: A Roadmap for Smart Manufacturing	Alok Kumar Jain	Wiley India	1st Edition, 2019
3	Big Data and Analytics in Manufacturing	Sudipta Roy, P.C. Mandal	Springer India	1st Edition, 2021
4	Artificial Intelligence in Industrial Applications"	Ajit Kumar	PHI Learning Pvt. Ltd.	1st Edition, 2019
5	Robotics and Automation: Principles and Applications	R.K. Mittal, I.J. Nagrath	Tata McGraw-Hill Education	2nd Edition, 2018
6	3D Printing and Additive Manufacturing: Principles and Applications	C.P. Kumar	IK International Publishing House	1st Edition, 2020
7	Cybersecurity in Smart Manufacturing"	Anil Kumar Singh	Wiley India	1st Edition, 2021
8	Sustainable Manufacturing and Operations	Ashok Kumar Sarkar	CRC Press India	1st Edition, 2019

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	INTRODUCTION TO INDUSTRY 4.0 AND INDUSTRIAL INTERNET OF THINGS Industry 4.0 https://archive.nptel.ac.in/courses/106/105/106105195/ IoT https://youtu.be/olZ3Iyib_wg CPS https://youtu.be/8wBQMbOntNc
2	AI https://youtu.be/zsE9EH0oJSk https://youtu.be/cTZVK7CK1gs
3	AM https://archive.nptel.ac.in/courses/112/103/112103306/
4	CS: https://youtu.be/KqSqyKwVuA8

SEMESTER S7

COMPUTER NETWORKS

Course Code	PEMRT751	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. Provide students with the knowledge to explain the roles and functions of various network protocols and to understand the OSI (Open Systems Interconnection) and TCP/IP models, along with their respective layers.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction – Uses of computer networks, Network hardware, Network software. Reference models – The OSI reference model, The TCP/IP reference model, Comparison of OSI and TCP/IP reference models. Physical Layer – Modes of communication, Physical topologies, Signal encoding, Repeaters and hub, Transmission media overview. Performance indicators – Bandwidth, Throughput, Latency, Queuing time, Bandwidth–Delay product.	9
2	Data link layer - Data link layer design issues, Error detection and correction, Sliding window protocols, High-Level Data Link Control(HDLC)protocol. Medium Access Control (MAC) sublayer –Channel allocation problem, Multiple access protocols, Ethernet, Wireless LANs - 802.11, Bridges & switches - Bridges from 802.x to 802.y, Repeaters, Hubs, Bridges, Switches, Routers and Gateways.	9
3	Network layer design issues. Routing algorithms - The Optimality Principle, Shortest path routing, Flooding, Distance Vector Routing, Link State	9

	Routing, Multicast routing, Routing for mobile hosts. Congestion control algorithms. Quality of Service (QoS) - requirements, Techniques for achieving good QoS.	
4	Transport service – Services provided to the upper layers, Transport service primitives. User Datagram Protocol (UDP). Transmission Control Protocol (TCP) – Overview of TCP, TCP segment header, Connection establishment & release, Connection management modeling, TCP retransmission policy, TCP congestion control. Application Layer –File Transfer Protocol (FTP), Domain Name System (DNS), Electronic mail, Multipurpose Internet Mail Extension (MIME), Simple Network Management Protocol (SNMP), World Wide Web(WWW) – Architectural overview.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the features of computer networks, protocols, and network designmodels	K2
CO2	Describe the fundamental characteristics of the physical layer and identify the usage in network communication	K2
CO3	Explain the design issues of data link layer, link layer protocols, bridges and switches	K2
CO4	Illustrate wired LAN protocols (IEEE 802.3) and wireless LAN protocols (IEEE 802.11)	K2
CO5	Select appropriate routing algorithms, congestion control techniques, and Quality of Service requirements for a network	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2										2
CO2	2	2	2									2
CO3	2	2	2									2
CO4	2	2	2									2
CO5	2	2	2	1								2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Computer Networks	Andrew S. Tanenbaum	PHI (Prentice Hall India)	4th Edition, 2003
2	Data Communication and Networking	Behrouz A Forouzan	Tata McGraw Hill	4th Edition, 2006

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Computer Networks – A Systems Approach	Larry L Peterson and Bruce S Dave	Morgan Kaufmann	5th Edition, 2011
2	Computer Networking and the Internet	Fred Halsall	Addison-Wesley	5th Edition, 2005
3	Computer Networking: A Top-Down Approach	James F. Kurose, Keith W. Ross	Pearson	6th Edition, 2012
4	TCP/IP Illustrated, Volume 1: The Protocols	W. Richard Stevens	Addison-Wesley	2005

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc22_cs19/preview
2	https://nptel.ac.in/courses/106105082
3	https://onlinecourses.nptel.ac.in/noc22_ee61/preview
4	https://onlinecourses.swayam2.ac.in/cec21_cs04/preview

SEMESTER S7

SIX SIGMA

Course Code	PEMRT752	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. To provide a comprehensive understanding of quality management principles, including the evolution of Total Quality Management (TQM) and the contributions of key TQM pioneers, along with the application of quality tools and methodologies for continuous improvement.
2. To develop proficiency in Six Sigma methodologies and their practical applications in both manufacturing and service industries, emphasizing process capability, DMAIC phases, and achieving customer satisfaction through quality enhancement.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to the concept of quality - definition, quality quantified, dimensions of quality, TQM – definition, new and old cultures, Gurus of TQM – Deming, Juran, Shewhart, Crosby, Fiegenbaum, Ishikawa, Taguchi – their main contributions, Obstacles to TQM, Benefits of TQM.	9
2	TQM axioms – commitment, scientific knowledge and involvement, Deming's fourteen points on quality management, Quality Council, Quality statements, customer satisfaction – customer perception of quality, customer retention, Employee involvement - motivation, empowerment, team and teamwork, suggestion system.	9
3	Continuous process improvement - Juran's quality trilogy, Seven basic tools of quality – flow chart or run chart, check sheet, histogram, Preto diagram, cause and effect diagram, scatter diagram and control charts, Benchmarking, New management tools – affinity diagram, inter relationship digraphs, tree diagram, matrix diagram, prioritization matrices,	9

	Process Decision Program Chart (PDPC), activity network diagram	
4	Six sigma –meaning- History of six sigma- working of six sigma-Process capability – meaning, significance and measurement – Six sigma roles and responsibilities- DMAIC process- Define phase-Measure phase- Analyse phase- Improve phase- Control phase- Case examples of DMAIC. Six sigma in manufacturing industries- Six sigma in service industries.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand basic concepts of quality highlighting the contributions by quality gurus	K2
CO2	Acquire knowledge on TQM axioms and focus on customer-centric approach with team work	K2
CO3	Acquire knowledge on TQM axioms and focus on customer-centric approach with team work	K2
CO4	Understand the various quality management systems	K2
CO5	Understand basic concepts and DMAIC phase of Six Sigma	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1									3	2
CO2	2	1									3	2
CO3	3	2						2	3		3	2
CO4	2	1					2	3	2		3	2
CO5		3			3		2		2			2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Total Quality Management	Besterfield, D.H., Besterfield, C., Besterfield, G.H., Besterfield M	Pearson Education	5th Edition
2	The Management and Control of Quality	James, R.E., and William, M.L.	Cengage Learning	8th ed, 2010
3	Total Quality Management - Text and Cases	Janakiraman, B. and Gopal, R.K.	Prentice Hall India	1st edition, 2006

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	The Six Sigma Handbook	Pyzdek, T., and Keller, P	McGraw-Hill.	5th edition, 2018
2	Six Sigma Revealed	International Six sigma institute	International Six sigma institute	Free pdf

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	NPTEL, Total Quality Management - I, IIT Kanpur https://nptel.ac.in/courses/110104080
2	UDEMY, OHSAS 18001 (2.5 hours) https://www.udemy.com/complete-health-and-safety-auditor-course-ohsas-18001/
3	UDEMY, Quality Management System Auditor Course (3.5 hours) https://www.udemy.com/iso-90012015-quality-management-system-auditor-course/
4	NPTEL, Six Sigma, IIT Kharagpur https://nptel.ac.in/courses/110105123

SEMESTER S7
NANO ELECTRONICS

Course Code	PEMRT753	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	GCPHT121	Course Type	Theory

Course Objectives:

1. Provide students with a comprehensive understanding of the principles, technologies, and applications of nano-scale electronic devices and systems.
2. Develop a solid foundation in the fundamentals of nanotechnology and nanoelectronics and understand the implications of nanotechnology on future electronic systems and devices

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to nanotechnology, Limitations of conventional microelectronics, characteristic lengths in mesoscopic systems, Quantum mechanical coherence. Low dimensional structures - Quantum wells, wires and dots, Density of states of 1D and 2D nanostructures. Basic properties of square quantum wells of finite depth, parabolic and triangular quantum wells	9
2	Introduction to methods of fabrication of nano-layers: physical vapour deposition- evaporation & Sputtering, Chemical vapour deposition, Molecular Beam Epitaxy, Ion Implantation, Formation of Silicon Dioxide- dry and wet oxidation methods. Fabrication of nano particle- grinding with iron balls, laser ablation, reduction methods, sol gel, self assembly, precipitation of quantum dots.	9
3	Quantum wells, multiple quantum wells, Modulation doped quantum wells, concept of super lattices Kronig - Penney model of super lattice. Transport of charge in Nanostructures - Electron scattering mechanisms, Hot electrons, Resonant tunnelling transport, Coulomb blockade, Effect of magnetic field on a crystal. Aharonov-Bohm effect, the Shubnikov-de Hass effect.	9

4	Nanoelectronic devices - MODFETS, Single Electron Transistor, CNT transistors – Properties of graphene. Resonant tunnel effect, RTD, RTT, Hot electron transistors Quantum well laser, quantum dot LED, quantum dot laser	9
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain quantum mechanical effects associated with low dimensional semiconductors	K2
CO2	Explain different processes involved in the fabrication of nanoparticles and nanolayers	K2
CO3	Explain different techniques for characterizing nano layers and particles	K2
CO4	Explain the different transport mechanisms in nano structures	K2
CO5	Illustrate the operating principle of nanoscale electronic devices like SET, Resonant tunnelling devices, Quantum lasers etc.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											3
CO2	3											3
CO3	3											3
CO4	3											3
CO5	3											3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Nanotechnology for Microelectronics and optoelectronics	J.M. Martinez-Duart, R.J. Martin Palma, F. Agulle Rueda	Elsevier Science	1st Edition, 2006
2	Nanotechnology and Nanoelctronics	W.R. Fahrner	Springer	1st Edition, 2005

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Nanoscience & Technology,	Chattopadhyay, Banerjee	PHI	2nd Edition, 2012
2	Introduction to Nanotechnology	Charles P. Poole Jr., Frank J. Owens	John Wiley & Sons	2nd Edition, 2022
3	Fundamentals of Nanoelectronics	George W. Hanson	Pearson Education	1st Edition, 2009
4	Nanoelectronics and nanosystems	K. Goser, P. Glosekotter, J. Dienstuhl	Springer	1st Edition, 2004

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc23_ee35/preview
2	https://onlinecourses.nptel.ac.in/noc22_ee121/preview
3	https://onlinecourses.nptel.ac.in/noc23_ph45/preview
4	https://onlinecourses.nptel.ac.in/noc21_ee59/preview

SEMESTER S7
DYNAMICS & MACHINERY

Course Code	PEMRT756	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. To provide students with a comprehensive understanding of the static and dynamic force analysis of mechanisms, including the application of principles such as D'Alembert's and the design of flywheels for energy efficiency in various engines.
2. To equip students with the skills to analyze and balance rotating and reciprocating systems, understand the effects of gyroscopic forces, and solve vibration problems in single-degree-of-freedom systems, ensuring stability in engineering applications.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Static and dynamic force analysis of mechanisms (four bar linkages only)-graphical method, virtual work method -D'Alembert's principle-equivalent dynamic systems reciprocating engine force analysis.	9
2	Energy Efficiency in Electricity Utilization: Flywheels-turning moment diagrams for steam engines-four stroke internal combustion engine and multi cylinder engines-coefficient of fluctuation of speed-coefficient of fluctuation of energy-design of flywheels.	9
3	Balancing: static balancing-dynamic balancing-balancing of several masses revolving in a single plane-several masses in different parallel planes-balancing of single cylinder reciprocating engines.Gyroscopic couple-effects on the stability of automobiles-two wheeler and four wheeler, stability of ships and air crafts	9
4	Vibration-free vibration of single degree of freedom systems-equation of	9

	motion Newton's method-energy method-natural frequency-undamped and damped systems logarithmic decrement-forced vibration-response of SDOF systems to harmonic excitation-whirling of shaft-vibration absorber-transmissibility	
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	To perform static and dynamic force analysis of four-bar linkages using graphical and virtual work methods, including D'Alembert's principle for reciprocating engines.	K2
CO2	To design flywheels by analyzing turning moment diagrams and calculating fluctuations in speed and energy for various engines.	K3
CO3	To achieve static and dynamic balancing of rotating and reciprocating systems, ensuring smooth operation and stability of machinery.	K2
CO4	To analyze the effects of gyroscopic couples on vehicle stability and solve vibration problems in single-degree-of-freedom systems using various methods.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2									2
CO2	3	3	2									2
CO3	3	3	2									2
CO4	3	3	2									2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Theory of machines and mechanisms	Ballaney, P. L.	Khanna Publishers	2010
2	Theory of Machines	Rattan S S,	Tata McGraw-Hill Education	2005

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Kinematics and Dynamics of Machinery	Charles E Wilson and J Peter Sadler	Tata McGraw-Hill Education	2008
2	Theory of Mechanisms and Machines,	Amithabha Ghosh and Asok Kumar Malik,	East West Press	2011
3	Theory of Machines	Thomas Bevan	Pearson	2013

SEMESTER S7

DIGITAL IMAGE PROCESSING

Course Code	PEMRT755	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0 :0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Develop a comprehensive understanding of digital image processing techniques, including image enhancement, restoration, and compression, to effectively analyze and manipulate visual data.
2. Gain practical skills in implementing and evaluating various image segmentation methods, ensuring proficiency in both spatial and frequency domain approaches for diverse applications.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Digital Image Fundamentals: Steps in Digital Image Processing – Components – Elements of Visual Perception – simple image formation model. Relationships between pixels – Colour image fundamentals – RGB, HSI models, Review of matrix theory: row and column ordering- Toeplitz, Circulant matrices.	9
2	Image Enhancement: Spatial domain methods: point processing- intensity transformations, histogram processing. Spatial filtering- smoothing filters, sharpening filters. Frequency domain methods: low pass filtering, high pass filtering, homomorphic filter	9
3	Image Restoration:	9

	Degradation model, Unconstraint restoration and constraint restoration. Inverse filtering, Weiner filtering. Image Compression: Need for compression, Basics of lossless compression and lossy compression. Concept of transform coding, JPEG Image compression standard.	
4	Image segmentation: Classification of Image segmentation techniques, Region approach - Region Growing, Region Split & Merge. Segmentation based on thresholding. Edge based segmentation-edge detection, Hough transform.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

<i>Attendance</i>	<i>Internal Ex</i>	<i>Evaluate</i>	<i>Analyse</i>	<i>Total</i>
5	15	10	10	40

Criteria for Evaluation(Evaluate and Analyse): 20 marks

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	2 questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. - Each question carries 9 marks. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	To study the image processing fundamentals and review of matrix theory.	K2
CO2	To study and interpret the various image enhancement techniques	K2
CO3	Analyze the various concepts of restoration techniques for image processing.	K3
CO4	Analyze basic image compression techniques.	K3
CO5	Illustrate image segmentation techniques	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	-	1	-	-	-	-	-	-	2
CO2	3	3	2	-	1	-	-	-	-	-	-	2
CO3	3	3	2	-	1	-	-	-	-	-	-	2
CO4	3	3	3	-	1	-	-	-	-	-	-	2
CO5	3	3	3	-	1	-	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Digital Image Processing,	Rafel C Gonzalez, Richard E. Woods	Pearson Education	fourth edition 2018
2	Digital Image Processing	S Jayaraman, S Esakkirajan, T Veerakumar,	Tata Mc Graw Hill	2nd Edition 2020

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fundamentals of digital image processing	Anil K Jain	Pearson Education India	First Edition 2015
2	Digital Image Processing	Kenneth R Castleman	Pearson Education India	1st edition 2007
3	Digital Image Processing	William K. Pratt	John Wiley & Sons;	4th Edition 2007

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://www.youtube.com/watch?v=DSGHkvQBMbs&list=PLuv3GM6-gsE08DuaC6pFUvFaDZ7EnWGX8&index=1
2	https://www.youtube.com/watch?v=bJjgyTQ-BT4&list=PLuv3GM6-gsE08DuaC6pFUvFaDZ7EnWGX8&index=32
3	https://www.youtube.com/watch?v=MrNafUqh860&list=PLuv3GM6-gsE08DuaC6pFUvFaDZ7EnWGX8&index=42
4	https://www.youtube.com/watch?v=j3_Ck5oP5oI&list=PLuv3GM6-gsE08DuaC6pFUvFaDZ7EnWGX8&index=56

SEMESTER S7

BASICS OF ROBOTICS & AUTOMATION

Course Code	OEMRT721	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. To provide students with a comprehensive understanding of robotics, including robot structures, classifications, sensors, end effectors, kinematics, and robotic programming, along with the basics of PLC systems and their applications.
2. To equip students with practical skills in robotic motion control, sensor integration, and PLC programming, enabling effective automation and process control in industrial and research settings.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Robotics –Introduction –Basic Structure-Classification of Robot and Robotic System-Law of Robotics-Robot Motion- Wrist Configuration-Motion – Roll –Pitch-Yaw-Drives-Hydraulic Motors-DC Motor-Stepper Motor-PowerTransmission Systems	9
2	Sensors in Robotics: Position Sensor-Potentiometer-Encoders-LVDT-Velocity Sensor-Acceleration Sensor-Force- Pressure and Torque Sensor-Touch and Tactile Sensor-Proximity –Range and Sniff Sensor-Robot End Effectors-Types of End Effectors- Mechanical Gripper – Types of Gripper Mechanism.	9
3	Position Orientation-Frames-Mapping-Changing Description from Frames to Frames. Transformation arithmetic's-Translation-rotation-transformation- transforms equations- transformation of the vectors-. Introduction to manipulations Forward Kinematics and inverse	9

	Kinematics- D-H representation-Method of Robotic Programming (Qualitative Treatment Only).	
4	Basics of PLC-Advantage- Capabilities of PLC-Architecture of PLC - Types of PLC-Types of I/O Modules- Configuring of PLC. Simple process control programme using ladder logic- PLC arithmetic functions- Timer and Counters Requirement of communication networks in PLC – Connecting PLC to computer.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain robot actuators and controls	K2
CO2	Summarize robot sensors for robotic application	K2
CO3	Explain the kinematics of robots and adaptive control	K2
CO4	Understand the basics of Programming Logic Circuits and programming Programmable Logic Circuits	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1		1								
CO2	3	2	1	2								2
CO3	3	3										
CO4	3	3	2	2								2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Industrial Robotics-Technology, Programming and Applications	M.P Groover	McGraw-Hill USA	1986
2	Introduction to Robotics,	John Craig	Macmillan	1985
3	Process Control Instrumentation Tech	Curtis Johnson	8 TH Edition Prentice Hall	June 2005
4	Programmable Controllers	Petrezeulla	McGraw Hills	1989

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Linear Integrated circuits	D Roy Choudhury and shaail B. jain	New age international Pvt.Ltd	2003
2	Mechatronics	Boltans w	Pearson Education	6th edition, published in 2023

SEMESTER S7

INDUSTRIAL HYDRAULICS & PNEUMATICS

Course Code	OEMRT722	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. To understand various components of pneumatic and fluid power systems and their advantages
2. Applying the fundamental concepts of fluid power and pneumatic to obtain solution to engineering problems.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basics of fluid power system, Components, Advantages and applications of Fluid power systems Fluid properties: specific weight, density and specific gravity, Pascal's Law and its application Continuity equation, Conservation Of Energy, Frictional losses in hydraulic pipelines - laminar and turbulent flow, Fluid power symbols.	9
2	Hydraulic pumps: Gear, Vane and Piston pumps, Pump performance, Characteristics and Selection - Sizing of pumps Hydraulic Motors- Limited Rotation Hydraulic Motors, Gear Motors, Vane Motors, Piston Motors, Hydraulic Motor Theoretical Torque, Power, And Flow Rate, Hydraulic Motor Performance	9
3	Hydraulic Valves- Direction control valves: Three way valve, Four way valves, Check valve and shuttle valve Actuation mechanisms in DCV Pressure control valves: Pressure relief, Pressure Reducing, Counter balance, Sequencing and Unloading Valves Flow control valves and its types Proportional Valves Servo valves: Mechanical type and Electrohydraulic servo valves.	9

4	Pneumatics: Air compressors, air treatment: Air filters, Dryer, Lubricator, pressure regulation, Compressor types, Pneumatics and Hydraulics, Properties Of Air, The Perfect Gas Laws, Fluid Conditioners, Air Control Valves, Pneumatic Actuators, Fluid Logic Control Systems - Introduction To Boolean Algebra, Laws Of Boolean Algebra, Illustrative Examples Using Boolean Algebra	9
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand various components of fluid power systems, their advantages and select them for various applications.	K2
CO2	Design and simulate fluid power systems for automation of machine tools and other equipment's.	K3
CO3	Choose appropriate control valves for fluid power applications	K3
CO4	Select pneumatic components and fluid power actuators for low cost automation	K2
CO5	Apply the fundamental concepts of fluid power and to obtain solution to engineering problems.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1									3
CO2	3	2	2									3
CO3	3	2	2			2						3
CO4	3	2	2					1				3
CO5	3	2	2			2						3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Hydraulics and Pneumatics	A. Parr	Elsevier	1999
2	Fluid Power with Applications	Anthony Esposito	Pearson New International Edition- Pearson	2013
3	Pneumatic System-Principles and Maintenance	S.R.Majumdar	McGraw Hill	2017.
4	Hydraulic and Pneumatics control	R.Srinivasan	McGraw Hill	2008

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Hydraulic control systems	H. E. Merritt	John Wiley& Sons, NewYork	1967
2	Basic fluid power	D. A. Peace	Prentice Hall	1987
3	Power Hydraulic	M. J. Pinches	Longman	1988

SEMESTER 8

MECHATRONICS ENGINEERING

SEMESTER S8

BIO MECHATRONICS

Course Code	PEMRT861	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To introduce students to basic biomedical engineering technology and different biological signals, their acquisition, measurements and related constraints.
2. Understand principles, techniques, procedures, and equipment used in bio-medical instrumentation.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Bio potential and bio electrodes: Cell structure – electrode – electrode potential- resting and action potential –Refractory period- Electrode electrolyte interface, Half-cell potential, Polarisable and Non-polarisable electrodes - Skin electrode interface – Bioelectrodes: Types of electrodes- Micro electrodes, Depth and Needle electrodes, Surface electrodes- Equivalent circuits of electrodes	9
2	Biopotential Recorders: Characteristics of the recording system, Electrocardiogram - Einthoven triangle, Standard 12-lead configurations - ECG Machine-Electroencephalography (EEG) –Brain waves, waveforms & measurements, Evoked potentials, Sleep stages, 10-20 electrodes placement system for EEG – EEG machine	9

3	Measurement of Physiological Parameters: Blood Pressure measurement- Blood Flow Meters: Need for blood flow meters, Electromagnetic blood flowmeters- Applications of electromagnetic blood flowmeters, Ultrasonic blood flowmeter: Ultrasonic blood flowmeter based on transit time principle, Doppler blood flowmeters-laser based Doppler blood flowmeters-Cardiac output measurements.	9
4	Physiological Assist Devices: Pacemakers, Defibrillators, Nerve and muscle stimulators, Heart lung machine- ventilators-Basic ideas of CT scanner- MRI scanner and ultrasonic scanner-Devices to protect against electrical hazards- Microshock and macroshock, Centralized patient monitoring system.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the bio potential and bio electrodes	K2
CO2	Understand the Characteristics of the recording system	K2
CO3	Understand the measurement of Physiological Parameters	K2
CO4	Understand the Physiological Assist Devices	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2									2
CO2	3	3	2									2
CO3	3	3	2									2
CO4	3	3	2									2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Textbook of Medical Physiology	Arthur C.Guyton	PrismBooks(Pvt)Ltd &W.B.Saunders Company	12th edition,2012
2	Bio Medical Instrumentation	Dr.Arumugam M	Anuradha agencies Publications	2nd edition,2002
3	Handbook of Medical Instrumentation	Khandpur R S	Tata McGraw Hill, New Delhi.	2004

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Electronics in Medicine and Biomedical Instrumentation	Nandini K	jog PHI	Second Edition 2013
2	Instrument transducer – An introduction to their performance and design	Hermann K P Neubert	Clarendon Press	1975
3	Biomedical sensors – Fundamentals and application	Harry N Norton	Noyes Publications	1982
4	Principles of applied biomedical instrumentation	L.A Geddes , L.E. Baker	<u>Wiley india Pvt. Ltd</u>	third addition, 2008

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc22_bt33/preview
2	https://onlinecourses.nptel.ac.in/noc22_ee87/preview
3	https://onlinecourses.swayam2.ac.in/nou23_bt05/preview
4	https://nptel.ac.in/courses/102106094

SEMESTER S8

HYBRID & ELECTRIC VEHICLES

Course Code	PE44T862	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. Introduces the fundamental concepts of electric, hybrid and autonomous vehicles, drive trains, electrical machines used, energy storage devices, charging systems and different communication protocols.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Hybrid Electric Vehicles: History of hybrid and electric vehicles, social and environmental importance of hybrid and electric vehicles, impact of modern drive-trains on energy supplies. Conventional Vehicles: Basics of vehicle performance, vehicle power source characterization, transmission characteristics. Hybrid Electric Drive-trains: Basic concept of hybrid traction, introduction to various hybrid drive-train topologies, power flow control in hybrid drive-train topologies. Electric Drive-trains: Basic concept of electric traction, introduction to various electric drive-train topologies, power flow control in electric drive-train topologies.	9
2	Electric Propulsion unit: Introduction to electric components used in hybrid and electric vehicles. DC Drives: Review of Separately excited DC Motor control – Speed and torque equations - Independent control of orthogonal flux and torque -	9

	Closed loop control of speed and torque (block diagram only). PMSM Drives: PMSM motor basics – Independent control of orthogonal flux and torque (concept only)- Field Oriented Control (FOC) – Sensored and sensorless control (block diagram only).	
3	Energy Storage: Introduction to energy storage requirements in Hybrid and Electric Vehicles- Battery based energy storage systems,Battery Management System, Types of battery- Fuel Cell based energy storage systems- Supercapacitors-Hybridization of different energy storage devices. Overview of Electric Vehicle Battery Chargers - On-board chargers, Electric Vehicle Supply Equipment (EVSE) - Grid to EVSE to On-board chargers to battery pack power flow block schematic diagrams – Types of charging stations - AC Level 1 & 2, DC - Level 3 – V2G concept-Types of Connectors - CHAdMO, CCS Type1 and 2, GB/T - PIN diagrams and differences.	9
4	Sizing the drive system: Matching the electric machine and the internal combustion engine (ICE), Sizing the propulsion motor, sizing the power electronics. Vehicle Communication protocols : Need & requirements - Functions of Control Pilot (CP) and Proximity Pilot (PP) pins, Communication Protocols - CAN, LIN, FLEXRAY (Basics only)- Power line communication (PLC) in EV.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the basic concepts of Conventional, Electric, Hybrid EV and describe different configurations of electric and hybrid electric drive trains	K2
CO2	Discuss the propulsion unit for electric and hybrid vehicles	K2
CO3	Compare various energy storage and EV charging systems	K2
CO4	Select drive systems and various communication protocols for EV	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1					2					2
CO2	3	2										2
CO3	3	2										2
CO4	3	3	2									2
CO5	3	1	2									2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Hybrid Electric Vehicles – Principles and applications with practical perspectives	Chris Mi, M A Masrur, D W Gao	Wiley	2011
2	Electric and Hybrid Vehicles: Design Fundamentals	Iqbal Hussein	CRC Press	2003

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Electric Vehicle Technology Explained	James Larminie, John Lowry	Wiley	2003
2	Modern Electric, Hybrid Electric and Fuel Cell Vehicles: Fundamentals, Theory and Design,	Mehrdad Ehsani, YimiGao, Sebastian E. Gay, Ali Emadi	CRC Press	2004

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/108/103/108103009/
2	https://nptel.ac.in/courses/108/102/108102121/
3	https://nptel.ac.in/courses/108/106/108106170/
4	https://nptel.ac.in/courses/108/103/108103009/

SEMESTER S8

OPERATIONS MANAGEMENT

Course Code	PEMRT863	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. This course is designed to facilitate the students to acquire knowledge about operations management concepts, tools and techniques.
2. This course covers system concept of production, facility location analysis, facility layout, line balancing, demand forecasting, aggregate planning, material requirement planning and production scheduling.
3. It empowers the students to amalgamate their knowledge and thus inculcate the skills needed to apply these concepts, tools and techniques in industry.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Operations Management: Objectives, Operations Management: Functions and Scope, Types of Production Systems, process planning, make or buy decisions, process reengineering, Operations Strategy. Facility Planning, Factors Affecting Plant Location, Plant Location: Case Study, location analysis techniques – location factor rating–center of gravity technique – load distance technique.	9
2	Capacities and facilities: capacity planning, facility layout – objectives, types of layout – process layout – product layout – fixed position layout – cellular layout, systematic layout, planning, layout design procedures – CRAFT – CORELAP – ALDEP.	9

	Assembly line balancing: methods for line balancing – rank positional weight method –COMSOAL.	
3	Demand forecasting: need and uses of forecasting, components of forecasting demand, time series methods – moving average – weighted moving average – exponential smoothing – adjusted exponential smoothing – linear regression – seasonal adjustments, forecast accuracy. Aggregate planning: aggregate planning strategies	9
4	Materials requirement planning: objectives, master production schedule, bill of materials, evolution from MRP to manufacturing resource planning (MRP II). production scheduling: objectives – processing characteristics and constraints – performance measures, Gantt chart, single machine scheduling	9

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand operations, production system and perform facility location analysis.	K2
CO2	Impart knowledge of facility layout, layout planning and perform line balancing.	K2
CO3	Compute demand forecast and forecast accuracy.	K2
CO4	Understand the Knowledge of aggregate planning and materials requirement planning.	K2
CO5	Impart Knowledge on Production Scheduling techniques.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	-	-	-	1	-	1	-	1	2	2
CO2	3	3	2	1	2	-	-	-	-	1	2	2
CO3	3	3	2	1	1	-	-	-	-	1	2	2
CO4	3	3	2	-	1	-	-	-	-	1	2	2
CO5	3	3	2	1	1	-	-	-	-	1	2	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Operations Management	Roberta S. Russell and Bernard W. Taylor III	John Wiley & Sons, Inc., Seventh Edition	2011
2	Production and Operations Management	R. Paneerselvam	PHI	2010

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Operations Management	Heizer and Render	Pearson Education	Eleventh Edition
2	Operations Management: A Quantitative Approach	P. B. Mahapatra	PHI	2010
3	Quantitative Models in Operations and Supply Chain Management	G. Sreenivasan,	, PHI	SECOND EDITION

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/112/107/112107238/ https://youtu.be/_VJkKZFuRvE
2	https://youtu.be/pcpp2rmAMx8 https://youtu.be/-aGk5-yx340
3	https://youtu.be/Ic_El2DkpjA
4	https://youtu.be/1kU8HG5Y9Kc

SEMESTER S8

ERGONOMICS

Course Code	PEMRT864	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. Equip students with the knowledge and skills to design ergonomic workspaces that reduce the risk of musculoskeletal disorders, improve posture, and ensure physical well-being by understanding human anatomy, anthropometry, and biomechanics.
2. Enable students to design optimal lighting, acoustic, and visual environments to minimize visual fatigue, eye strain, and auditory discomfort, while improving overall comfort and productivity in various work settings.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Humanise work, Basis Body Mechanics, Anatomy of Spine and Pelvis for postures, Stability and Postural Adaptation, Musculoskeletal Disorders in Work Spaces, Behavioural Aspects of Postures, Sources of Human Variability, Principles of Applied Anthropometry, Application of Anthropometry in Design, Design for Everyone, Personal Space.	9
2	Design of Standing and Sitting, Workstation Design, Work Surface Design, Visual Display Units, Guidelines for Design of Static and Repetitive Works, Ergonomic Interventions, Handle Design, Keyboard Design, Anatomy and Biomechanics of Manual Handling, Prevention of Injuries in Manual Handling, Design of Manual Handling Task, Carrying.	9
3	Muscle Structure - Function and Capacity, Factors Affecting Physical Work Capacity, Applied Physiology in Work Space, Measurement of	9

	Physiological Cost of Work, Comfort and the Indoor Climate, Protection Against Extreme Climates.	
4	Eye Vision, Measurement of Light, Lighting Design Consideration, Visual Fatigue, Eye Strain and Near Work, Indoor Lighting – Psychological Aspects, Measurement of Sound, Ear Protection, Design of Acoustic Environment, Industrial Noise Control, Design of Visual Displays, Design of Audio Displays, Design of Controls, Combining Displays and Controls, Virtual Environment.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand how to design workspaces that reduce musculoskeletal disorders and enhance productivity.	K2
CO2	Understand principles for creating effective and comfortable standing, sitting, and manual handling workstations.	K2
CO3	Understand how to optimize lighting, acoustics, and visual environments to minimize fatigue and discomfort.	K2
CO4	Understand how to measure physiological work costs and design workspaces that ensure comfort and accommodate diverse human needs.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	3	2									3
CO2	3	3	2									3
CO3	3	3	2									3
CO4	3	3	2									3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to human factors and ergonomics	R.S Bridger	CRC Press Tylor and Francis group	4th edition, 2018
2	ERGONOMICS in the Automotive Design Process	Vivek D. Bhise	CRC press Tylor and francis group	2nd edition, 2024

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Human factors method for improving performance in the process industries	Dennis Attwood , et.al	John wiley and sons Inc. Publications	2006
2	A guide to human factors and ergonomics	Martin Helander	CRC press Tylor and francis group	2nd edition, 2005

SEMESTER S8

INDUSTRIAL INSTRUMENTATION

Course Code	PEMRT865	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs.30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To familiarize the students about various types of temperature, pressure, level, pH, density, viscosity, turbidity and conductivity measurement systems

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Measurement of Temperature Definitions and units- Standards of temperature – Thermometry and pyrometry- Thermocouples- Peltier effect- Thomson effect- Seebeck effect- Types of thermocouples (Chromel- copper. Chromel-alumel.copper- constantan, iron-constantan, platinum-rhodium- platinum) Cold junction compensation-Signal conditioning for thermocouple- Thermowells— Thermopiles- Resistance thermometers- Principle of operation- Platinum RTD- Construction of RTDs- Lead compensation- 3wire method – 4 wire method- Thermistors – Quartz crystal sensors- Expansion thermometers- Bimetallic thermometers – Liquid filled thermometers- Gas filled thermometers- Vapor pressure thermometers- Pyrometry- Stefan Boltzmann’s law- Black body radiation- Total radiation pyrometers-Optical radiation pyrometers-Disappearing filament photo electric pyrometer Conductivity Measurement Theory of operation, Two electrode cells, Four electrode measurement, electrodeless cell	9
2	Measurement of Pressure Introduction – units and definitions – standards of pressure – bellows type pressure sensors: motion balance and force balance pressure sensors, C-	9

	bourdon, spiral and helical bourdon pressure sensors-Diaphragm or capsule type sensors: motion balance and force balance pressure sensors-Differential pressure elements: dry, force balance and dry, motion balance-Electronic pressure sensors: strain gauge, Capacitance, Potentiometric, Resonant wire, Piezoelectric, Magnetic and Optical transducers, High-Pressure sensors: Dead-Weight piston gauge, Bulk modulus cells, Manometers: U tube manometer – inclined manometer- ring balanced type manometer, Micromanometers. Vacuum pressure measuring elements- McLeod gauge – ionization gauges – Knudsen gauge – momentum transfer gauges- thermal conductivity gauges – Pirani's gauge.	
3	Level measurement Bubblers-Capacitance level sensors-Level switches; field effect and diaphragm switch, Differential pressure level detector: electronic d/p transmitters, displacer level detectors: torque tube displacer, interface measurement, Float level devices: float level switch ,Laser level detectors: triangular measurement, time-of-reflection measurement. Level gauges: Flat glass gauge, magnetic level gauge, remote reading gauge, Optical level devices; Light reflection, transmission and refraction. Radar level transmitters: time-of flight determination, radiation level sensor; continuous level measurement, Ultrasonic level measurement, Resistance tape level measurement, Rotating paddle switch	9
4	Density Measurement Displacement and float type, hydrometer, hydrostatic densitometers. Radiation densitometer, vibrating densitometer, gas densitometers. pH measurement Theory-Measurement electrodes -Fibre-optic pH measurement-digital pH meters- amplifiers for pH electrodes. Viscosity measurement Laboratory types: Rotational, Saybolt, Redwood, Falling ball and Capillary tube viscometers. Industrial types –Capillary type of viscosity measurements- Falling piston-float viscometer –Oscillating blade. Turbidity measurement Forward scattering type, scattered light detectors, backscatter turbidity analyzer	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

<i>Attendance</i>	<i>Internal Ex</i>	<i>Evaluate</i>	<i>Analyse</i>	<i>Total</i>
5	15	10	10	40

Criteria for Evaluation(Evaluate and Analyse): 20 marks

A micro project with a maximum of 3 students to design and implement a Soft Computing System or model should be given, and the evaluation should be evaluating Bloom's knowledge level. Micro-projects related to Instrumentation and Control Engineering with Society, Sustainability, and ethics in content should be given more weightage.

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 3 sub divisions. Each question carries 9 marks. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain working principle and measurement techniques of various types of temperature measuring and conductivity measuring instruments	K2
CO2	Discuss working principle and measurement techniques of various types of pressure measuring instruments.	K2
CO3	Explain working principle and measurement techniques of various types of level measuring instruments	K2
CO4	Describe working principle and measurement techniques of various types of density, pH, viscosity and turbidity measuring instruments.	K2
CO5	Design and develop an industrial instrumentation project	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											3
CO2	3											3
CO3	3											3
CO4	3											3
CO5	3	3	2	1	3				3	3	2	3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Instrument Engineers Handbook	Liptak.B.G	Chilton book.co.	Fourth,Volume 1, 2003
2	Instrument technology	Jones.E.B	Scientific Pub Elsevier	Volume1,Edition3, 1974
3	Instrumentation Reference Book	Walt Boyes		4/e.2010
4	Applied Instrumentation in Process Industries	Andrew and Williams	Gulf publications	Volume 1,2.Edition2,1979

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Mechanical and Industrial Measurements	Jain.R.K	Khanna Publishers	Edition11,1995
2	Principles of Industrial Instrumentation	Patranabis.D	Tata McGraw Hill	Edition2,2008
3	Measurement System - Application and Design	Doebelin.E.O	McGraw hill	Edition4,1990

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/content/storage2/112/103/112103261/MP4/mod10lec28.mp4 https://archive.nptel.ac.in/content/storage2/112/103/112103261/MP4/mod10lec29.mp4 https://archive.nptel.ac.in/content/storage2/112/103/112103261/MP4/mod10lec27.mp4
2	https://archive.nptel.ac.in/content/storage2/112/103/112103261/MP4/mod08lec21.mp4 https://archive.nptel.ac.in/content/storage2/112/103/112103261/MP4/mod08lec22.mp4 https://archive.nptel.ac.in/content/storage2/112/103/112103261/MP4/mod08lec23.mp4
3	https://archive.nptel.ac.in/content/storage/103/105/103105130/MP4/mod10lec46.mp4 https://archive.nptel.ac.in/content/storage/103/105/103105130/MP4/mod10lec47.mp4 https://archive.nptel.ac.in/content/storage/103/105/103105130/MP4/mod10lec48.mp4 https://archive.nptel.ac.in/content/storage/103/105/103105130/MP4/mod10lec49.mp4
4	https://archive.nptel.ac.in/content/storage/103/105/103105130/MP4/mod11lec54.mp4 https://archive.nptel.ac.in/content/storage/103/105/103105130/MP4/mod11lec55.mp4

SEMESTER S8

AUTOTRONICS

Course Code	OEMRT831	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To understand the fundamentals and control of automobiles using electronic equipment.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Automotive fundamentals: The engine, components, drive train, starting & charging systems operation, ignition system, suspension systems, brakes, steering system. Sensors and actuators for power train and chassis systems: Introduction, working principle, throttle position, Manifold absolute pressure (MAP), lambda sensor, detonation, emission.	9
2	SI Engine Management: Feedback carburettor system, throttle body injection and multi point fuel injection system, injection system controls, advantage of electronic ignition systems, three way catalytic converter, group and sequential injection techniques. Working of the fuel system components. Advantages of electronic ignition systems CI Engine Management: Parameters affecting combustion, noise and emissions in CI engines. Pilot, main, advanced, post injection and retarded post injection layout of the Common Rail Fuel Injection system, working of components like fuel injector, fuel pump, rail pressure limiter, flow limiter, EGR valve control in electronically controlled systems.	9
3	Safety and comfort : Antilock Braking System (ABS), Electronic Stability Control (ESC), Roll Stability Control (RSC), traction control system,	9

	adaptive cruise control, electric power steering, electronic clutch, electronic suspension system, airbags. Electric vehicles and hybrid vehicles: Introduction, electric vehicle development, system layout, basic system components, fuel cell, electric vehicle, hybrid vehicle: series hybrid vehicle, parallel hybrid vehicle, CNG-electric hybrid vehicle.	
4	Vehicle Intelligence: Introduction, basic structure, vision based autonomous road vehicles, architecture for dynamic vision system, features, applications, application of mobile robot vision to a vehicle information system, object detection, collision warning and avoidance system, low tire pressure warning system.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Acquire knowledge on various electronic controls, sensors and actuators used in automobiles	K2
CO2	Obtain an overview of components, sub-systems, and basics of engine management employed in today's automotive industry.	K2
CO3	Apply fundamental knowledge for design of safety systems like airbags, ABS, etc. in automobiles.	K3
CO4	Comprehend the basic system components and their functions of electric and hybrid vehicles. Acquire a fundamental knowledge on intelligence vehicle systems.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2										
CO2	3	2										
CO3	3	2	2						1			
CO4	3	2	2						1			2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Automobile electrical and electronic systems	Tom Denton	BH Publications	3rd Ed., 2004
2	Automobile Electrical & Electronic Equipments	Young, Griffiths	Butterworths	
3	Automotive Electronics Design Fundamental	Najamuz Zaman	Springer,	1st Ed., 2015
4	Fundamentals of Motor Vehicle Technology on Chassis and Body Electronics	Hillier's	Nelson	5th Ed.

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Automotive Technology	Jack Erjavec, Robert Scharff	Delmar publications Inc	1992
2	Electric and Hybrid-electric vehicles	Ronald K. Jorgen	SAE	2002
3	Vision-based Vehicle Guidance	Ichiro Masaki	Springer Verlag, Newyork	1992
4	Class room manual for automotive service and system	Jay Webster	Delmer	1995

SEMESTER S6

ADVANCED AUTOMATION

Course Code	OEMRT832	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. Make students familiar with automation and control technologies in modern manufacturing.
2. Determine the modern application of automation systems in manufacturing industry.
3. Provide knowledge on the elements of modern manufacturing systems.
4. Examine the mechanisms of CMM and FMS.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Production system facilities-Manufacturing support systems - Automation in production systems-manual labor in production systems - automation principles and strategies USA principle-ten strategies of Automation and Production Systems - Automation Migration strategy-manufacturing industries and products-manufacturing operations-processing and assembly operations-product / production relationships. Classification of manufacturing systems	9
2	Elements of an automated system-advanced automation functions - safety monitoring-maintenance and repair diagnostics - capabilities of computer control and levels of industrial process control-computer process monitoring-direct digital control-numerical control and robotics-PLC. Components of a manufacturing system - production machines - material handling system.	9
3	Automation levels - part or product variety - Type I type II and type III	9

	manufacturing systems. Part families - parts classification and coding-features and examples of part classification and coding systems - production flow analysis-cellular manufacturing - application of group technology. Inspection metrology-coordinate measuring machines - CMM construction - CMM operation and planning - CMM applications and benefits.	
4	Machine vision - image acquisition and digitizing - machine vision applications. Flexible manufacturing systems - types of FMS - FMS components - workstations - material handling and storage systems - computer control systems - human resources - FMS applications and benefits - FMS planning and implementation issues - FMS planning and design issues - FMS operational issues. Comparison of lean and agile manufacturing.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the principles of automation systems and to determine the relationship between product and production.	K2
CO2	Analyse the different elements of automation system and to find the importance of control systems in automation.	K3
CO3	Classify the manufacturing systems, what are the components of manufacturing systems and understand about CMM.	K2
CO4	Explain about Machine vision in manufacturing system and FMS.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1		1								
CO2	3	2	1	2								2
CO3	3	3										
CO4	3	2	3	2								2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Automation, Production Systems and Computer – Integrated Manufacturing	Mikell P Groover,	Pearson Education	2001

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	CAD/CAM and CIM	Radhakrishnan, P Subramanian	Wiley Eastern	2003
2	Advamced Automation Techniques in Adaptive material processing	XiaoQi Chen, Rajagopalan Devanathan, Aik Meng Fong	World Scientific, first edition	2002
3	Principles of automation and Advanced Manufacturing System	Dr. KC Jain, Sanjay Jain	Khanna Publishers	2004